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Assessing the Impact of Intercropping on Crop Type Classification
Using Combined Multispectral and Synthetic Aperture Radar
Satellite Imagery, Nyagatare, Rwanda.

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Abstract

Accurate crop type classification maps are an important tool for agricultural and food security monitoring, especially in sub-Saharan Africa. Traditional field survey methods are time-consuming and labour-intensive, making it challenging to collect data at the frequency and spatial resolution required for effective agricultural monitoring. Satellite-based crop type classification can offer a cost-effective alternative, however the high prevalence of intercropping present in sub-Saharan Africa poses significant challenges. This dissertation focuses on classifying agricultural land and crop types using combined multispectral and Synthetic Aperture Radar (SAR) satellite imagery in Nyagatare District, Rwanda – representative of a high cloud cover, topographically complex tropical agricultural area with small field sizes and a high prevalence of intercropping. The final Random Forest model achieved a test accuracy of 94.2%, Kappa score of 0.932, and weighted F1 score of 0.941. Among the main crop types present in the study area, Rice had an F1 score of 0.947, Bananas 0.811, and Dryland Cereals 0.737. Sensitivity analysis was undertaken to assess the impact of SAR and spectral index feature-level data fusion on results, with the inclusion of VV SAR data in particular improving classification performance for texturally distinct crop types. This dissertation highlights a gap in the literature regarding the impact of intercropping on classification accuracy. The impact was quantified by crop type, with reductions in F1 score ranging from 0.315 to 0.494 for the main crop types present, indicating that traditional crop type classification methodologies can produce highly misleading performance metrics in areas with prevalent intercropping. General recommendations are then made in relation to the methods used for classifying agricultural land and agricultural land by crop type in the representative study area.

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Chapter 1 – Introduction

1.1 Rwandan Agriculture

Rwanda is known as “the land of a thousand hills” and is situated in sub-Saharan Africa. The country has a fragmented agricultural system with a high prevalence of subsistence farming and intercropping in primarily small fields on hilly terrain (FAO., 2024) (NISR., 2021a). Rwandan agriculture is heavily rain dependant, with the country experiencing heavy rainfall between September and June and a following dry season from July to August, with two main cultivable seasons: Season A running from September to February and Season B running from March to June (NISR., 2021b). Agriculture comprises a large portion of the Rwandan economy, employing around 70% of the total population, comprising approximately 33% of national GDP, and contributing 63% to the value of Rwanda’s exports (FAO., 2024) (MINAGRI., 2024). To monitor agricultural production in the country, the National Institute of Statistics of Rwanda (NISR) run a Seasonal Agricultural Survey (SAS) program which captures information about agricultural practices and crop production for each agricultural season (NISR., 2021b).

1.2 Satellite Remote Sensing in Agriculture

Satellite remote sensing is the process of capturing information about the earth’s surface via satellite-based sensors. This technology has been utilised extensively for agricultural monitoring (Weiss et al., 2020), including crop stress detection (Garbulsky et al., 2011), crop phenology monitoring (Bolton & Friedl., 2013), crop yield prediction (Lobell et al., 2015), and crop type classification (Defourny et al., 2019). Crop type classification can provide cost-effective and high-resolution data on the distribution of crop types and has been studied extensively academically (Orynbaikyzy et al., 2019). This technology has been operationalised in both the USA’s Crop Data Layer (CDL) program (Boryan et al., 2011) and the EU’s Monitoring Agricultural Resources (MARS) program (Van der Velde et al., 2019), with these programs facilitating crop type monitoring and crop yield forecasting, providing information for agricultural policy monitoring and development.

1.3 Justification

Despite rapid and continuous improvements in both agricultural productivity and food security in much of the developing world over the past 50 years, many countries in Sub-Saharan Africa have experienced the inverse of this trend in the past decade (Wudil et

al., 2022). Food insecurity and malnutrition are major factors in high infant mortality rates and poor health outcomes in the region (Beyene., 2023), and are detrimental to countries in the region developing economically (Wudil et al., 2022). Timely high-accuracy crop production information is needed to monitor food security and inform agricultural policies, with crop type classification using remote sensing being a key tool which can help facilitate this (Fritz et al., 2013). Operationalised crop type classification would be particularly valuable for Rwanda given the high prevalence of subsistence farming and the country's economic reliance on agriculture (FAO., 2024) (MINAGRI., 2024). Traditional field surveys such as Rwanda's SAS are time-consuming and labour-intensive, making it challenging to collect data at the frequency and spatial resolution needed for effective in-season agricultural monitoring (Hegarty-Craver et al., 2020). Satellite remote sensing offers a cost-effective alternative, providing spatially disaggregated, high spatial and temporal resolution data on the distribution of crop types. This information is important for Rwanda to effectively plan resource allocation, monitor food security, and inform agricultural and economic policies as it aims to transform its largely subsistence-based agricultural sector into a more market-oriented and value-adding industry (MINAGRI., 2018).

1.4 Gaps in the Literature

From a review of the literature on crop type classification using remote sensing, there were two key gaps identified which this dissertation aims to fill:

1. Firstly, there is a locational gap in the literature. Sub-Saharan Africa is an underserved region for crop type classification studies broadly, especially Rwanda. Sub-Saharan Africa has specific challenges which make undertaking crop type classification more difficult than in other regions, including high cloud cover, small field sizes, and a high prevalence of intercropping. However, given the ubiquity of subsistence farming in the region and the economic reliance on agriculture, the region arguably has the most to gain from high-accuracy crop type classification. Therefore, this dissertation and its focus on region specific crop type classification methodologies and challenges in Rwanda, should help fill this locational gap.
2. Secondly, there is a knowledge gap in the literature relating to the impact of intercropping on crop type classification accuracy. Intercropping poses significant methodological challenges for crop type classification, including

complicating the choice of model classes, hindering the production of adequate training data, and reducing model classification accuracies. Many studies in areas where intercropping is present reference intercropping as a reason for low classification accuracies and inter-class confusion, however from a review of the literature, no studies were found which aim to quantify this reduction in accuracy for their respective study areas. The Seasonal Agricultural Survey (SAS) dataset used in this dissertation, which contains widespread intercropping, provides a unique opportunity to study the impact of intercropping on crop type classification accuracy and therefore this dissertation should help fill this knowledge gap.

1.5 Research Questions

There are three research questions this dissertation aims to answer:

1. How accurately can crop type classification using satellite remote sensing be performed in the representative study area?
2. What is the impact of intercropping on crop type classification accuracy within the representative study area?
3. What are the most important methodological decisions, including the impact of Synthetic Aperture Radar (SAR) and spectral index feature-level data fusion, for practical implementation of crop type classification in the representative study area?

1.6 Objectives

The following series of objectives were developed to answer these research questions:

- To perform data linking, data cleaning, and exploratory data analysis of SAS data to identify suitable class groupings for classification.
- To create supplementary remotely sensed ground truth data from high-resolution satellite imagery and combine it with SAS data.
- To identify medium-resolution low-cloud satellite imagery of the study area and generate cloud-free composite imagery for modelling.
- To perform feature-level data fusion to combine optical satellite imagery, radar satellite imagery, and optically derived spectral indices.
- To sample every pixel of the composite imagery falling within the combined SAS agricultural plots to create train-validation and test datasets.

- To create a Random Forest crop type classification model, perform backward selection based on feature importance, and perform hyperparameter tuning.
- To classify every pixel within the study area using the final model and create a crop type classification map.
- To use the final model architecture trained on pure cropped data and tested on both pure and intercropped data to quantify the impact of intercropping on crop type classification in the study area.
- To perform sensitivity analysis on the final model to determine the impact of data-fused features on classification accuracy in the study area, including SAR and spectral indices.

Additionally, while achieving these objectives, a key aim of this dissertation was to use only freely available software and data, apart from SAS data, to outline a process which would not result in any additional software or data costs for NISR.

1.7 Dissertation Structure

This dissertation will firstly consider the relevant literature on crop type classification methodologies using satellite imagery and intercropping in the Literature Review, with gaps in the literature this dissertation aims to fill identified. Secondly, there will be a Study Area section detailing the relevant agricultural and environmental characteristics of the study area. Then the Methodology section will provide an explanation and justification of the methods used in relation to the study area and literature. The methods used for answering each research question will be presented, including analysis of the best performing model, analysis on the impact of intercropping, and sensitivity analysis of data-fused features. Then the findings from both the best performing model, intercropping analysis, and sensitivity analysis will be recorded in the Results section. The Discussion section will include interpretation and analysis of these results, and finally the Conclusion will summarise the findings of this dissertation.

Chapter 2 – Literature Review

2.1 Background

This literature review aims to assess various methods of crop type classification using satellite imagery, identify existing research on the impact of intercropping on classification accuracy, and identify gaps in the literature that this dissertation seeks to address.

2.1.1 Satellite Remote Sensing

Satellite remote sensing is the process of gathering information about the Earth's surface from space using satellites (Campbell & Wynne., 2011). Remote sensing systems are categorised as active or passive. Active sensors, like Synthetic Aperture Radar (SAR), emit their own electromagnetic energy and measure the reflected signal (Curlander & McDonough., 1991), while passive sensors, such as optical multispectral imagers, detect naturally reflected or emitted energy from the Earth's surface (Campbell & Wynne., 2011). Multispectral imagery captures reflected solar radiation in multiple spectral bands, typically covering visible and infrared wavelengths (Richards., 2022). SAR emits microwave signals, measuring the intensity and phase of the reflected energy, enabling it to penetrate clouds and operate independently of solar illumination (Ulaby et al., 1986). Remote sensing has been utilised in a vast array of use cases including urban monitoring (Liu et al., 2015), disaster monitoring (Voigt et al., 2007), marine monitoring (Dohan et al., 2010), vegetation monitoring (Gašparović et al., 2018), and land use mapping (Phiri et al., 2020), among other applications.

2.1.2 Satellite Remote Sensing in Agriculture

Satellite remote sensing has been utilised extensively for agricultural monitoring (Weiss et al., 2020), including biomass estimation (Dong et al., 2016), soil moisture assessment (Hassan-Esfahani et al., 2015), crop stress detection (Garbulsky et al., 2011), crop phenology monitoring (Bolton & Friedl., 2013), crop yield prediction (Lobell et al., 2015), and crop type classification (Defourny et al., 2019). The integration of optical and SAR data has become increasingly common, leveraging the complementary strengths of both technologies for improved agricultural monitoring (Weiss et al., 2020), and the arrival of cloud computing for remote sensing through Google Earth Engine has eased the technical difficulties associated with multi-sensor methods (Gorelick et al., 2017).

2.2 Crop Type Classification

2.2.1 Optical

Optical satellite remote sensing has been extensively utilised for crop type classification. Traditionally, Landsat and MODIS were the primary optical sensors used, due to their wide coverage and long-term data archives (Pluto-Kossakowska, 2021), and have been shown to produce high accuracy crop type classification (Cai et al., 2018) (Wardlow et al., 2007). Landsat imagery has been successfully operationalised in the USA's Crop Data Layer (CDL) program (Boryan et al., 2011), facilitating high-accuracy crop type monitoring and crop yield forecasting, providing information for agricultural policy monitoring and development. However, in recent years, Sentinel-2 has also emerged as a common sensor for crop type classification, due to its improved spatial and temporal resolution (Mahlayeye et al., 2022) (Orynbaikyzy et al., 2019), and has been shown to produce high accuracy crop type mapping (Feng et al., 2019) (Saini & Ghosh, 2018) (Vuolo et al., 2018) even with small field sizes (Maolan et al., 2024). Sentinel-2 imagery has been successfully operationalised in the EU's Monitoring Agricultural Resources (MARS) program (Van der Velde et al., 2019), providing information for agricultural policy monitoring and development. However, optical sensors have limitations in distinguishing certain crop types and are hindered by cloud cover, which can result in data gaps during critical crop growth stages, potentially reducing classification accuracy, especially in high cloud cover regions (McNairn et al., 2009) (Orynbaikyzy et al., 2019).

2.2.2 Synthetic Aperture Radar

Synthetic Aperture Radar (SAR) satellite remote sensing has been increasingly utilised for crop type classification. Traditionally, RADARSAT-2 and ERS were the primary SAR sensors used and have been shown to produce high accuracy crop type classification (Chakraborty et al., 1997) (Jiao et al., 2014). However, in recent years, Sentinel-1 has also emerged as a common SAR sensor for crop type classification, due to its improved temporal resolution (Orynbaikyzy et al., 2019), and has been shown to produce high accuracy crop type mapping (Arias et al., 2020) (Kenduiywo et al., 2018) (Xu et al., 2019). Sentinel-1 imagery has been successfully operationalised in the EU's Monitoring Agricultural Resources (MARS) program (Van der Velde et al., 2019), providing information for agricultural policy monitoring and development. SAR sensors have advantages over optical sensors in distinguishing certain crop types based on their

structural properties and are largely unaffected by cloud cover, which allows for consistent data acquisition during critical crop growth stages (Orynbaikyzy et al., 2019). However, SAR sensors have limitations in distinguishing crop types with similar structural properties and can be affected by soil moisture and roughness, potentially reducing classification accuracy especially in areas with high soil variability (Mahlayeye et al., 2022) (Orynbaikyzy et al., 2019).

2.2.3 Combined Optical and SAR

Combined optical and SAR satellite remote sensing has been increasingly utilised for crop type classification, combining the advantages of both individual approaches. In a review of the literature, Orynbaikyzy et al., 2019 found integrations of Landsat with RADARSAT to be the most commonly used combined approach, which has been shown to produce high accuracy crop type classification (Park et al., 2018) (Skakun et al., 2015). However, in recent years, the combination of Sentinel-2 and Sentinel-1 has also emerged as a common approach for crop type classification, due to their improved spatial and temporal resolutions and complementary nature (Orynbaikyzy et al., 2019), and has been shown to produce high accuracy crop type mapping (Kussul et al., 2018) (Orynbaikyzy et al., 2020) (Sonobe et al., 2017). Combined optical and SAR approaches have advantages in distinguishing a wider range of crop types by leveraging both spectral and structural properties (Orynbaikyzy et al., 2019), and have been shown to increase classification accuracy over single sensor methods (Kussul et al., 2018) (Orynbaikyzy et al., 2020) (Sonobe et al., 2017). However, combined approaches can be computationally intensive and require careful feature selection to avoid data redundancy (Orynbaikyzy et al., 2019).

2.3 Intercropping

2.3.1 Definitions

Cropping patterns can be broadly categorised into single cropping (pure cropping) and multiple cropping, with multiple cropping further divided into sequential cropping and intercropping (Mahlayeye et al., 2022). Sequential cropping is defined as the practice of growing two or more crops consecutively in one growing season (Mahlayeye et al., 2022). Intercropping is defined as the practice of growing two or more crops simultaneously in the same field (Bégué et al., 2018). There are several types of intercropping, including mixed, row, and strip cropping, which differ in the spatial arrangement of crops within a field (Mahlayeye et al., 2022), detailed in Figure. 2.3.1.

Farmers adopt intercropping for various reasons, including maximising land use, and increasing crop and income diversity to mitigating the risk of yield loss (Giller et al., 2021) (Himmelstein et al., 2017) (Zuo et al., 2013). Intercropping offers numerous additional benefits such as weed reduction, pest and disease control, soil nutrient preservation, reduced rainwater runoff, and enhanced biodiversity (Bybee-Finley & Ryan., 2018) (Liu & Chen., 2019) (Sun et al., 2019a) (Yang et al., 2020) (Zaefarian & Rezvani., 2016). On average, intercropping can increase crop yields by 23% compared to pure cropping (Himmelstein et al., 2017). Intercropping is particularly prevalent in Africa, where it often forms the backbone of agricultural systems (Garrett et al., 2020).

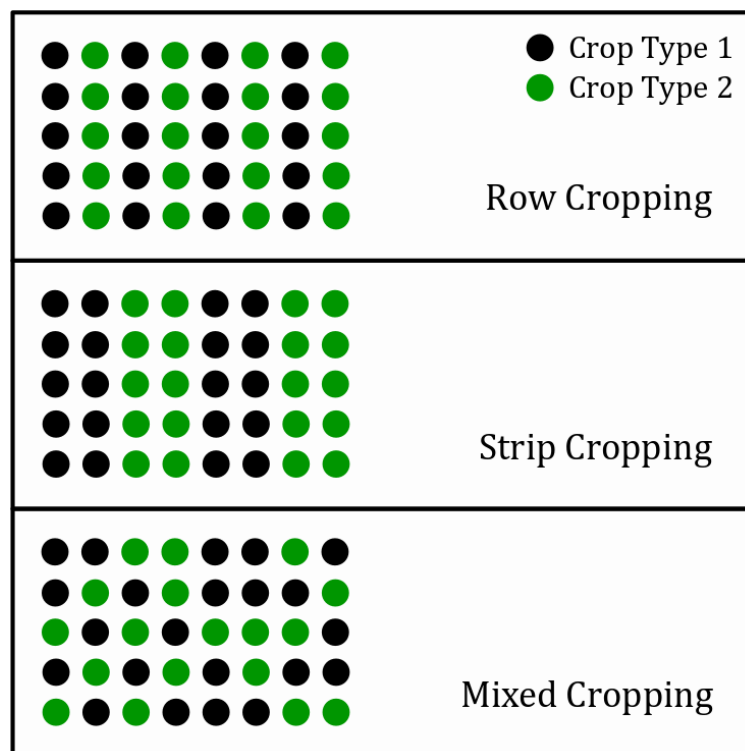


Figure. 2.3.1. A figure showing different types of single time point intercropping for two crop types.

2.3.2 Cropping Pattern Classification

Satellite remote sensing has been increasingly utilised for cropping pattern classification, including intercropping, although research on intercropping specifically remains limited (Mahlayeye et al., 2022). Optical sensors, particularly Sentinel-2 and Landsat, have been the primary tools for mapping intercropping (Gumma et al., 2022) (Luciani et al., 2019). These sensors have enabled the classification of maize, wheat, and peas intercropped with additional crops, with studies highlighting the importance of Short-Wave Infrared (SWIR) bands for crop identification (Feng et al., 2019) (Gumma et al., 2022) (Luciani et al., 2019). Additionally, red-edge bands have proven effective in

mapping maize-bean intercropping (Richard et al., 2017). While optical sensors dominate intercropping research, there is growing interest in using SAR sensors (Mahlayeye et al., 2022), which offer complementary information on crop structure and moisture content (McNairn et al., 2009) (Veloso et al., 2017). Although SAR has been less utilised for intercropping specifically, it has been utilised to map various cropping patterns (Kenduiwo et al., 2018) (Minh et al., 2019) (Useye & Chen., 2019). While still largely unexplored, in a review of the cropping pattern classification literature, Mahlayeye et al., 2022 identified combined optical-SAR methods as a key area for future research.

2.3.3 Quantifying the Impact of Intercropping

From a review of the crop type classification literature, it was determined that studies in areas where intercropping is present often reference intercropping as a reason for low classification accuracies and inter-class confusion (Forkuor et al., 2014) (Hegarty-Craver et al., 2020) (Ibrahim et al., 2021). However, no studies were found which aim to quantify these reductions in accuracy for their respective study areas. This is problematic as studies in heavily intercropped regions which train and test on pure cropped data (or even specific intercropped combinations) may therefore produce test accuracies not representative of their models' performance on their respective study areas as a whole. The reduction in accuracy as a result of intercropping does not appear to have been studied in any regions and therefore possible reductions in accuracy for studies in heavily intercropped regions are not known. There is therefore a knowledge gap in the crop type classification literature on the impact of intercropping on crop classification accuracies.

2.4 Machine Learning Models for Crop Type Classification

In a review of the literature, Pluto-Kossakowska., 2021 identified Support Vector Machines (SVM) (Cortes & Vapnik., 1995), Artificial Neural Networks (ANNs) (Rumelhart et al., 1986), and Random Forest (Breiman., 2001) as the three most commonly used machine learning algorithms for crop type classification. Performance comparisons between these algorithms in the literature are often contradictory across studies, however each algorithm has situational advantages and disadvantages.

2.4.1 Support Vector Machines

Support Vector Machines (SVMs) are a supervised machine learning algorithm which aims to find the optimal hyperplane that separates different classes in a high-

dimensional feature space (Cortes & Vapnik., 1995). SVMs have been used extensively for crop type classification using optical, radar, and combined optical-radar satellite data (McNairn et al., 2009) (Orynbaikyzy et al., 2019), often outperforming other classifiers (Nitze et al., 2012). Advantages of SVMs include their ability to handle high-dimensional data and their effectiveness with limited training data (Mountrakis et al., 2011). However, disadvantages are that SVMs can be sensitive to the choice of kernel and features, have a multitude of parameters to optimise, and may struggle with large datasets due to high computational requirements (Mountrakis et al., 2011).

2.4.2 Artificial Neural Networks

Artificial Neural Networks (ANNs) are machine learning models consisting of interconnected nodes organised in layers (Rumelhart et al., 1986). ANNs have been used extensively for crop type classification using optical, radar, and combined optical-radar satellite data (Orynbaikyzy et al., 2019). In a review of the literature Pluto-Kossakowska., 2021 found that ANN classifiers on average achieve slightly superior results compared to other classifiers for crop type mapping (Mazzia et al., 2020) (Skakun et al., 2015). Advantages of ANNs include their ability to handle complex, non-linear relationships in data and their capacity to learn from large datasets (Kussul et al., 2017). However, disadvantages include the long training times required to build and optimise the network, the need for large training datasets, and the potential for overfitting (Maxwell et al., 2018) (Pluto-Kossakowska., 2021).

2.4.3 Random Forest

Random Forest is an ensemble learning method that constructs multiple decision trees with each tree created using its own randomly generated subset of the training data and a random subset of features (Breiman., 2001). The majority vote of all the trees is then used to assign final class predictions. Random Forest has been used extensively for crop type classification using optical, radar, and combined optical-radar satellite data (Orynbaikyzy et al., 2019), often outperforming other classifiers, especially when dealing with complex and heterogeneous datasets typical in crop type classification studies (Espinell et al., 2024) (Ok et al., 2012) (Saini & Ghosh., 2018). Advantages of Random Forest include its general robustness to overfitting, ability to handle high-dimensional data, minimal parameter tuning, capacity to assess feature importance, low computational requirements, and resistance to “noise” often prevalent in crop type reference data (Belgiu & Drăguț., 2016) (Maxwell et al., 2018) (Pluto-Kossakowska.,

2021) (Tatsumi et al., 2015). However, disadvantages include the algorithm's tendency to overfit with very noisy datasets, a lack of accuracy increase with increased training data and its limited ability to extrapolate beyond the range of the training data (Belgiu & Drăguț, 2016) (Maxwell et al., 2018) (Pluto-Kossakowska, 2021).

2.5 Ground Truth Data Methods

Ground truth reference data quality plays a crucial role in supervised crop type classification, significantly influencing the accuracy and reliability of results. Field surveys remain the primary source, with researchers gathering observations during site visits (Asgarian et al., 2016) (Immitzer et al., 2016) (Inglada et al., 2016), however this method is expensive. Farmer declarations (Abdikan et al., 2015), agricultural company records (Foerster et al., 2012), and governmental agricultural records and statistics (Lussem et al., 2016) have also been used. Additionally, visual interpretation of very high-resolution optical satellite imagery (Park & Im., 2016) and UAV-based optical imagery (Hegarty-Craver et al., 2020) have been utilised to derive reference data.

2.6 Multitemporal versus Single Image Classification

There are two methods of crop type classification in terms of the temporal dimension of the satellite imagery used: single time point and multitemporal classification. Single time point classification relies on imagery acquired at a specific date, with the dates usually selected to maximise spectral and/or radiometric separability between crops (Orynbaikyzy et al., 2019). Different crop types have variable spectral and radiometric separability throughout the growing season (Foerster et al., 2012) (Panigrahy & Sharma, 1996), and therefore the selected image dates are highly dependent on study area context. Single image methods can be effective but have limitations due to their inability to capture crop phenology throughout a growing season. In contrast, multitemporal classification utilises satellite image time series to monitor crop development throughout the growing season and can effectively capture phenological differences between crop types (Mahlayeye et al., 2022). Multitemporal methods can achieve higher crop type classification accuracies than single image classification (Sun et al., 2019b) (Vuolo et al., 2018). However, this approach is predicated on the availability of cloud-free satellite image time series which can be challenging in certain regions (Whitcraft et al., 2015).

2.7 Spectral Indices

Spectral indices are mathematical combinations of reflectance values from different spectral bands that enhance the visibility of specific land cover features, especially vegetation (Huete et al., 2002). In crop type classification, the most commonly used indices are the Normalised Difference Vegetation Index (NDVI), Enhanced Vegetation Index (EVI), Normalised Difference Water Index (NDWI), and Simple Ratio (SR) (Orynbaikyzy et al., 2019), with additional indices such as Soil-Adjusted Vegetation Index (SAVI) (Khosravi et al., 2018) and Green Normalised Difference Vegetation Index (GNDVI) (Karimi et al., 2022) also used. These indices leverage the reflective properties of crops in the visible and near-infrared wavelengths to provide information on vegetation condition, chlorophyll content, and plant water content to aid in crop type classification (Orynbaikyzy et al., 2019). NDVI and EVI have been found to be particularly effective for crop type classification (Peña-Barragán et al., 2011) (Vidican et al., 2023) (Wardlow et al., 2006).

2.8 Data Fusion Methods

In a review of the literature Orynbaikyzy et al., 2019 detailed three types of data fusion methods for combining remote sensing data to increase crop type classification accuracy: pixel-level, feature-level, and decision-level data fusion. Pixel-level data fusion is when the individual pixel values from multiple datasets are combined at a low processing level to create new pixel values, often to enhance the spatial resolution of the imagery. Feature-level data fusion is when features or variables extracted from the original datasets, such as spectral bands, vegetation indices, or radar backscatter information, are combined at an intermediate processing level pre-modelling. Decision-level data fusion is when the results of individual classifications from multiple datasets are combined at a higher processing level, with outcomes merged to produce a final classification map. Pixel-level (Gibril et al., 2017) (Haldar et al., 2012) (Sukawattanavijit & Chen., 2015), feature-level (Forkuor et al., 2014) (Mansaray et al., 2017) (Torbick et al., 2017), and decision-level data fusion methods (Ban et al., 2010) (Waske & van der Linden., 2008) have all been shown to increase crop type classification accuracy over non-data fusion approaches.

2.9 Feature Selection Methods

Feature selection is a critical step in crop type classification as it helps identify the most informative spectral and radiometric variables while reducing data redundancy and

computational requirements. Several approaches have been employed for feature selection in crop type classification studies including local sensitivity analysis (Zhou et al., 2017), principal component analysis (Paul & Kumar., 2019), transformed divergence separability (Dusseux et al., 2014), Jeffries-Matusita distance (Erasmí & Twele., 2009), and combined distance and feature correlation-based approaches (Khosravi et al., 2018). However, Random Forest based methods allow for the use of in-built feature importance estimated during model training to rank and select the most relevant features, providing an efficient and effective approach for feature selection in crop type classification (Belgiu & Drăguț., 2016) (Inglada et al., 2016).

2.10 Pixel-based versus Object-based Classification

The two main methods for analysing remote sensing imagery for crop type classification are pixel-based and object-based classification (Pluto-Kossakowska., 2021). Pixel-based methods classify each individual pixel based solely on its spectral or radiometric information, while object-based approaches first segment the image into homogeneous objects before classification (Orynbaikyzy et al., 2019). Object-based classification has some key advantages over pixel-based methods. By considering contextual information like the shape, size, and spatial relationships between objects, object-based approaches can better represent landscape features and reduce the "salt and pepper" effect common in pixel-based results due to misclassified pixels (Orynbaikyzy et al., 2019). For example, object-based methods can easily distinguish roads from bare agricultural fields by analysing shape characteristics (Ban et al., 2010). Several studies have found that object-based classification methods result in higher crop type classification accuracy (Castillejo-González et al., 2009) (Gibril et al., 2017) (Salehi et al., 2017), with other studies finding similar accuracies between methods (Duro et al., 2012) (Esetlili et al., 2018). However, the effectiveness of object-based classification depends heavily on the quality of image segmentation (Shackelford & Davis., 2003), which can be challenging in complex, heterogeneous agricultural landscapes with small field sizes (Forkuor., 2015).

2.11 Crop Type Classification Challenges

2.11.1 Intercropping

Intercropping presents significant challenges for crop type classification. In intercropped fields, multiple crop types grow simultaneously, creating complex spectral and spatial patterns that are difficult to distinguish and classify accurately (Bégué et al.,

2018) (Mahlayeye et al., 2022). The spatial resolution of satellite sensors is often insufficient to resolve individual crop rows or patches within intercropped fields resulting in mixed pixels and spectral signatures that confound classification algorithms (Forkuor et al., 2014) (Hegarty-Craver et al., 2020) (Ibrahim et al., 2021). Additionally, the temporal dynamics of intercropped systems, where different crops may have overlapping or asynchronous growth cycles, can complicate the use of multitemporal methods for crop type classification (Mahlayeye et al., 2022). These factors make it challenging to develop robust and accurate classification methods for intercropped agricultural landscapes, particularly when using lower resolution satellite imagery.

2.11.2 Crop Spectral and Radiometric Distinguishability

Spectral and textural similarities between different crops can make crop type classification challenging. Crops with similar phenological cycles or structural properties often exhibit comparable spectral signatures and backscatter characteristics in optical and SAR data respectively (Orynbaikyzy et al., 2019). This can lead to misclassification, especially in agricultural landscapes with complex cropping patterns such as intercropping (Luciani et al., 2019). For example, maize and sorghum can have similar spectral profiles (Luciani et al., 2019), while crops with comparable canopy structures can produce similar SAR backscatter values (Steele-Dunne et al., 2017).

2.11.3 High Cloud Cover

High cloud cover regions present significant challenges for crop type classification, especially when relying solely on optical satellite imagery. Persistent cloud cover can result in data gaps and inconsistent time series, limiting the ability to capture crucial phenological stages of crops throughout the growing season (Chen et al., 2018) (Ibrahim et al., 2021). While SAR sensors can penetrate clouds, providing all-weather monitoring capability, they lack the spectral information provided by optical sensors that is often crucial for crop discrimination (McNairn & Brisco., 2004). This is especially problematic in tropical regions, where cloud cover is often persistent and coincides with critical crop growth periods (Ibrahim et al., 2021) (Torbick et al., 2017).

2.11.4 Small Field Sizes

Complex agricultural regions characterised by small field sizes present challenges for crop type classification. The spatial resolution of commonly used satellite sensors, such as Landsat (30m) and MODIS (250m), is often insufficient to accurately capture the heterogeneity of small agricultural plots (Jain et al., 2013). This limitation can lead to

mixed pixels containing multiple crop types or non-crop features, complicating spectral analysis and reducing classification accuracy (Jain et al., 2013). While medium-resolution sensors like Sentinel-2 (10m) offer improved spatial detail, they may still struggle with very small or irregularly shaped fields common in smallholder farming systems especially in Africa (Defourny et al., 2019) (Forkuor et al., 2014) (Ibrahim et al., 2021). Intercropping practices further exacerbate these challenges by creating intricate spectral and spatial patterns that are difficult to discern and classify accurately (Forkuor et al., 2014) (Ibrahim et al., 2021).

2.11.5 Complex Topography

Agricultural regions in hilly and mountainous areas present significant challenges for crop type classification. The topographic complexity introduces issues such as differing illumination which can alter spectral signatures (Ren et al., 2022) (Wang et al., 2012), SAR data distortion (Ren et al., 2022) (Wang et al., 2012), and terrace farming creating unique spatial patterns that may not be well-captured by standard classification approaches (Liu et al., 2020). These challenges are further compounded by intercropping and small fields sizes, often present in topographically complex agricultural regions (Liu et al., 2020).

2.12 Study Area Locations

Crop type classification studies have been undertaken at continental (d'Andrimont et al., 2021), national (Defourny et al., 2019), regional (Kussul et al., 2018), and local (Conrad et al., 2010) scales across the world. However, in a review of the literature on crop type classification using combined optical and radar data, Orynbaikyzy., 2019 found only three out of seventy-five papers covering the topic were undertaken in Africa, with none in East Africa. Rwanda is underserved for crop type classification studies, especially those using combined optical and radar data (Choukri et al., 2024). In the review of the literature for this dissertation only two crop type classification studies using satellite remote sensing were found focussing on Rwanda (Hegarty-Craver et al., 2020) (Polly et al., 2019), with an additional two using UAV-based methods (Kilwenge et al., 2021) (Chew et al., 2020).

2.13 Studies in Rwanda

2.13.1 UAV-based Studies

Kilwenge et al., 2021 used high-resolution multispectral UAV imagery with survey- and UAV-based ground truth data to map banana-cultivated land across four villages in

Rwanda. The study compared three pixel-based supervised classification models: SVM, classification and regression trees, and Random Forest. The Random Forest model consistently outperformed the others, achieving over 95% overall accuracy and a kappa score above 0.94. The study highlighted the challenges of classifying the complex mosaiced farming landscape in Rwanda such as small field sizes, spectral similarity between crops, and the high prevalence of intercropping. The study however does not explicitly detail how intercropping was dealt with in the methodology, and this is a key limitation as bananas are often intercropped across Rwanda (NISR., 2021a).

Chew et al., 2020 used UAV-based optical imagery, UAV-acquired ground truth data, and object-based deep neural networks to classify bananas, maize, and legumes, achieving an overall test F1 score of 0.86, with individual class F1 scores ranging from 0.49 for legumes to 0.96 for bananas. The study highlighted challenges in crop classification for Rwanda, including the high prevalence of intense intercropping systems, small plot sizes, and high within-class crop type heterogeneity. The study dealt with intercropping by having image labellers chose a single majority class for each image as opposed to identifying all crops present, with the intercropping present potentially reducing classification accuracy.

2.13.2 Satellite-based Studies

Polly et al., 2019 used Sentinel-2 optical and Sentinel-1 SAR satellite imagery with UAV-acquired ground truth data to classify maize in Rwanda. Polly et al created separate pixel-based Random Forest models for Sentinel-1 and Sentinel-2 data, achieving higher accuracy with the Sentinel-2 model at 75% user's accuracy for maize, compared to the Sentinel-1 model at <50% user's accuracy. Neither a confusion matrix or producer's accuracy results are provided and therefore it is not possible to determine Macro, Weighted, or by class F1 scores. The study highlighted challenges for crop type classification in Rwanda, including significant cloud cover and the prevalence of small, often intercropped plots. The study however does not explicitly detail how intercropping was dealt with in the methodology and both trains and tests models on pure cropped data. This is a key limitation as maize is heavily intercropped with a wide variety of different crops across Rwanda (NISR., 2021a).

Hegarty-Craver et al., 2020 used combined Sentinel-2 optical and Sentinel-1 SAR satellite imagery with UAV-acquired ground truth data to classify crop types in Rwanda. UAV images corresponding to Sentinel-2 pixels were labelled based on the dominant

landcover (at least 75% of the pixel) with cells not meeting this threshold and cells qualitatively determined to be “poor” examples not included for training and testing. Hegarty-Craver et al developed a pixel-based Random Forest model to classify seven land cover categories, including four key crops. The model achieved an overall accuracy of 83%, with an F1 score of 0.881 for Maize, 0.696 for Beans, 0.632 for Cassava, and 0.733 for Bananas. However, the study used limited reference data at only 123 test set crop data points for the whole of Rwanda including only fifteen for Bananas. The study highlighted challenges for crop type classification in Rwanda, including the spectral similarity between crop types and the prevalence of intercropping. The authors compared their model results to Rwanda's Seasonal Agricultural Survey, finding good agreement for maize and beans but discrepancies for cassava and bananas. The study emphasized the need for larger ground truth datasets and additional categories to better capture intercropping practices.

There is therefore a locational gap in the crop type classification literature for Rwanda with only two studies found undertaking satellite-based crop type classification, and only a single study doing so for multiple crop types. There is also a locational-methodological gap, with no studies explicitly detailing how they dealt with the high prevalence of intercropping for model evaluation, or quantifying the impact of intercropping on classification results.

2.14 Summary of Gaps in the Literature

Firstly, there is a locational gap in the literature. Sub-Saharan Africa is an underserved region for crop type classification studies broadly, especially Rwanda. Sub-Saharan Africa has specific challenges which make undertaking crop type classification more difficult than in other regions, including high cloud cover, small field sizes, and a high prevalence of intercropping. Therefore, this dissertation will focus on region specific crop type classification methodologies and challenges in Rwanda.

Secondly, there is a knowledge gap in the literature relating to the impact of intercropping on crop type classification accuracy. Many studies in areas where intercropping is present reference intercropping as a reason for low classification accuracies and inter-class confusion, however no studies were found which aim to quantify this reduction in accuracy for their respective study areas. Therefore, this dissertation will focus on quantifying the impact of intercropping on crop type classification accuracy in Rwanda.

Chapter 3 – Study Area

3.1 Location

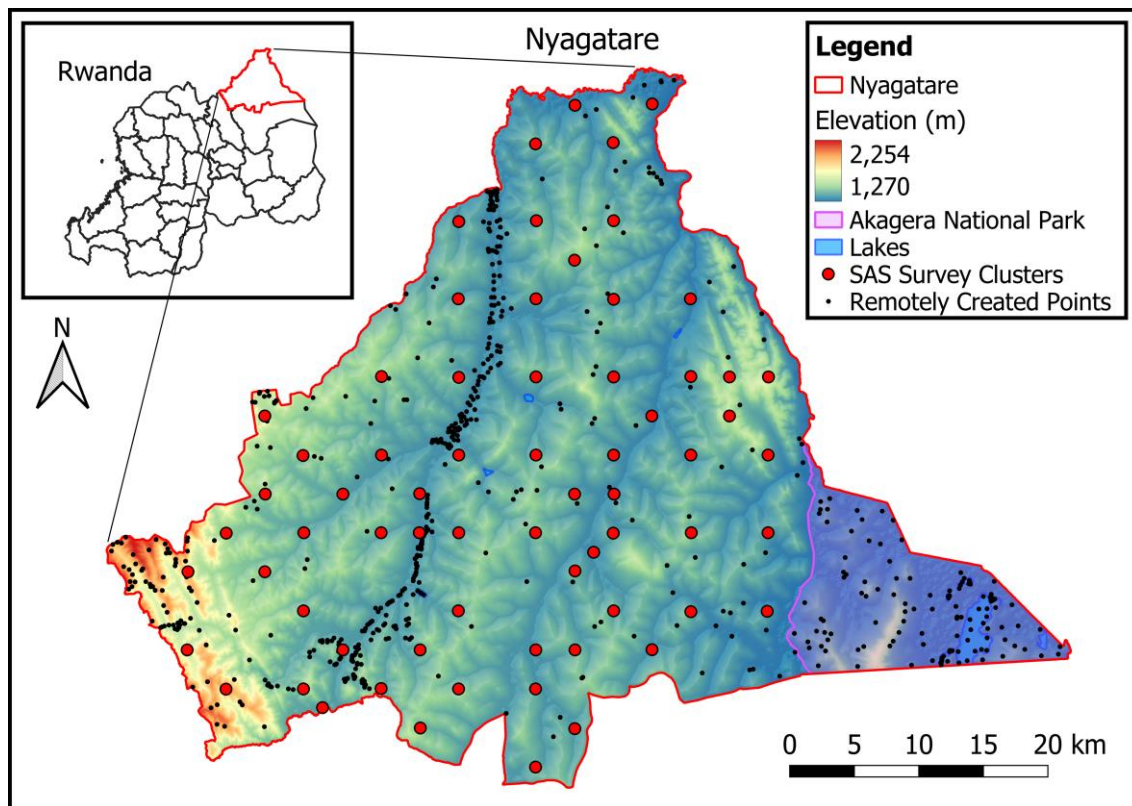


Figure. 3.1. A map showing the study area location and topography, Nyagatare District, Eastern Province, Rwanda. The location of Seasonal Agricultural Survey point clusters and remotely created ground truth data points are shown, discussed fully in Section 4.1.1.

The study area this dissertation will focus on is the district of Nyagatare, Eastern Province, Rwanda (Figure. 3.1.). Nyagatare is located in the Northeast of Rwanda, bordering Uganda and Tanzania.

3.2 Agricultural and Environmental Characteristics

Nyagatare is the largest district by land area in Rwanda at over 1,900km² (NISR., 2021a), and the second largest by population at over 650,000 people (NISR., 2022). Nyagatare is the district with the largest agricultural land area at over 1450km² and highest percentage of agricultural land at 76.5% (NISR., 2021a). Rwanda, including Nyagatare, experiences a large amount of cloud cover especially during the two wet agricultural seasons, Season A and Season B (Weather Spark., 2024). Rwanda is known as “the land of a thousand hills” and is topographically complex, with an estimated 90% of cropland based on slopes between 5% and 55% (FAO., 2024). Nyagatare is slightly less mountainous than Rwanda as a whole but is still topographically complex as shown

in Figure. 3.1. Rwanda, including Nyagatare, has small farms and fields, with an average farm size of 0.4Ha for Rwanda and 0.7Ha for Nyagatare (NISR., 2021c). Rwanda, including Nyagatare, has a high prevalence of intercropping. Rwanda as a whole has 28.3% of cultivated land pure cropped and 71.7% intercropped (NISR., 2021a). Nyagatare has similar with 29.9% of cultivated land pure cropped and 70.1% intercropped (NISR., 2021a). The study area of Nyagatare is therefore representative of a high cloud cover, topographically complex tropical agricultural area with small field sizes and a high prevalence of intercropping.

3.3 Agricultural Practices – Examples

Nyagatare and the two surrounding districts Gicumbi and Gatsibo were visited to observe the agricultural practices present and qualitatively inform methodological decisions for this dissertation. Figure. 3.3. presents photography covering a range of cropping practices present in these three districts to aid the readers' understanding. Gicumbi and Gatsibo both have broadly similar agricultural practices to Nyagatare.



Figure. 3.3.A. A photo showing hilly terrain terraced to facilitate agriculture. There are small (often less than 0.1Ha) heavily intercropped fields containing cereals, bananas, tubers, legumes and pulses. There is hill-based forestry in the foreground. Gicumbi District, Northern Province.

Photograph taken 23rd May 2024. Agricultural Season B.



Figure. 3.3.B. A photo showing a river valley used for agriculture. There are small pure cropped and intercropped fields (less than 0.1Ha) primarily containing potatoes, sweet potatoes and bush beans. There is hill-based forestry in the background. Gicumbi District, Northern Province.

Photograph taken 23rd May 2024. Agricultural Season B.



Figure. 3.3.C. A photo showing hilly terrain used for agriculture. There are small (~0.1 to 0.2Ha) pure cropped fields of bananas, and some hill-based forestry in the background. Nyagatare District, Eastern Province.

Photograph taken 24th May 2024. Agricultural Season B.



Figure. 3.3.D. A photo showing a large river valley (~0.5km across here) used for agriculture. There is a small, intercropped field of maize and bush beans in the foreground and large fields of rice in the background. Nyagatare District, Eastern Province.

Photograph taken 24th May 2024. Agricultural Season B.



Figure. 3.3.E. A photo showing very small agricultural plots next to a house. There are pure cropped bananas and sugar cane. Gatsibo District, Eastern Province.

Photograph taken 24th May 2024. Agricultural Season B.



Figure. 3.3.F. A photo showing me standing in a large river valley (~0.6km across here) used for agriculture. There are small, pure cropped fields of maize and sweet potatoes in the foreground and large fields of rice in the background. Nyagatare District, Eastern Province.

Photograph taken 24th May 2024. Agricultural Season B.

Figure. 3.3. A figure presenting photographs showing a range of agricultural practices present in Nyagatare and the two adjacent districts, Gicumbi and Gatsibo.

Chapter 4 – Methodology

4.1 Data

4.1.1 Seasonal Agricultural Survey Data

4.1.1.1 Agricultural Plot Data

Agricultural plot survey data was obtained for Season A 2021, collected as part of the National Institute of Statistics of Rwanda Seasonal Agricultural Survey (SAS) program (NISR., 2021a). The SAS uses a multiple-frame hierarchical sampling methodology with area frame national scale grid sampling used to determine where sample clusters will be taken from, and area frame grid sampling then taken at 60m intervals within each square cluster for a total of twenty-five sampled points per cluster. It is determined if a sampled point is within an agricultural plot, and based on discussions with farmers the extent of an agricultural plot is then measured using GPS devices, with associated information about the plot collected from surveys filled out jointly with the farmer. When a point does not fall within an agricultural plot, a sample of the non-agricultural land use type present is taken. Additionally, a list frame of large-scale farmers, with at least ten hectares of agricultural land is sampled to supplement the area frame. The resulting data consists of shapefiles of agricultural plots surveyed with associated plot information, including what crops are present within each plot and the land use for non-agricultural plots. The Season A 2021 data contained 634 plots for Nyagatare.

4.1.1.2 Supplementary Agricultural Plot Data

SAS plot data for Nyagatare was supplemented with additional remotely created ground truth data using high-resolution WorldView satellite imagery from Google Earth Pro. This was done to increase the amount of train and test data available for Nyagatare and increase the number of samples for prevalent classes which had not been picked up consistently by grid sampling e.g. Rice. Data from between October 2020 and January 2021 was used for manual plot polygon creation. It was assessed that only certain crop types and land use classes could be accurately manually identified remotely, including Bananas, Bare Earth and Urban, Grassland, Forestry, Rice, Water, and Wetland. Based on exploratory analysis of the SAS data and having visited Nyagatare District and observed the cropping patterns present it was determined that rice was not intercropped and that bananas were either not intercropped and planted close together or intercropped and planted further apart to allow for other crops to grow in between. Based on cross

referencing locations visited in Nyagatare with WorldView satellite imagery for the visit time period it was determined that pure cropped bananas were able to be accurately manually remotely identified and therefore data was able to be created for this class. This supplementary data consisted of an additional 610 plots in total for Nyagatare, detailed in Table. 4.4.4.4.

4.1.2 Sentinel-2 Data

Sentinel-2 multispectral satellite imagery data collected as part of the European Space Agency's Copernicus program (ESA., 2024) was used for this dissertation, accessed via Google Earth Engine (GEE) (Gorelick et al., 2017). Sentinel-2 collects information on multiple parts of the electromagnetic spectrum, including Visible, Near Infrared (NIR), and Short-Wave Infrared (SWIR) spectral bands. The spectral bands available in Sentinel-2 data are detailed in Table. 4.1.2. Sentinel-2 data was chosen as it is freely available, medium-resolution, and has been used extensively for crop type classification.

Band	Resolution (m)	Central Wavelength (nm)	Description
B1	60m	443nm	Coastal Aerosol
B2	10m	490nm	Blue
B3	10m	560nm	Green
B4	10m	665nm	Red
B5	20m	705nm	Vegetation Red Edge
B6	20m	740nm	Vegetation Red Edge
B7	20m	783nm	Vegetation Red Edge
B8	10m	842nm	NIR
B8A	20m	865nm	Narrow NIR
B9	60m	940nm	SWIR – Water Vapour
B10	60m	1375nm	SWIR – Cirrus
B11	20m	1610nm	SWIR
B12	20m	2190nm	SWIR

Table. 4.1.2. A table detailing the spectral bands available in Sentinel-2 data.

4.1.3 Sentinel-1 Data

Sentinel-1 Synthetic Aperture Radar (SAR) satellite imagery data collected as part of the European Space Agency's Copernicus program (ESA., 2024) was used for this dissertation, also accessed via GEE. Sentinel-1 collects information on how radar emitted from the satellite bounces back. The Sentinel-1 data products available in GEE are detailed in Table. 4.1.3. GRD data for VV and VH polarisations was the only data

available via GEE for the study time period and therefore only backscatter data was used as opposed to phase data. Sentinel-1 data was chosen as it is freely available, medium-resolution, and has been used extensively for crop type classification.

Product	Polarisation	Resolution	Description
GRD (Ground Range Detected)	VV, VH, HH, HV	~10m	Backscatter amplitude information, ortho-corrected, terrain corrected
SLC (Single Look Complex)	VV, VH, HH, HV	Variable	Backscatter amplitude and phase information
RTC (Radiometrically Terrain Corrected)	VV, VH, HH, HV	~10m	Backscatter amplitude information, enhanced terrain correction

Table. 4.1.3. A table detailing the data products available from Sentinel-1 via GEE.

4.2 Study Area and Time Period Selection

4.2.1 Nyagatare

Nyagatare was chosen as the focus of this dissertation due to being the district with the largest agricultural land area and highest percentage of agricultural land area (NISR., 2021a), therefore being both agriculturally and economically important for Rwanda. Nyagatare was chosen as it has a wide variety of crop types present, including all main crop types covered in the SAS and similar rates of intercropping to those found in Rwanda as a whole (NISR., 2021a). Therefore, agriculturally Nyagatare is a reasonably good representation of Rwanda as a whole allowing for wider crop type classification recommendations to be made for the whole country from the representative study area.

4.2.2 Season A 2021

Agricultural Season A was chosen as the focus of this dissertation due to it coinciding with the main rainy period and being the most important season both agriculturally and economically. Season A has the most favourable growing conditions and is when farmers typically plant their most important staple crops (NISR., 2021b) and therefore being able to monitor crop types during this season is the main priority for the National Institute of Statistics of Rwanda. 2021 was chosen due to it being after the SAS updated methodology was introduced in 2020 and being the most recent data provided by NISR for undertaking this proof-of-concept dissertation.

4.3 Model Selection

A Random Forest model was chosen, with all implementations using the randomForest package in R (Liaw & Wiener., 2002). Random Forest was chosen due to its general robustness to overfitting and “noise”, ability to handle high-dimensional data, low computational requirements, and in-built capacity to assess feature importance, all useful benefits for crop type classification (Belgiu & Drăguț., 2016) (Maxwell et al., 2018) (Pluto-Kossakowska., 2021) (Tatsumi et al., 2015).

4.4 Model Creation

4.4.1 Initial Feature Selection

4.4.1.1 Sentinel-2 Bands

All Sentinel-2 bands with a resolution of 20m or less were selected initially for inclusion to retain the additional red-edge and SWIR information useful for crop differentiation in intercropped systems (Feng et al., 2019) (Richard et al., 2017), and to reduce the time and space complexity of modelling compared to using 10m bands. Being selective with initial band inclusion was not required due to Random Forests in-built feature importance for final feature selection (Belgiu & Drăguț., 2016).

4.4.1.2 Spectral Indices

NDVI, EVI, NDWI, SR, SAVI, and GNDVI were identified as frequently used spectral indices for crop type classification from the literature, and therefore were initially selected for inclusion. Given that NDWI and GNDVI are perfectly negatively correlated due to their antithetical equations, only NDWI was included to prevent data redundancy. Equations for calculating all included indices are detailed in Figure. 4.4.1.2.

$$\begin{aligned} NDVI &= \frac{B8 - B4}{B8 + B4} \\ EVI &= 2.5 \times \frac{B8 - B4}{B8 + 6 \times B4 - 7.5 \times B2 + 1} \\ NDWI &= \frac{B3 - B8}{B3 + B8} \\ SR &= \frac{B8}{B4} \\ SAVI &= \frac{(B8 - B4) \times (1.5)}{B8 + B4 + 0.5} \\ GNDVI &= \frac{B8 - B3}{B8 + B3} \end{aligned}$$

Figure. 4.4.1.2. A figure showing the equations for calculating Normalised Difference Vegetation Index (NDVI), Normalised Difference Water Index (NDWI), Simple Ratio (SR), Soil-Adjusted Vegetation Index (SAVI), and Green Normalised Difference Vegetation Index (GNDVI) from Sentinel-2 bands.

4.4.1.3 Sentinel-1 SAR

VV and VH backscatter GLM data were selected initially for inclusion due to being the only Sentinel-1 data products available for the study area and time period through GEE. Being selective with initial Sentinel-1 data inclusion was not required due to Random Forests in-built feature importance for final feature selection (Belgiu & Drăguț., 2016).

4.4.1.4 Initial Features

All initial features selected for modelling are detailed in Table. 4.4.1.4.

Feature	Source
Band 2	Sentinel-2
Band 3	Sentinel-2
Band 4	Sentinel-2
Band 5	Sentinel-2
Band 6	Sentinel-2
Band 7	Sentinel-2
Band 8	Sentinel-2
Band 8A	Sentinel-2
Band 11	Sentinel-2
Band 12	Sentinel-2
VV	Sentinel-1
VH	Sentinel-1
NDVI	Calculated from Sentinel-2 bands
EVI	Calculated from Sentinel-2 bands
NDWI	Calculated from Sentinel-2 bands
SR	Calculated from Sentinel-2 bands
SAVI	Calculated from Sentinel-2 bands

Table. 4.4.1.4. A table detailing the initial features selected for modelling.

4.4.2 Cloud-free Image Generation

Sentinel-2 imagery of the study area from September 2020 to February 2021 was manually assessed in GEE to determine the prevalence of cloud cover throughout agricultural Season A. It was determined there was such a high prevalence of cloud cover that it would not be possible to undertake analysis using multitemporal classification techniques. This necessitated a single image classification approach. Sentinel-2 imagery of the study area for January 2021 was manually assessed to identify the most cloud free imagery. January 2021 was chosen due to it being at the end of Season A but prior to harvest to ensure that crops were well developed and at their most spectrally distinguishable (Vuolo et al., 2018). Four scenes from the 7th and 27th of

January were determined to be the lowest cloud imagery available. Two scenes from the 7th and two scenes from the 27th were mosaiced respectively to ensure full coverage of the study area on both dates. These mosaics were not completely cloud free and therefore creating composite imagery was required. A series of composite imagery generation methods for selecting cloud-free pixels were applied to the imagery and assessed qualitatively for how successfully they removed cloud. Qualitatively it was determined that a method of creating a pixel quality score from Normalised Difference Vegetation Index (NDVI) minus Normalised Difference Snow Index (NDSI) resulted in the best cloud-free image generation. Equations for calculating these indices and the pixel quality score used are detailed in Figure. 4.4.2.A. The resulting composite image is shown in Figure. 4.4.2.B.

$$NDVI = \frac{B8 - B4}{B8 + B4}$$

$$NDSI = \frac{B3 - B11}{B3 + B11}$$

$$Pixel\ Quality = NDVI - NDSI$$

Figure. 4.4.2.A. A figure showing the equations for calculating Normalised Difference Vegetation Index (NDVI) and Normalised Difference Snow Index (NDSI) from Sentinel-2 bands, and the equation for calculating pixel quality used for pixel selection in composite image generation.

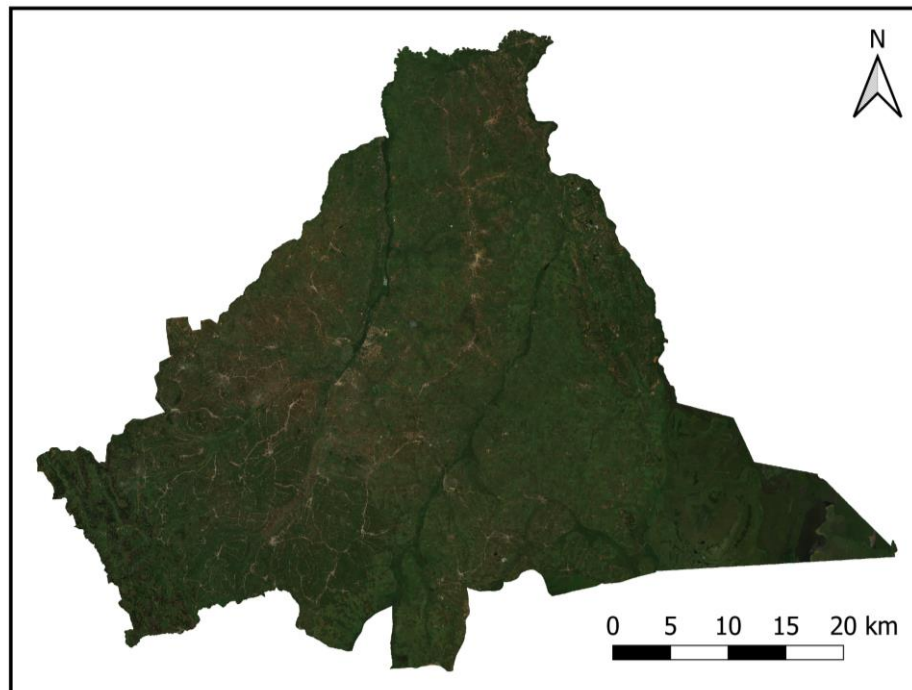


Figure. 4.4.2.B. A map showing the composite image of the study area generated using imagery from the 7th and 27th of January 2021, visualised using real colour RGB bands.

4.4.3 Data Preprocessing

Feature-level data fusion was used to join the composite Sentinel-2 imagery, Sentinel-2 derived spectral indices, and Sentinel-1 imagery for the study area in GEE. Sentinel-1 imagery from the 25th of January 2021 was used as it was captured within the date range used for Sentinel-2 composite imagery generation. All data was combined to match the spatial resolution and pixel locations of Sentinel-2 data. Spectral indices derived from Sentinel-2 data automatically matched Sentinel-2 pixels, however Sentinel-1 data did not and required resampling, with pixel values calculated as the weighted average of native Sentinel-1 pixels fully and partially within each 20m Sentinel-2 pixel. This combined imagery was then sampled to keep only pixels fully within the SAS plot shapefiles to create the overall reference dataset for model training and testing. The resulting data contained rows with reflectance, spectral index, and backscatter values for each sampled pixel and the associated SAS information for the plot each pixel came from. Plots too small to have a single 20m by 20m pixel fully within them were filtered out, resulting in 30 plots not used. A flowchart detailing the full data preprocessing methodology is provided in Appendix A. All analysis following this section was undertaken in R.

4.4.4 Class Selection

4.4.4.1 Exploratory Data Analysis

The choice of classes was determined through both exploratory data analysis of SAS data and the crop type aggregations used for reporting in SAS publications (NISR., 2021a). It was determined that there was such a high prevalence of intercropping (Table. 4.4.4.1.A.) (Table. 4.4.4.1.B.) and such a large number of unique intercropping combinations present in the SAS data (149 total) (Table. 4.4.4.1.C.) that it would not be possible to classify all individual crop types (25 total) or intercropping combinations. For example, there were 70 plots of pure cropped Maize, 35 plots of pure cropped Sorghum, 36 plots containing only Maize and Sorghum, and a further 54 plots containing Maize, Sorghum, and at least one other crop. This makes individually classifying Maize and Sorghum extremely challenging. It was therefore decided that crop type classes would be aggregated based on those used in SAS publications: Cereals, Tubers and Roots, Bananas, Legumes and Pulses, and Vegetables and Fruits, with two exceptions. Firstly, Rice was determined to not be intercropped and the distribution of spectral reflectance, backscatter, and index values were visibly distinct from other

Cereals in the SAS grouping and therefore Cereals was split into Dryland Cereals and Rice. Secondly, the crops contained in the Vegetables and Fruits grouping were not prevalent enough in the Nyagatare data to use this class for modelling, at only 5 plots. Non-agricultural classes were chosen qualitatively based upon having observed the land uses present when visiting Nyagatare and the non-agricultural plot data available from the SAS. The final classes chosen for modelling are detailed in Table. 4.4.4.1.D.

Cropping Practice	No. of SAS Plots
Intercropping	385
Pure Cropping	160

Table. 4.4.4.1.A. A table detailing the prevalence of cropping practices from SAS plot data, not including remotely created plots.

No. of Crop Types Present in a Plot	No. of SAS Plots
1	160
2	151
3	113
4	66
5	44
6	10
7	1

Table. 4.4.4.1.B. A table detailing the prevalence of different numbers of crops present in a plot from SAS plot data, not including remotely created plots.

Intercropping Combination	No. of SAS Plots
Maize	70
Bush bean, Maize	48
Bush bean, Cassava, Maize	37
Maize, Sorghum	36
Sorghum	35
Climbing bean	13
Banana for beer, Bush bean, Cooking banana, Dessert banana, Maize	10
Bush bean, Maize, Sorghum	10
Cooking banana, Dessert banana	10
Banana for beer, Cooking banana, Dessert banana	9

Table. 4.4.4.1.C. A table detailing the top 10 most prevalent intercropping combinations from SAS plot data, not including remotely created plots. Note this table only contains 10 out of the 149 unique intercropping combinations present in the data.

Class	Class Type
Bananas	Cropland
Dryland Cereals	Cropland
Legumes and Pulses	Cropland
Rice	Cropland
Tubers and Roots	Cropland
Bare Earth and Built Up	Non-agricultural
Forest	Non-agricultural
Grassland	Both Agricultural and Non-agricultural
Water	Non-agricultural
Wetland	Non-agricultural

Table. 4.4.4.1.D. *A table detailing the final classes selected for modelling.*

4.4.4.2 Main Crop Type Definition

SAS plot data used for this dissertation details all crop types present in a plot but does not include the relative proportion crops within each plot. There are 149 different intercropping combinations present in the data and therefore, a practical assumption for the main crop type within a plot had to be made to facilitate crop type classification. The definition of the main crop type in a plot used for this dissertation is the crop type present in a plot that is logically most spectrally visible, primarily based on crop height and size. For example, in the case of a plot containing Bananas intercropped with Tubers and Roots, Bananas are taller and more visible to a satellite and therefore considered the main crop. This qualitative assumption was made based upon observing agricultural plots and intercropping patterns while visiting Nyagatare. For the purposes of this dissertation, crop types have been considered to have priority for being the main crop in the following order: Bananas, Dryland Cereals, Legumes and Pulses, and Tubers and Roots, with Rice not intercropped and therefore not requiring this assumption.

4.4.4.3 Text Mining Agricultural Plot Classes

Regex-based text mining of a structured “crops present” text field was then utilised to classify the main crop type present in a plot and whether intercropping was present. Firstly, each “crops present” row was regex searched to check for all crops present in a plot and determine which individual crop present had the highest priority in terms of being the main crop. Each plot was then assigned to the class associated with the highest priority crop present. Then each row was regex searched to check if the remaining crops present were all from the class assigned. If they were, then the plot was assigned as a pure cropped plot. If they were not, then the plot was assigned as an

intercropped plot. For example, a plot containing “Cooking Bananas” and “Dessert Bananas” would be assigned to the Bananas class and be assigned as pure cropped, and a plot containing “Bush bean”, “Cassava”, “Maize”, and “Sorghum” would be assigned to the Dryland Cereals class and be assigned as intercropped.

4.4.4.4 Distribution of Classes

The overall distribution of class types by plot, by pixel and by SAS vs remotely created data are detailed in Table. 4.4.4.4.

Class	No. of SAS Plots	No. of Remotely Created Plots	No. of Plots Total	No. of SAS Pixels	No. of Remotely Created Pixels	No. of Pixels Total
Bananas	144	100	244	1,363	1,257	2,620
Dryland Cereals	343	0	343	3,123	0	3,123
Legumes and Pulses	38	0	38	146	0	146
Rice	8	150	158	21	11,008	11,029
Tubers and Roots	12	0	12	49	0	49
Bare Earth and Built Up	46	130	176	146	8,112	8,258
Forest	12	130	142	92	9,932	10,024
Grassland	0	25	25	0	6,189	6,189
Water	1	50	51	2	7,274	7,276
Wetland	0	25	25	0	7,067	7,067

Table. 4.4.4.4. A table detailing the distribution of classes in the combined SAS data by plot, by pixel, and by SAS vs remotely created data. Note there are 604 total SAS plots as opposed to the initial 634 due to 30 plots not containing a full pixel.

4.4.5 Train-validation and Test Sets

A 70/30 train-validation/test split was chosen to ensure sufficient observations in the test set to give a good idea of how the models would perform on unseen data, and to ensure the train-validation set remained large enough that cross-validation results were meaningful. Due to the class imbalance present in the dataset, detailed in Table. 4.4.4.4, stratified random sampling without replacement by crop type class was used to sample 30% of the data into the test set to ensure the test set was representative of the whole reference dataset in terms of the distribution of crop type classes. The remaining 70% of the reference data constituted the train-validation set.

4.4.6 5-fold Cross-validation

The train-validation set was used to perform 5-fold cross validation at multiple stages of the final model creation process, with each fold used once as the validation set and the remaining folds used as the training set, repeated once for each fold, with performance metrics averaged across the 5 runs. The data was partitioned into folds using random sampling without replacement, with 5 folds chosen as a trade-off between reducing time complexity, reducing issues resulting from class imbalance, and having enough folds to get meaningful results for how the model would perform on unseen data.

Despite greater computational requirements, cross-validation was chosen as opposed to Random Forests in-built out-of-bag (OOB) error as OOB error estimates can produce biased results (Matthew., 2011).

4.4.7 Final Feature Selection

Backward selection using feature importance was then performed. First, model training using 5-fold cross-validation was run with all features, and the feature with the lowest feature importance was removed prior to the next model training. This process of removing the lowest importance feature and rerunning model training using cross-validation was repeated until only a single feature remained. Classification accuracy for unseen data was estimated from cross-validation for every model combination, with the feature combination included in the best performing model chosen as the final model features. Backward selection using feature importance was chosen to reduce time complexity, as opposed to full backward selection or best subset selection which were prohibitively time complex and redundant due to the in-built Random Forest feature importance. The results from backward selection found a model using B2, B3, B4, B5, B6, B7, B8, B11, B12, VV, NDVI, and NDWI as predictor features resulted in the best estimated accuracy, with full results detailed in Table. 4.4.7.

Number of Features	Estimated Accuracy (%)	Included Features
17	92.7	B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12, VV, VH, NDVI, EVI, NDWI, SR, SAVI
16	92.8	B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12, VV, VH, NDVI, EVI, NDWI, SR
15	92.6	B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12, VV, NDVI, EVI, NDWI, SR
14	92.7	B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12, VV, NDVI, NDWI, SR
13	92.8	B2, B3, B4, B5, B6, B7, B8, B11, B12, VV, NDVI, NDWI, SR
12	92.9	B2, B3, B4, B5, B6, B7, B8, B11, B12, VV, NDVI, NDWI
11	92.7	B2, B3, B4, B5, B6, B7, B8, B11, B12, VV, NDWI
10	92.5	B2, B3, B4, B5, B6, B8, B11, B12, VV, NDWI
9	92.4	B3, B4, B5, B6, B8, B11, B12, VV, NDWI
8	91.2	B3, B4, B5, B6, B8, B11, B12, NDWI
7	91.2	B3, B4, B5, B6, B11, B12, NDWI
6	91.1	B3, B4, B5, B11, B12, NDWI
5	89.2	B4, B5, B11, B12, NDWI
4	86.5	B4, B11, B12, NDWI
3	80	B4, B11, B12
2	74.1	B4, B12
1	62.5	B12

Table. 4.4.7. A table detailing the results of backward feature selection using feature importance. The bolded row corresponds to the highest estimated accuracy.

4.4.8 Hyperparameter Tuning

Hyperparameter tuning was performed on the best model identified from backward selection, to select mtry and ntree parameters. mtry refers to the number of features randomly selected at each split for decision trees, while ntree is the total number of decision trees in the ensemble. Hyperparameter tuning was performed in two stages due to the prohibitive time complexity of tuning for all combinations of both parameters. For mtry, values between 1 and 12 (all initial features) were tested, with the default value for ntree of 500 selected (Liaw & Wiener., 2002), to find the optimal mtry value in terms of estimated classification accuracy for unseen data. Classification accuracy for unseen data was estimated using 5-fold cross-validation for each run. The optimal mtry value was determined to be 7, with full mtry tuning results detailed in Table. 4.4.8.A. For ntree, values of 500, 1000, 2000, and 4000 were tested with the

optimal mtry value of 7 selected to determine if increasing the ntree value improved cross-validation accuracy as per Scornet., 2017. Increasing ntree did not increase cross-validation accuracy and therefore the ntree value was left at 500, with full ntree tuning results detailed in Table. 4.4.8.B.

mtry Value	Estimated Accuracy (%)
1	91.6
2	92.5
3	92.9
4	92.9
5	93.0
6	93.0
7	93.1
8	93.0
9	93.0
10	93.0
11	92.8
12	92.7

Table. 4.4.8.A. A table detailing the results of mtry hyperparameter tuning. The bolded row resulted in the highest estimated accuracy.

ntree Value	Estimated Accuracy (%)
500	93.1
1000	93.1
2000	93.1
4000	93.1

Table. 4.4.8.B. A table detailing the results of ntree hyperparameter tuning.

4.4.9 Final Model Specification

Based on backward selection using feature importance and hyperparameter tuning, with 5-fold cross-validation at each step, the best model in terms of estimated model accuracy on unseen data was identified. This Random Forest model included B2, B3, B4, B5, B6, B7, B8, B11, B12, VV, NDVI, and NDWI as predictor features, with hyperparameter values of 7 and 500 for mtry and ntree respectively.

4.4.10 Final Model Training and Testing

The final model was then trained on the full train-validation set and tested on the test set using predictor feature values to predict the crop type class of every pixel in the test set. Results are detailed in Section 5.1.

4.5 Crop Type Map Creation

The final model, having been validated and tested, was then trained on the overall reference dataset (including both train-validation and test sets) to utilise all available training data for crop type classification. The final model was then applied to the entire Nyagatare data-fused composite image using predictor feature values to predict the crop type class of every pixel. Results are detailed in Section 5.2.

4.6 Intercropping Analysis

4.6.1 Intercropping Model Training

The intercropping model used the same model specification as the final model, detailed in Section 4.4.9. However, the intercropping model was trained and tested on different data. The overall reference dataset was split into a pure cropped pixels dataset and an intercropped pixels dataset based on if intercropping was present in a pixel, as determined by the regex-based text mining. The pure cropped dataset was then split in two using a 70/30 train/test split to ensure sufficient observations in the pure cropped test set to give a good idea of how the model would perform on unseen pure cropped data. Due to the class imbalance present in the pure cropped dataset, stratified random sampling without replacement by crop type class was used to sample 30% of the data into the pure cropped test set to ensure the test set was representative of the pure cropped dataset in terms of the distribution of crop type classes. The remaining 70% of the pure cropped data constituted the train set. Additionally, the intercropped pixels dataset was used as a second test set to facilitate comparison between model results when trained and tested on pure cropped data versus results when trained on pure cropped data and tested on intercropped data. Non-crop classes were included in the train and test sets using stratified random sampling, with the non-crop data in both test sets being identical to allow meaningful comparison of crop classification results.

4.6.2 Intercropping Model Testing

The intercropping model was then tested on both the pure cropped and intercropped test sets using predictor feature values to predict the crop type class of every pixel in the respective test sets. Results are detailed in Section 5.3.

4.7 Sensitivity Analysis

4.7.1 SAR

Sensitivity analysis was undertaken on the final model (from Section 4.4.10) to quantify the impact of the removal of SAR on test accuracy.

4.7.1.1 VV

The final model minus VV was trained on the full train set and tested on the test set using predictor feature values to predict the crop type class of every pixel in the test set. This was done to quantify the impact of VV inclusion on classification accuracy. Results are detailed in Section 5.4.1.

4.7.2 Spectral Indices

Sensitivity analysis was undertaken on the final model (from Section 4.4.10) to quantify the impact of the removal of spectral indices on test accuracy.

4.7.2.1 NDVI

The final model minus NDVI was trained on the full train set and tested on the test set using predictor feature values to predict the crop type class of every pixel in the test set. This was done to quantify the impact of NDVI inclusion on classification accuracy. Results are detailed in Section 5.4.2.

4.7.2.2 NDWI

The final model minus NDWI was trained on the full train set and tested on the test set using predictor feature values to predict the crop type class of every pixel in the test set. This was done to quantify the impact of NDWI inclusion on classification accuracy. Results are detailed in Section 5.4.3.

Chapter 5 – Results

5.1 Final Model

This section details the overall performance metrics, confusion matrix, and performance metrics by class from the final model testing, as detailed in Section 4.4.10. It also details feature importance derived from the final model cross-validation.

Overall Performance Metric	Score
Accuracy	94.2%
Kappa Score	0.932
Macro Precision	0.850
Macro Recall	0.751
Macro F1 Score	0.764
Weighted F1 Score	0.941

Table. 5.1.A. A table detailing the overall performance metrics from the final model testing.

Predicted Class	True Class									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bananas (1)	617	12	57	22	2	5	14	5	0	2
Bare Earth and Built Up (2)	25	2,410	49	2	6	0	3	1	0	0
Dryland Cereals (3)	90	39	687	12	29	26	39	5	0	0
Forest (4)	30	0	9	2,917	6	0	18	0	0	19
Grassland (5)	14	6	70	3	1,763	4	58	0	0	5
Legumes and Pulses (6)	0	0	1	0	0	5	0	0	0	0
Rice (7)	3	11	51	32	28	4	3,115	3	2	22
Tubers and Roots (8)	1	0	0	0	0	0	1	1	0	0
Water (9)	0	0	0	0	0	0	0	0	2,181	0
Wetland (10)	6	0	13	20	23	0	61	0	0	2,073

Table. 5.1.B. A table detailing the confusion matrix from the final model testing.

Class	Precision	Recall	F1 Score
Bananas	0.838	0.785	0.811
Bare Earth and Built Up	0.966	0.973	0.969
Dryland Cereals	0.741	0.733	0.737
Forest	0.973	0.970	0.971
Grassland	0.917	0.949	0.933
Legumes and Pulses	0.833	0.114	0.200
Rice	0.952	0.941	0.947
Tubers and Roots	0.333	0.067	0.111
Water	1.000	0.999	1.000
Wetland	0.944	0.977	0.960

Table. 5.1.C. A table detailing the by class performance metrics from the final model testing.

Feature	Feature Importance
B2	2,015
B3	2,982
B4	5,159
B5	2,335
B6	1,864
B7	2,263
B8	2,995
B11	3,589
B12	5,762
VV	1,661
NDVI	1,149
NDWI	1,645

Table. 5.1.D. A table detailing the feature importance derived from the final model cross-validation.

5.2 Classified Crop Type Map

This section presents the final classified crop type map, as detailed in Section 4.5. It also details the area estimates for each class compared to the SAS area estimates.

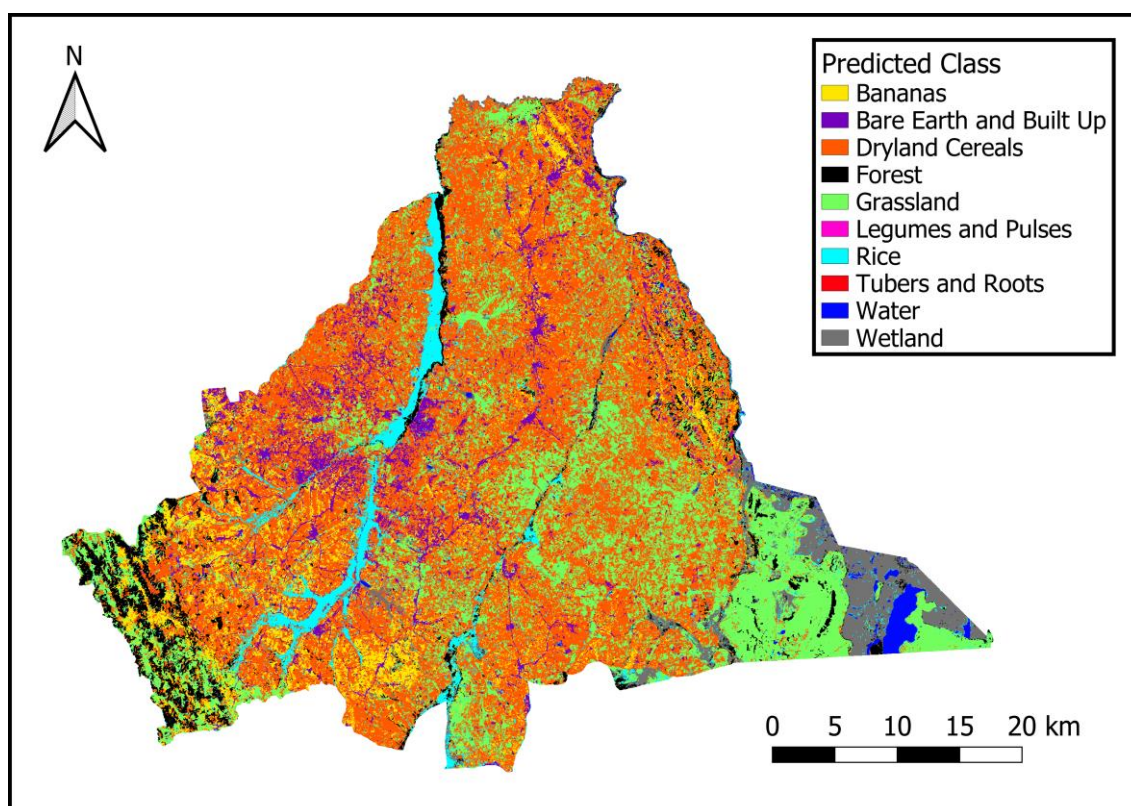


Figure. 5.2. A map showing the predicted classes for Nyagatare using the final model trained on all available data.

Class	Predicted Area (Ha)	SAS Estimated Area (Ha)
Bananas	14,301	11,274
Bare Earth and Built Up	13,566	NA
Dryland Cereals	77,600	41,963
Forest	9,159	NA
Grassland	52,745	NA
Legumes and Pulses	8	22,380
Rice	13,354	1,537
Tubers and Roots	6	6,319
Water	1,436	NA
Wetland	10,945	NA
Total Cropland	105,269	84,697

Table. 5.2. A table detailing the estimated area for each class from the classified crop type map alongside the estimated area for each class from the 2021 Season A SAS publication.

5.3 Intercropping Model

5.3.1 Pure Cropped Testing

This section details the overall performance metrics, confusion matrix, and performance metrics by class from the intercropping model tested on pure cropped data, as detailed in Section 4.6.

Performance Metric	Score
Accuracy	96.4%
Kappa Score	0.958
Macro Precision	0.915
Macro Recall	0.852
Macro F1 Score	0.877
Weighted F1 Score	0.964

Table. 5.3.1.A. A table detailing the overall performance metrics from the intercropping model tested on pure cropped data.

Predicted Class	True Class									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bananas (1)	419	3	7	17	1	0	4	0	0	1
Bare Earth and Built Up (2)	21	2,439	13	3	3	0	6	0	0	0
Dryland Cereals (3)	2	14	311	7	7	8	12	4	0	0
Forest (4)	21	0	16	2,920	9	1	16	2	0	31
Grassland (5)	4	5	23	6	1,787	3	49	0	0	6
Legumes and Pulses (6)	1	0	2	0	0	19	0	0	0	0
Rice (7)	1	17	17	29	31	5	3,167	2	2	27
Tubers and Roots (8)	0	0	1	1	0	0	1	7	0	0
Water (9)	0	0	0	0	0	0	0	0	2,181	0
Wetland (10)	0	0	6	25	19	0	54	0	0	2,056

Table. 5.3.1.B. A table detailing the confusion matrix from the intercropping model tested on pure cropped data.

Class	Precision	Recall	F1 Score
Bananas	0.927	0.893	0.910
Bare Earth and Built Up	0.981	0.984	0.983
Dryland Cereals	0.852	0.785	0.817
Forest	0.968	0.971	0.969
Grassland	0.949	0.962	0.956
Legumes and Pulses	0.864	0.528	0.655
Rice	0.960	0.957	0.959
Tubers and Roots	0.700	0.467	0.560
Water	1.000	0.999	1.000
Wetland	0.952	0.969	0.961

Table. 5.3.1.C. A table detailing the by class performance metrics from the intercropping model tested on pure cropped data.

5.3.2 Intercropped Testing

This section details the confusion matrix, and performance metrics by class from the intercropping model tested on intercropped data, as detailed in Section 4.6. Overall performance metrics are not detailed as there is no intercropped Rice in the overall dataset and therefore no Rice data points in the intercropped test set, resulting in biased (or non-calculatable) overall performance metrics.

Predicted Class	True Class									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bananas (1)	321	3	140	17	1	5	0	0	0	1
Bare Earth and Built Up (2)	153	2,439	367	3	3	1	0	0	0	0
Dryland Cereals (3)	282	14	711	7	7	10	0	0	0	0
Forest (4)	78	0	32	2,920	9	2	0	1	0	31
Grassland (5)	103	5	296	6	1,787	4	0	0	0	6
Legumes and Pulses (6)	3	0	9	0	0	1	0	0	0	0
Rice (7)	89	17	217	29	31	3	0	1	2	27
Tubers and Roots (8)	5	0	0	1	0	3	0	0	0	0
Water (9)	0	0	0	0	0	0	0	0	2,181	0
Wetland (10)	23	0	31	25	19	0	0	0	0	2,056

Table. 5.3.2.A. A table detailing the confusion matrix from the intercropping model tested on intercropped data. Note there are no Rice class test data points as rice is not intercropped.

Class	Precision	Recall	F1 Score
Bananas	0.658	0.304	0.416
Bare Earth and Built Up	0.822	0.984	0.896
Dryland Cereals	0.690	0.394	0.502
Forest	0.950	0.971	0.960
Grassland	0.810	0.962	0.879
Legumes and Pulses	0.077	0.034	0.048
Rice	0.000	NA	NA
Tubers and Roots	0.000	0.000	0.000
Water	1.000	0.999	1.000
Wetland	0.955	0.969	0.962

Table. 5.3.2.B. A table detailing the by class performance metrics from the intercropping model tested on intercropped data. Note that recall and F1 score are NA for Rice as there are no intercropped Rice test data. Note that the F1 score for Tubers and Roots is 0 as opposed to the mathematically correct NA as per convention for recording F1 scores when precision and recall are both 0.

5.3.3 Pure and Intercropped Testing Comparison

This section details the F1 score by class from the intercropping model tested on both pure cropped and intercropped data, including the resulting differences in F1 scores.

Class	Pure Cropped F1 Score	Intercropped F1 Score	Difference
Bananas	0.910	0.416	-0.494
Bare Earth and Built Up	0.983	0.896	-0.087
Dryland Cereals	0.817	0.502	-0.315
Forest	0.969	0.960	-0.009
Grassland	0.956	0.879	-0.077
Legumes and Pulses	0.655	0.048	-0.607
Rice	0.959	NA	NA
Tubers and Roots	0.560	0.000	-0.560
Water	1.000	1.000	0.000
Wetland	0.961	0.962	0.001

Table. 5.3.3. A table detailing the by class performance metrics from the intercropping model tested on pure and intercropped data. Note that F1 score and difference in F1 scores are NA for Rice as there is no intercropped Rice test data.

5.4 Sensitivity Analysis

5.4.1 VV

This section details the overall performance metrics, confusion matrix, and performance metrics by class from the final model excluding VV testing, as detailed in Section 4.7.1.1.

Performance Metric	Score
Accuracy	92.9%
Kappa Score	0.917
Macro Precision	0.889
Macro Recall	0.724
Macro F1 Score	0.737
Weighted F1 Score	0.926

Table. 5.4.1.A. A table detailing the overall performance metrics from the final model excluding VV testing.

Predicted Class	True Class									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bananas (1)	517	13	110	24	3	5	31	3	0	2
Bare Earth and Built Up (2)	29	2,414	46	2	3	0	5	1	0	0
Dryland Cereals (3)	126	33	607	9	35	27	39	5	0	0
Forest (4)	34	0	9	2,901	6	0	36	1	0	25
Grassland (5)	14	8	73	7	1,758	3	59	0	0	2
Legumes and Pulses (6)	0	0	1	0	0	3	0	0	0	0
Rice (7)	61	10	79	40	28	6	3,093	4	2	21
Tubers and Roots (8)	0	0	0	0	0	0	0	1	0	0
Water (9)	0	0	0	0	0	0	0	0	2,181	0
Wetland (10)	5	0	12	25	24	0	46	0	0	2,071

Table. 5.4.1.B. A table detailing the confusion matrix from the final model excluding VV testing.

Class	Precision	Recall	F1 Score
Bananas	0.730	0.658	0.692
Bare Earth and Built Up	0.966	0.974	0.970
Dryland Cereals	0.689	0.648	0.668
Forest	0.963	0.964	0.964
Grassland	0.914	0.947	0.930
Legumes and Pulses	0.750	0.068	0.125
Rice	0.925	0.935	0.930
Tubers and Roots	1.000	0.067	0.125
Water	1.000	0.999	1.000
Wetland	0.949	0.976	0.962

Table. 5.4.1.C. A table detailing the by class performance metrics from the final model excluding VV testing.

5.4.2 NDVI

This section details the overall performance metrics, confusion matrix, and performance metrics by class from the final model excluding NDVI testing, as detailed in Section 4.7.2.1.

Performance Metric	Score
Accuracy	94.0%
Kappa Score	0.930
Macro Precision	0.847
Macro Recall	0.746
Macro F1 Score	0.760
Weighted F1 Score	0.939

Table. 5.4.2.A. A table detailing the overall performance metrics from the final model excluding NDVI testing.

Predicted Class	True Class									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bananas (1)	598	9	71	22	1	6	13	2	0	2
Bare Earth and Built Up (2)	24	2,411	47	2	5	0	3	1	0	0
Dryland Cereals (3)	100	39	664	13	29	25	39	6	0	0
Forest (4)	30	0	10	2,913	6	0	20	2	0	17
Grassland (5)	14	7	74	4	1,763	4	55	0	0	2
Legumes and Pulses (6)	0	0	1	0	0	5	0	0	0	0
Rice (7)	14	12	56	32	30	4	3,120	3	2	20
Tubers and Roots (8)	1	0	0	0	0	0	1	1	0	0
Water (9)	0	0	0	0	0	0	0	0	2,181	0
Wetland (10)	5	0	14	22	23	0	58	0	0	2,080

Table. 5.4.2.B. A table detailing the confusion matrix from the final model excluding NDVI testing.

Class	Precision	Recall	F1 Score
Bananas	0.826	0.761	0.792
Bare Earth and Built Up	0.967	0.973	0.970
Dryland Cereals	0.726	0.709	0.717
Forest	0.972	0.968	0.970
Grassland	0.917	0.949	0.933
Legumes and Pulses	0.833	0.114	0.200
Rice	0.947	0.943	0.945
Tubers and Roots	0.333	0.067	0.111
Water	1.000	0.999	1.000
Wetland	0.945	0.981	0.962

Table. 5.4.2.C. A table detailing the by class performance metrics from the final model excluding NDVI testing.

5.4.3 NDWI

This section details the overall performance metrics, confusion matrix, and performance metrics by class from the final model excluding NDWI testing, as detailed in Section 4.7.2.2.

Performance Metric	Score
Accuracy	93.9%
Kappa Score	0.928
Macro Precision	0.845
Macro Recall	0.745
Macro F1 Score	0.758
Weighted F1 Score	0.937

Table. 5.4.3.A. A table detailing the overall performance metrics from the final model excluding NDWI testing.

Predicted Class	True Class									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Bananas (1)	591	12	69	24	2	5	14	5	0	2
Bare Earth and Built Up (2)	24	2,410	47	2	5	0	4	1	0	0
Dryland Cereals (3)	107	39	665	13	32	25	38	5	0	0
Forest (4)	28	0	9	2,908	9	0	20	0	0	19
Grassland (5)	15	6	69	5	1,759	4	53	0	0	3
Legumes and Pulses (6)	0	0	1	0	0	5	0	0	0	0
Rice (7)	14	11	61	31	28	5	3,118	3	2	23
Tubers and Roots (8)	1	0	0	0	0	0	1	1	0	0
Water (9)	0	0	0	0	0	0	0	0	2,181	0
Wetland (10)	6	0	16	25	22	0	61	0	0	2,074

Table. 5.4.3.B. A table detailing the confusion matrix from the final model excluding NDWI testing.

Class	Precision	Recall	F1 Score
Bananas	0.816	0.752	0.783
Bare Earth and Built Up	0.967	0.973	0.970
Dryland Cereals	0.720	0.710	0.715
Forest	0.972	0.967	0.969
Grassland	0.919	0.947	0.933
Legumes and Pulses	0.833	0.114	0.200
Rice	0.946	0.942	0.944
Tubers and Roots	0.333	0.067	0.111
Water	1.000	0.999	1.000
Wetland	0.941	0.978	0.959

Table. 5.4.3.C. A table detailing the by class performance metrics from the final model excluding NDWI testing.

Chapter 6 – Discussion

6.1 Final Model

The final model had good overall performance (Table. 5.1.A.), with a test accuracy of 94.2%, macro F1 score of 0.764, and a weighted F1 score of 0.941. The 94.2% test accuracy of the model was higher than the 93.1% estimated accuracy from cross-validation. This is likely due to the differing sampling strategies used, with stratified random sampling used for the train/test split and random sampling into folds for cross-validation. As a result, the test set had a guaranteed representative distribution of crop types present in the data, while each individual test fold for cross-validation did not. The discrepancy between macro and weighted F1 scores is a result of the large class imbalance present in the dataset (Chicco & Jurman., 2020), with smaller classes performing worse than larger classes. However, given the number of non-crop classes and their associated data points, the overall performance metrics do not fully represent the model performance for crop type classification, and therefore by class performance metrics will primarily be used for assessing model performance.

For crop type classes (Table. 5.1.C.), Rice had an F1 score of 0.947, Bananas 0.811, Dryland Cereals 0.737, Legumes and Pulses 0.200, and Tubers and Roots 0.111. Rice classification performed well, with only a small proportion of Rice points misclassified as primarily Wetland and Grassland. Rice classification performance was similar to that achieved by other optical, SAR, and optical-SAR combined studies from different regions (Bazzi et al., 2019) (Cai et al., 2019) (Park et al., 2018) (Yang et al., 2018) (Zhou et al., 2016). Bananas classification also performed well, with Bananas points misclassified primarily as Dryland Cereals, likely as a result of the crops comprising these classes often being intercropped. Bananas classification performance was similar to that achieved by another optical-SAR combined study in sub-Saharan Africa (Alabi et al., 2022), however from the literature the primary methodology for banana classification studies appears to be UAV-based classification, and Alabi et al., 2022 in the same study achieved higher classification accuracy using UAV methods. Compared to UAV-based studies, Bananas classification performance was slightly worse, however these studies trained and tested on pure cropped data, so this is expected given the prevalence of intercropping (Kilwenge et al., 2021) (Selvaraj et al., 2020). Dryland Cereals classification also performed well, with misclassified Dryland Cereals points spread

fairly evenly across Bananas, Bare Earth and Built Up, Grassland, and Rice. While not directly comparable, Dryland Cereals classification performance was similar to that achieved for pure cropped and intercropped Maize by an optical study in sub-Saharan Africa (Ibrahim et al., 2021). Both Legumes and Pulses, and Tubers and Roots classification performances were poor, with the majority of points for both these classes classified as Dryland Cereals, likely as a result of the crops comprising these classes being heavily intercropped. All non-crop classes performed well, with F1 scores ranging from 0.933 to 1.000 (rounded up). Therefore, when assessed using the test dataset, the methodology outlined in this dissertation was determined to be effective for classification of both non-crop classes and certain crop classes, including Rice, Bananas, and Dryland Cereals. However, as part of this dissertation, potential broad issues were identified relating to how crop type classification studies in areas where intercropping is present evaluate their results, and the final model results should be interpreted in the context of these additional findings, detailed in Section 6.3.

Feature importance for distinguishing classes was derived from the final model cross-validation (Table. 5.1.D.), with the highest importance features determined to be Sentinel-2 Band 12 (B12), Sentinel-2 Band 4 (B4), and Sentinel-2 Band 11 (B11). The Short-Wave Infrared (SWIR) bands (B12 and B11) having high feature importance for crop type classification is consistent with the literature (Feng et al., 2019), although it is worth noting that feature importance values also factor in non-crop classes. Therefore, when undertaking crop type classification studies in areas similar to the representative study area, a careful decision is required regarding the resolution of Sentinel-2 bands chosen for modelling. Choosing to use 20m resolution allows for the inclusion of SWIR bands which were determined to have high feature importance, and choosing to use only 10m bands could potentially better capture the heterogeneous agricultural systems present, including small fields. While outside the scope of this dissertation, analysis of using a 10m resolution with 20m bands resampled to 10m pixels was undertaken to determine if this method biased results. Forward Selection using estimated accuracy from cross-validation was undertaken for both 10m and 20m resolutions, and it was determined that 20m bands had biased feature importance and overall results compared to 10m bands when run at 10m resolution. This was determined to be as a result of undertaking a pixel-based sampling approach, with 20m pixel values biasing 10m pixel values located within them. Further research would be

required to outline an appropriate methodology of using 20m band information at a 10m modelling resolution for the study area. For example, it might be effective to train a model using a 20m resolution, ensuring that all 20m pixels are fully within plots to prevent field edge mixed-pixel bias in training, and then apply the resulting model to the study area at a 10m resolution to potentially better classify small fields.

6.2 Classified Crop Type Map

Qualitative assessment of the classified crop type map (Figure. 5.2.) was undertaken, and it was assessed to perform reasonably well when compared to both WorldView satellite imagery for the study time period and to the broad crop type trends observed throughout Nyagatare when visiting. Several key trends observed when visiting were reflected in the map, including a higher prevalence of Bananas in the West and Southwest, the presence of Rice throughout the main river valley, and a higher prevalence of Dryland Cereals and Grassland in the North, Central, and East.

Additionally, Akagera National Park in the East contained limited misclassified pixels with some small Wetland and Grassland areas misclassified as Rice and Dryland Cereals respectively. Non-crop classes such as Bare Earth and Built Up, Forest, and Water were all assessed to be classified well when compared to WorldView imagery for the study time period.

The estimated areas for overall cropland and individual crop classes from the classified crop type map were compared to the estimated areas from the 2021 Season A Seasonal Agricultural Survey (SAS) (NISR., 2021a) (Table. 5.2.). The final model overestimated total cropland area by 24.3%. The area for Rice was overestimated by 768.8%, however it is worth noting that potential significant underestimation of the SAS estimated area for Rice in Nyagatare was identified and will be discussed further in Section 6.6. The area for Bananas was overestimated by 26.8%. The area for Dryland Cereals was overestimated by 84.9%, with the areas for Legumes and Pulses, and Tubers and Roots massively underestimated. However, given the extremely high prevalence of intercropping and confusion between these three classes, a comparison was made of the three classes grouped. This resulted in only a 9.8% overestimate in area for the grouped class, indicating that grouping these classes for classification and using historical statistics on the proportion of crop types within this grouped class might provide a more practical approach for crop area estimation. Therefore, both the overall cropland area and the area of most individual crop classes was overestimated when compared to

the SAS estimates, and for potential future work, efforts to create more non-crop class reference data could result in closer results. However, it is worth noting that the SAS crop area values are themselves estimates and the methodology outlined in this dissertation could provide additional area estimates as a form of sense checking SAS results, discussed further in Section 6.6.

6.3 Intercropping Analysis

The intercropping model tested on pure cropped data had good overall performance (Table. 5.3.1.A.) with a test accuracy of 96.4%, macro F1 score of 0.877, and a weighted F1 score of 0.964. These overall performance metrics are not directly comparable to those of the final model due differences in the train and test data (Section 4.6), however they are indicative of increased overall model performance. Overall performance metrics for the intercropping model tested on intercropped data are not provided as there is no intercropped Rice in the overall dataset and therefore no Rice data points in the intercropped test set, resulting in biased (or non-calculatable) overall performance metrics. Comparison of crop class performance metrics (not including Rice) is the only viable way of quantifying the impact of intercropping on crop type classification in this case. For the intercropping model tested on pure cropped data (Table. 5.3.1.C.), Bananas had an F1 score of 0.910, Dryland Cereals 0.817, Legumes and Pulses 0.655, and Tubers and Roots 0.560. For the intercropping model tested on intercropped data (Table. 5.3.1.B.), Bananas had an F1 score of 0.416, Dryland Cereals 0.502, Legumes and Pulses 0.048, and Tubers and Roots 0.000. Therefore, Legumes and Pulses had a 0.607 reduction in F1 score, Tubers and Roots a 0.560 reduction, Bananas a 0.494 reduction, and Dryland Cereals a 0.315 reduction (Table. 5.3.3.). All crop classes show a large reduction in classification performance when tested on intercropped data in the study area. This indicates that training and testing on pure cropped data for crop type classification in areas with a high prevalence of intercropping produces misleading performance metrics, with model accuracies appearing far higher than what would be representative of actual model performance. However, given there are only small test samples for Legumes and Pulses, and Tubers and Roots, the difference in performance metrics for these classes should be interpreted with appropriate scepticism. Generally, model performance metrics can be influenced greatly by variations in class prevalence and differing levels of reference standard, especially in remote sensing (Foody et al., 2023). This dissertation argues that this is even more of an issue in areas with a high

prevalence of intercropping, as choosing a reference standard which will produce representative results of how a model will perform on unseen data is extremely challenging, with traditional reference standards and performance metrics producing misleading results when tested on pure cropped data for the representative study area. Additionally, in areas similar to the representative study area, which are heterogeneous agriculturally with a wide variety and differing prevalence of crop types, this further exacerbates performance metric issues (Foody et al., 2023). Therefore, further research in this area should be conducted to identify appropriate reference standards to help ensure crop type classification studies in heterogeneous agricultural systems such as those with a high prevalence of intercropping produce meaningful performance metrics.

6.4 Sensitivity Analysis

6.4.1 VV

The sensitivity analysis undertaken by removing VV SAR as a model feature (Table. 5.4.1.A.) resulted in a test accuracy of 92.9%, macro F1 score of 0.737, and a weighted F1 score of 0.926. This represents a 1.3 percentage point reduction in test accuracy, 0.027 reduction in macro F1 score, and a 0.015 reduction in weighted F1 score. For crop type classes (Table. 5.4.1.C.), Bananas had a 0.119 reduction in F1 score, Legumes and Pulses a 0.075 reduction, Dryland Cereals a 0.069 reduction, Rice a 0.017 reduction, and Tubers and Roots a 0.014 increase. There was an increase in misclassification of Bananas, primarily as Dryland Cereals and Rice. There was an increase in misclassification of Dryland Cereals, primarily as Bananas and Rice. This is logical, as these crop classes are distinct texturally and this will no longer be captured without the inclusion of VV. While the removal of VV only reduced overall accuracy a small amount, this reduction in accuracy was disproportionately experienced as confusion between crop classes, and therefore the inclusion of VV does assist in increasing crop type classification accuracy specifically.

6.4.2 NDVI

The sensitivity analysis undertaken by removing NDVI as a model feature (Table. 5.4.2.A.) resulted in a test accuracy of 94.0%, macro F1 score of 0.760, and a weighted F1 score of 0.939. This represents a 0.2 percentage point reduction in test accuracy, 0.004 reduction in macro F1 score, and a 0.002 reduction in weighted F1 score. For crop type classes (Table. 5.4.2.C.), reductions in F1 score ranged from 0.020 to no reduction. Both reductions in overall and by class performance metrics were small and therefore

the removal of NDVI does not have a large impact on classification results, likely due to NDVI being highly correlated with certain features still included in the final model such as NDWI, detailed in Appendix B.

6.4.3 NDWI

The sensitivity analysis undertaken by removing NDWI as a model feature (Table. 5.4.3.A.) resulted in a test accuracy of 93.9%, macro F1 score of 0.758, and a weighted F1 score of 0.937. This represents a 0.3 percentage point reduction in test accuracy, 0.006 reduction in macro F1 score, and a 0.004 reduction in weighted F1 score. For crop type classes (Table. 5.4.3.C.), reductions in F1 score ranged from 0.028 to no reduction. Both reductions in overall and by class performance metrics were small and therefore the removal of NDWI does not have a large impact on classification results, likely due to NDWI being highly correlated with certain features still included in the final model such as NDVI, detailed in Appendix B.

6.5 Limitations

Two potential limitations of the methodology used for this dissertation have already been outlined above, being that small test samples for Legumes and Pulses, and Tubers and Roots means that strong conclusions cannot be drawn for these classes, and that a 20m spatial resolution for modelling potentially does not adequately capture the agricultural heterogeneity of the study area, primarily the smallest fields. In addition to these, other potential limitations are detailed below.

One key limitation of the methodology used for this dissertation is the assumption regarding the main crop type in a plot, detailed in Section 4.4.4.2. A practical assumption for the priority for which crops are the main crop type in a plot was necessitated by the lack of data on the proportion of each crop present in a plot. This resulted in crops which are commonly intercropped being grouped into classes with the highest priority. This is reflected in both the small number of data points and the poor classification performance for both the Legumes and Pulses, and Tubers and Roots classes, both of which are heavily intercropped with higher priority crop classes. However, while not available for this dissertation, data on the proportions of crops present in each plot is captured by the SAS (NISR., 2021a) and potential methods for how this additional data could be used to mitigate this methodological limitation for operational use are detailed in Section 6.6.

Another limitation of the methodology used for this dissertation is that bias in model performance metrics could be present as a result of using every pixel within each SAS plot to maximise the available reference data for modelling. Pixels within the same plot are logically more likely to have similar spectral and radiometric values, as they likely contain similar proportions and density of crop type(s) present. This could potentially bias results, as although the model is tested on unseen data, these data values logically could be more similar than data from completely unsampled plots, with model performance metrics potentially overestimated. This bias is logically less of an issue the more plots and the more data points you have, adding additional weight to the argument that strong conclusions cannot be drawn about the Legumes and Pulses, and Tubers and Roots classes. Additionally, given that adjacent pixel values can influence each other due to the mechanics of how satellites capture spectral information (Fisher, 1997), this could add further bias. Further research would be required to assess how this impacts model performance and whether it does bias performance metrics as it is also logically possible that taking the mean or median of pixel values within each plot could lose crucial information on within plot heterogeneity important for classification, alongside the clear downside of having far fewer reference data points.

Finally, another limitation of the methodology used for this dissertation is the potential locational bias present due to the methodology used for remotely sampling Grassland and Wetland ground truth data points. Data points were primarily created for these two classes within Akagera National Park as this is an area where crops are not grown. This ensured that these two classes could be accurately identified from Worldview imagery, however the performance metrics calculated for these classes could have a locational bias and might overestimate classification performance for these classes for the study area as a whole. To mitigate this limitation, additional non-crop class data could be collected as part of the SAS, however this carries with it additional costs.

6.6 Operational Challenges, Recommendations, and Future Work

A key operational challenge is to mitigate the limitations of the main crop type assumption, detailed in Section 6.5. Given that data on the proportions of crops present in each plot is captured by the SAS (NISR., 2021a), accessing and including this additional data would allow different class groupings and assumptions to be tested. This could allow for a less imbalanced split of crop classes to be created and allow for rule-based class groupings for specific intercropping combinations based on their

proportions in a plot. For example, if a plot has more than 20% of a crop in it, the crop could be included for the grouping and a plot could be classified as “Intercropped Dryland Cereals, Legumes and Pulses”, “Intercropped Bananas and Dryland Cereals”, etc. A rule-based approach like this may provide aggregated enough classes to facilitate classification, however implementing this approach for each individual crop or intercropping combination would still not be possible given the large number of combinations present (Table. 4.4.4.1.C.). For an operational methodology not utilising this additional data, it is recommended that a broader aggregation approach is used. Given the extensive intercropping of Dryland Cereals, Legumes and Pulses, and Tubers and Roots, it is recommended that these categories are grouped into a broad “Heavily Intercropped” class. Grouping these classes for classification and using historical statistics on the distribution of crop types within this grouped class could provide a practical approach for crop area estimation for Rwanda specifically, leveraging their extensive historical SAS data. This could also be applied by using the historical distribution of individual crop types present to get individual crop area estimates. While this disaggregation methodology is backwards looking in nature, it might provide high enough accuracy results for practical decision making and should be studied further. One operational question is whether to compare estimated crop areas to SAS estimated areas as the final point of reference for assessing classification performance. It was determined that the SAS area of Rice specifically could be underestimated in Nyagatare for Season A 2021. This was identified by analysing WorldView imagery for the study time period, analysing the locations of SAS clusters, and from having visited Nyagatare, albeit not in Season A 2021. The main river valley contained primarily rice at each location visited in Season B 2024, and this seemed to be the case for both Season A 2021 and Season B 2024 from WorldView imagery. Only a single SAS Season A 2021 sampling cluster captured any rice in Nyagatare and this could potentially explain the discrepancy between the area of Rice estimated and observed, and the SAS area estimates. Therefore, if this theory is correct, it indicates that there could be operational value in using the methodology outlined in this dissertation as a second independent method for calculating crop area statistics from the same plot information to cross evaluate and sense check results between the two methods.

Another operational challenge is the creation of cloud-free Sentinel-2 imagery for different agricultural seasons and for the whole of Rwanda. For Season A 2021 only two

dates were determined to have broadly cloud free imagery and this is not necessarily guaranteed for other agricultural seasons. This means that a larger number of images could be required to generate composite imagery, reducing the consistency of band values in said imagery and potentially reducing classification accuracy (Man et al., 2018). This adds additional weight to the methodological choice of using all SAS pixels, as this logically could better capture this variance. Testing the impact of composite imagery vs a single scene on classification accuracy was not undertaken for this dissertation due to some plots being under clouded areas. However, further work could include manually filtering out plots under cloud and running a sensitivity analysis for composite vs single date imagery to quantify the impact of composite imagery on classification results. The cloud removal methodology used for this dissertation would also need to be tested on more cloud dense imagery from additional agricultural seasons to determine if the approach is effective broadly. One potential solution to this issue not explored in this dissertation is to use feature-level data fusion to combine multitemporal SAR imagery from different periods of the agricultural season. This approach is largely unaffected by cloud and has proved effective for capturing crops differing phenology throughout a season (Inglada et al., 2016).

An additional aim of this dissertation was to use only freely available software and data, apart from SAS data. To this end, Google Earth Engine (GEE), JavaScript, R, QGIS, Google Earth Pro (GEP), WorldView data (via GEP), and Sentinel-2 and Sentinel-1 data (via GEE) were utilised, all being freely available, including GEE as Rwanda is categorised as a Least Developed Country (Google., 2024) (UN., 2023). One operational challenge is that there are compute limits on free-use GEE, meaning that for this dissertation all model training, testing, and map classification had to be undertaken using R on a local PC. If this methodology were operationalised for Rwanda as a whole, either a local computer running for a relatively long time period, a paid cloud computing resource such as Amazon EC2, or paid GEE would be required. However, this process should be resilient and remain freely available for the foreseeable future if using local compute, as no plans have been announced by any of the companies or organisations involved to gate access, but removal of free access does still remain an operational risk.

Additionally, before operationalising the methodology outlined in this dissertation, it would need to be tested on Rwanda as a whole, on Season B, and on data for additional years to check the methodology generalises well both locationally and temporally.

Chapter 7 – Conclusion

This dissertation focused on crop type classification using combined multispectral and Synthetic Aperture Radar (SAR) satellite imagery in Nyagatare District, Rwanda, representative of a high cloud cover, topographically complex tropical agricultural area with small field sizes and a high prevalence of intercropping. The final Random Forest model achieved an overall test accuracy of 94.2%, Kappa score of 0.932, and weighted F1 score of 0.941. For crop type classes, Rice had an F1 score of 0.947, Bananas 0.811, Dryland Cereals 0.737, Legumes and Pulses 0.200, and Tubers and Roots 0.111.

Therefore, it was determined that data-fused optical-SAR satellite-based crop type classification was able to produce high accuracy results for certain crop types, including Rice, Bananas, and Dryland Cereals, and for all non-crop classes. However, the final model showed poor performance for classifying Legumes and Pulses, and Tubers and Roots, likely as a result of the crops comprising these classes being heavily intercropped with Dryland Cereals. Random Forest feature importance indicated that the 20m Short-Wave Infrared (SWIR) and 10m Red Sentinel-2 bands (B12, B11, and B4) were especially important for distinguishing between classes. Sensitivity analysis for the removal of VV SAR from modelling resulted in decreased classification performance. This reduction in classification performance was disproportionately experienced as confusion between specific texturally distinct crop classes, including Bananas, Dryland Cereals, and Rice, and therefore the inclusion of VV SAR does assist in increasing crop type classification accuracy in the study area. The removal of NDVI and NDWI spectral indices had minimal impact on classification accuracy when removed individually, likely as a result of being highly correlated with certain features still included in the final model. Therefore, this dissertation and its focus in region specific methodologies and challenges in the representative study area, should help fill the locational gap in the literature on crop type classification studies in sub-Saharan Africa, and in Rwanda.

In a review of the literature on crop type classification using satellite remote sensing, it was identified that there is a knowledge gap relating to the impact of intercropping on crop type classification accuracy. Intercropping poses significant methodological challenges for crop type classification, with many studies in areas where intercropping is present referencing intercropping as a reason for low classification accuracies and inter-class confusion. However, no studies were found which aim to quantify this

reduction in accuracy for their respective study areas. The Seasonal Agricultural Survey (SAS) dataset used in this dissertation, which contains ubiquitous intercropping, provided a unique opportunity to study the impact of intercropping on crop type classification accuracy. The impact of intercropping on classification accuracy was quantified by crop type for the representative study area, with reductions in F1 score ranging from 0.315 to 0.494 for the main crop types present. This indicates that training and testing on pure cropped data in areas with a high prevalence of intercropping produces misleading performance metrics, with model accuracies appearing far higher than what would be representative of actual model performance. Therefore, further research in this area should be conducted to identify appropriate reference standards to help ensure crop type classification studies in areas with a high prevalence of intercropping produce meaningful performance metrics.

Limitations of this dissertation include the assumption for defining main crop type present in a plot, the limited data points available for certain crop type classes, and potential bias in performance metrics as a result of the pixel-based sampling strategy. Recommendations for how to mitigate these limitations in potential future work are detailed in the Discussion.

Challenges and recommendations were outlined for operationalising the methodology detailed in this dissertation for Rwanda. Operational crop type classification would be particularly valuable for Rwanda given the high prevalence of subsistence farming and the country's economic reliance on agriculture. The main operational challenges identified include mitigating the limitations of the main crop type assumption and generating cloud-free composite imagery. A potential process for using additional SAS data not available for this dissertation was outlined to mitigate the main crop type limitation, and potential future work for operationalising the methodology detailed in this dissertation was highlighted in the Discussion.

In conclusion, this dissertation helps to fill both a locational and knowledge gap in the literature for crop type classification using satellite remote sensing and has highlighted key areas of further research. Despite its limitations, this dissertation has detailed a methodology for crop type classification in Rwanda using only freely available data and software (excluding SAS data), and has outlined areas for further methodological refinement and potential future work.

Appendices

Appendix A – Data Preprocessing Methodology Flowchart

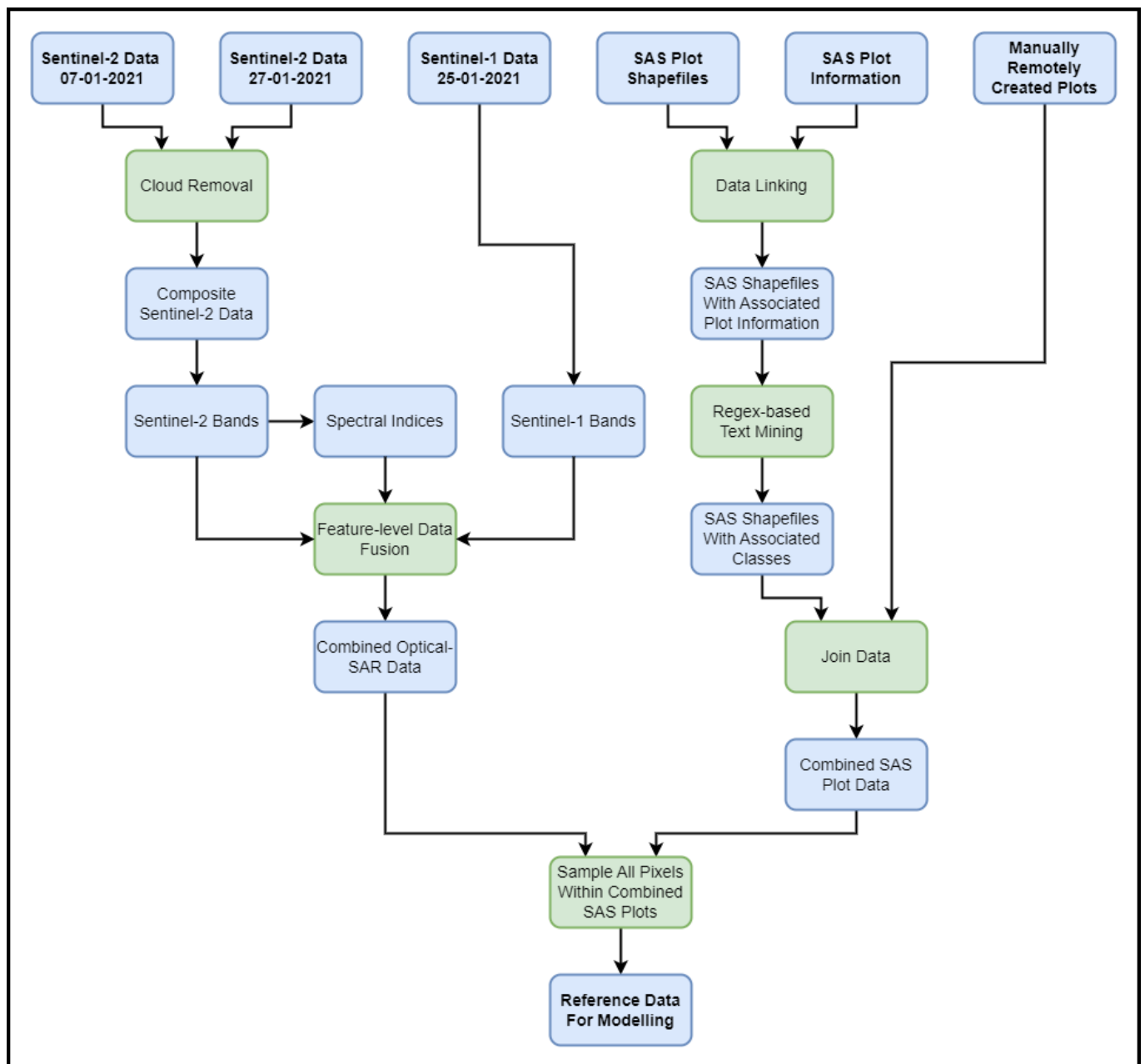


Figure. A. A flowchart detailing the data processing methodology used to create the reference dataset used for model training, validation, and testing. Blue boxes represent datasets, including initial raw datasets, intermediary datasets, and the final reference dataset. Green boxes represent data processing steps, with only the main data processing steps included and smaller processing steps, such as calculating spectral indices from Sentinel-2 bands, not included for brevity. Initial and final datasets are denoted in bold.

Appendix B – Initial Features Correlation Pairs Plot

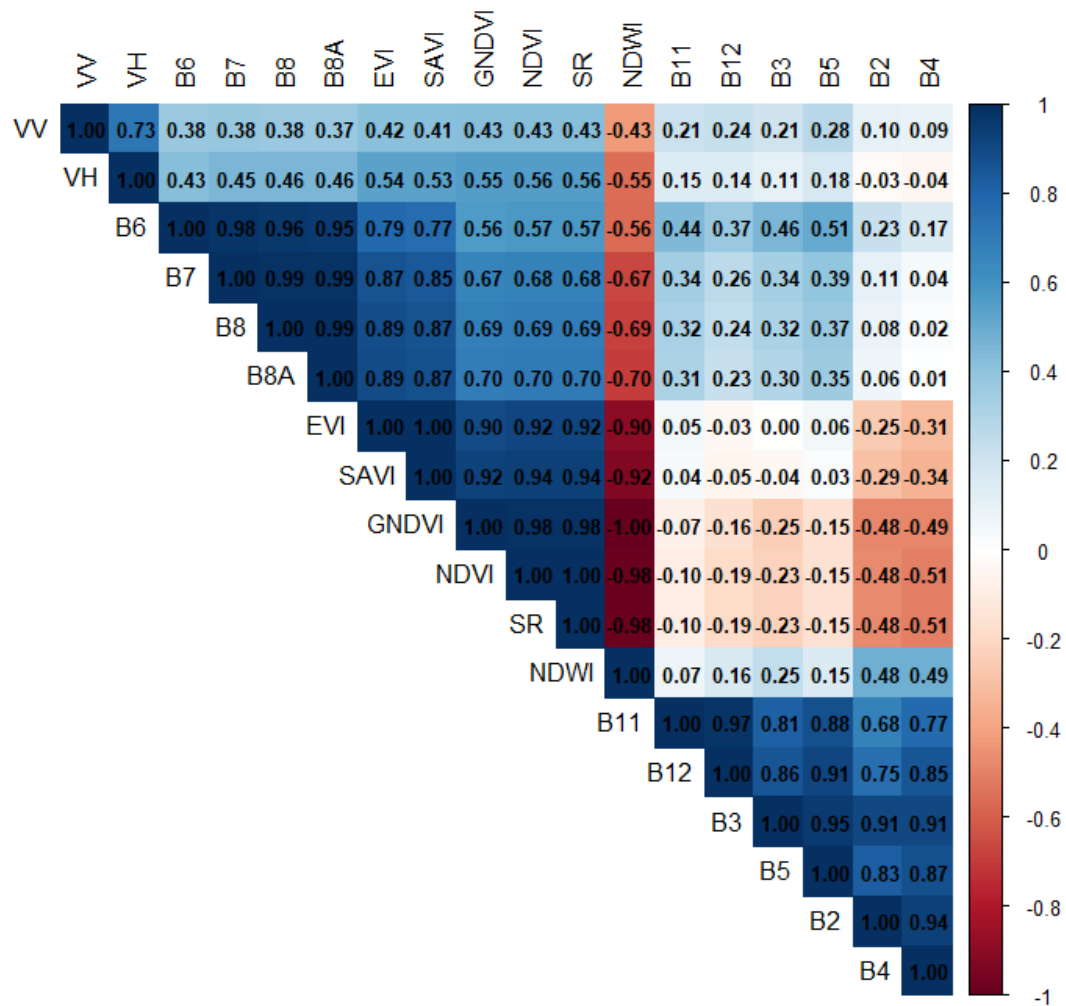


Figure. B. A correlation pairs plot detailing the Spearman's Rank correlation between each of the initial features selected. Spearman's Rank was chosen as not all features are normally distributed.

Appendix C – JavaScript Code

```
// Cloud Removal Testing
// Flags
var image_type = 'RGB';
var filter_by_rwanda_or_district = "Rwanda";
var district_name = "Nyagatare";
var zoom_level = 9;
var load_districts = false;
var load_nayagatare = false;
var run_classification = false;
var smooth_classification = true;
var run_confusion_matrix = false;
var model_type = 'randomForest';
var filter_bands = false;
var bands = ['B4', 'B3', 'B2'];

// Date range
var startDate = '2020-09-01';
var endDate = '2021-02-01';

if (filter_by_rwanda_or_district === "Rwanda") {

  // Import country borders
  var countries = ee.FeatureCollection('USDOS/LSIB_SIMPLE/2017');

  // Filter for Rwanda
  var study_area = countries.filter(ee.Filter.eq('country_co', 'RW'));

} else if (filter_by_rwanda_or_district === 'Region') {

  // Import region borders
  var regions = ee.FeatureCollection('users/alsadair-breasley/rwanda_regions');

  // Filter for specific region
  var regionFeature = ee.Feature(regions.filter(ee.Filter.eq("ADM1_EN", district_name)).first());
  var study_area = regionFeature.geometry();

} else if (filter_by_rwanda_or_district === "District") {

  // Import district borders
  var districts = ee.FeatureCollection('users/alsadair-breasley/rwanda_districts');

  // Filter for specific district
  var districtFeature = ee.Feature(districts.filter(ee.Filter.eq("ADM2_EN", district_name)).first());
  var study_area = districtFeature.geometry();

}

// Function to mask clouds and shadows using the Scene Classification Layer (SCL)
function maskS2CloudsAndShadows(image) {

  var scl = image.select('SCL');

  // Mask out clouds and shadows
  var cloudShadowBitMask = scl.eq(3) // Shadow
    .or(scl.eq(9)) // Cirrus clouds
    .or(scl.eq(10)); // Clouds with high probability

  // Invert mask i.e. select clear conditions
  var mask = cloudShadowBitMask.not();

  // Apply the mask to the image, scale reflectance, select RGB bands
  return image.updateMask(mask).divide(10000);

}

if (image_type === 'RGB') {

  // Load Sentinel-2 Level-2A data and apply cloud filter
  var S2 = ee.ImageCollection('COPERNICUS/S2_SR')
```



```

        .filterDate(startDate, endDate)
        .filterBounds(study_area)
        .map(function(image) {return image.clip(study_area)}) // Clipped here and later to ease commenting out and running
        .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 30));

// Apply cloud and shadow masking function
var cloudShadowMasked = S2.map(maskS2CloudsAndShadows);

// Compute median of images
var medianComposite = cloudShadowMasked.median();

// Clip by Rwanda boundary
var medianCompositeClip = medianComposite.clip(study_area);

// Visualisation parameters for RGB composite
var visParams = {
  min: 0.0,
  max: 0.3,
  bands: ['B4', 'B3', 'B2']
};

// Add RGB composite layer to map
Map.addLayer(study_area, {color: 'FF0000'}, 'Border');
Map.centerObject(study_area, zoom_level);
Map.addLayer(medianCompositeClip, visParams, 'RGB composite');
} else if (image_type === 'NIR_R_G') {

// Load Sentinel-2 Level-2A data and apply cloud filter
var S2 = ee.ImageCollection('COPERNICUS/S2_SR')
  .filterDate(startDate, endDate)
  .filterBounds(study_area)
  .map(function(image) {return image.clip(study_area)}) // Clipped here and later to ease commenting out and running
  .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 30));

// Apply cloud and shadow masking function
var cloudShadowMasked = S2.map(maskS2CloudsAndShadows);

// Compute median of images
var medianComposite = cloudShadowMasked.median();

// Clip by Rwanda boundary
var medianCompositeClip = medianComposite.clip(study_area);

// Visualisation parameters for RGB composite
var visParams = {
  min: 0.0,
  max: 0.3,
  bands: ['B8', 'B4', 'B3']
};

// Add NIR G B composite layer to map
Map.addLayer(study_area, {color: '0000FF'}, 'Border');
Map.centerObject(study_area, zoom_level);
Map.addLayer(medianCompositeClip, visParams, 'False colour composite');
} else if (image_type === 'NIR_SWIR1_R') {

// Load Sentinel-2 Level-2A data and apply cloud filter
var S2 = ee.ImageCollection('COPERNICUS/S2_SR')
  .filterDate(startDate, endDate)
  .filterBounds(study_area)
  .map(function(image) {return image.clip(study_area)}) // Clipped here and later to ease commenting out and running
  .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 30));

// Apply cloud and shadow masking function
var cloudShadowMasked = S2.map(maskS2CloudsAndShadows);

// Compute median of images
var medianComposite = cloudShadowMasked.median();

// Clip by Rwanda boundary
var medianCompositeClip = medianComposite.clip(study_area);

```

```

// Visualisation parameters for RGB composite
var visParams = {
  min: 0.0,
  max: 0.3,
  bands: ['B8', 'B11', 'B4']
};

// Add NIR SWIR1 R composite layer to map
Map.addLayer(study_area, {color: '0000FF'}, 'Border');
Map.centerObject(study_area, zoom_level);
Map.addLayer(medianCompositeClip, visParams, 'False colour composite');

} else if (image_type === 'SWIR2_SWIR1_NIR') {

  // Load Sentinel-2 Level-2A data and apply cloud filter
  var S2 = ee.ImageCollection('COPERNICUS/S2_SR')
    .filterDate(startDate, endDate)
    .filterBounds(study_area)
    .map(function(image) {return image.clip(study_area)}) // Clipped here and later to ease commenting out and running
    .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 30));

  // Apply cloud and shadow masking function
  var cloudShadowMasked = S2.map(maskS2CloudsAndShadows);

  // Compute median of images
  var medianComposite = cloudShadowMasked.median();

  // Clip by Rwanda boundary
  var medianCompositeClip = medianComposite.clip(study_area);

  // Visualisation parameters for RGB composite
  var visParams = {
    min: 0.0,
    max: 0.3,
    bands: ['B12', 'B11', 'B8']
  };

  // Add SWIR2 SWIR1 NIR composite layer to map
  Map.addLayer(study_area, {color: 'FF0000'}, 'Border');
  Map.centerObject(study_area, zoom_level);
  Map.addLayer(medianCompositeClip, visParams, 'False colour composite');

} else if (image_type === 'NDVI') {

  // Load Sentinel-2 Level-2A data and apply cloud filter
  var S2 = ee.ImageCollection('COPERNICUS/S2_SR')
    .filterDate(startDate, endDate)
    .filterBounds(study_area)
    .map(function(image) {return image.clip(study_area)}) // Clipped here and later to ease commenting out and running
    .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 30));

  // Apply cloud and shadow masking function
  var cloudShadowMasked = S2.map(maskS2CloudsAndShadows);

  // Compute median of images
  var medianComposite = cloudShadowMasked.median();

  // Clip by Rwanda boundary
  var medianCompositeClip = medianComposite.clip(study_area);

  // Compute NDVI using NIR (B8) and Red (B4) bands
  var ndvi = medianCompositeClip.normalizedDifference(['B8', 'B4']).rename('NDVI');

  // Visualisation parameters for NDVI
  var visNDVI = {
    min: 0,
    max: 1,
    palette: ['white', 'green']
  };

  // Add NDVI layer to map
  Map.addLayer(study_area, {color: 'FF0000'}, 'Border');
  Map.centerObject(study_area, zoom_level);
  Map.addLayer(ndvi, visNDVI, 'NDVI');

```

```

} else if (image_type === 'NDWI') {

  // Load Sentinel-2 Level-2A data and apply cloud filter
  var S2 = ee.ImageCollection('COPERNICUS/S2_SR')
    .filterDate(startDate, endDate)
    .filterBounds(study_area)
    .map(function(image) {return image.clip(study_area)}) // Clipped here and later to ease commenting out and running
    .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 30));

  // Apply cloud and shadow masking function
  var cloudShadowMasked = S2.map(maskS2CloudsAndShadows);

  // Compute median of images
  var medianComposite = cloudShadowMasked.median();

  // Clip by Rwanda boundary
  var medianCompositeClip = medianComposite.clip(study_area);

  // Compute NDWI using Green (B3) and NIR (B8) bands
  var ndwi = medianCompositeClip.normalizedDifference(['B3', 'B8']).rename('NDWI');

  // Visualisation parameters for NDWI
  var visNDWI = {
    min: -1,
    max: 1,
    palette: ['00FF00', 'FFFFFF', '0000FF']
  };

  // Add NDWI layer to map
  Map.addLayer(study_area, {color: 'FF0000'}, 'Border');
  Map.centerObject(study_area, zoom_level);
  Map.addLayer(ndwi, visNDWI, 'NDWI');

} else if (image_type === 'MNDBI') {

  // Load Sentinel-2 Level-2A data and apply cloud filter
  var S2 = ee.ImageCollection('COPERNICUS/S2_SR')
    .filterDate(startDate, endDate)
    .filterBounds(study_area)
    .map(function(image) {return image.clip(study_area)}) // Clipped here and later to ease commenting out and running
    .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 30));

  // Apply cloud and shadow masking function
  var cloudShadowMasked = S2.map(maskS2CloudsAndShadows);

  // Compute median of images
  var medianComposite = cloudShadowMasked.median();

  // Clip by Rwanda boundary
  var medianCompositeClip = medianComposite.clip(study_area);

  // Compute MNDBI
  var mnbi = medianComposite.expression(
    '(SWIR - NIR) / (SWIR + NIR)', {
      'SWIR': medianComposite.select('B11'),
      'NIR': medianComposite.select('B8')
    }).rename('MNDBI');

  // Visualization parameters for MNDBI
  var visParams = {
    min: -0.5,
    max: 0.5,
    palette: ['blue', 'white', 'red']
  };

  // Add MNDBI layer to map
  // Map.addLayer(study_area, {color: 'FF0000'}, 'Border');
  Map.centerObject(study_area, zoom_level);
  Map.addLayer(mnbi, visParams, 'MNDBI');

} else if (image_type === 'RGB_highest_NDVI') {

```

```

// Define a function that will add an NDVI band
var addNDVI = function(image) {
  var ndvi = image.normalizedDifference(['B8', 'B4']).rename('NDVI');
  return image.addBands(ndvi);
};

// Load Sentinel-2 Level-2A data and apply cloud filter
var S2 = ee.ImageCollection('COPERNICUS/S2_SR')
  .filterDate(startDate, endDate)
  .filterBounds(study_area)
  .map(function(image) {return image.clip(study_area)}) // Clipped here and later to ease commenting out and running
  .map(addNDVI); // Add NDVI layer

// Apply cloud and shadow masking function
var cloudShadowMasked = S2.map(maskS2CloudsAndShadows);

// Select each pixel with the highest NDVI value
var highestNDVI = cloudShadowMasked.qualityMosaic('NDVI');

// Clip by Rwanda boundary
var medianCompositeClip = highestNDVI.clip(study_area);

// Visualisation parameters for RGB composite
var visParams = {
  min: 0.0,
  max: 0.3,
  bands: ['B4', 'B3', 'B2']
};

// Add RGB composite layer to map
Map.addLayer(study_area, {color: 'FF0000'}, 'Border');
Map.centerObject(study_area, zoom_level);
Map.addLayer(medianCompositeClip, visParams, 'RGB composite');
}

// Load districts shape file
if (load_districts === true) {

  var districts = ee.FeatureCollection('users/aldasair-breasley/rwanda_districts');

  var visParams = {
    color: 'FF0000'
  };

  Map.addLayer(districts, visParams, 'Rwanda Districts');
}

// Load districts shape file
if (load_nayagatare === true) {

  var districts = ee.FeatureCollection('users/aldasair-breasley/rwanda_districts');

  var nayagatare = ee.Feature(districts.limit(6).toList(6).get(5));

  var visParams = {
    color: 'FF0000'
  };

  Map.addLayer(nayagatare, visParams, 'Nayagatare');
}

// Filter for specific bands
if (filter_bands === true) {

  // Define desired bands
  var filter_bands = bands;

} else {

  // Define desired bands as all bands
  var filter_bands = medianCompositeClip.bandNames();
}

```

```

}

// Run classification
if (run_classification === true) {

  if (model_type === 'cart') {

    // Decision Tree

    // Define training features
    var trainFeatures = water_class.merge(bare_earth_class).merge(vegetation_class);

    // Define training data
    // Sample pixel values for each band at training data points
    var trainData = medianCompositeClip.sampleRegions({
      collection: trainFeatures,
      properties: ['land_cover'],
      scale: 10 // Resolution
    });

    // Train classification model
    var cartModel = ee.Classifier.smileCart().train({
      features: trainData,
      classProperty: 'land_cover',
      inputProperties: filter_bands
    });

    // Classify image
    var classifiedImage = medianCompositeClip.select(filter_bands).classify(cartModel);

    // Smooth classification
    if (smooth_classification === true) {

      var classifiedImage = classifiedImage.focal_mode({
        radius: 1,
        kernelType: 'square',
        units: 'pixels'
      });

    }

    // Add classified layer to map
    Map.addLayer(classifiedImage, {min: 0, max: 2, palette: ['blue', 'red', 'green']}, 'Classified Image');

  } else if (model_type === 'randomForest') {

    // Random Forest

    // Define training features
    var trainFeatures = water_class.merge(bare_earth_class).merge(vegetation_class);

    // Define training data
    // Sample pixel values for each band at training data points
    var trainData = medianCompositeClip.sampleRegions({
      collection: trainFeatures,
      properties: ['land_cover'],
      scale: 10 // Resolution
    });

    // Train classification model
    var rfModel = ee.Classifier.smileRandomForest(10).train({
      features: trainData,
      classProperty: 'land_cover',
      inputProperties: filter_bands
    });

    // Classify image
    var classifiedImage = medianCompositeClip.select(filter_bands).classify(rfModel);

    // Smooth classification
    if (smooth_classification === true) {

      var classifiedImage = classifiedImage.focal_mode({

```

```

    radius: 1,
    kernelType: 'square',
    units: 'pixels'
  });
}

// Add classified layer to map
Map.addLayer(classifiedImage, {min: 0, max: 2, palette: ['blue', 'red', 'green']}, 'Classified Image');

} else if (model_type === 'svm') {

  // Support Vector Machines

  // Define training features
  var trainFeatures = water_class.merge(bare_earth_class).merge(vegetation_class);

  // Define training data
  // Sample pixel values for each band at training data points
  var trainData = medianCompositeClip.sampleRegions({
    collection: trainFeatures,
    properties: ['land_cover'],
    scale: 10 // Resolution
  });

  // Train classification model
  var svmModel = ee.Classifier.libsvm({
    kernelType: 'RBF',
    gamma: 0.5,
    cost: 10
  }).train({
    features: trainData,
    classProperty: 'land_cover',
    inputProperties: filter_bands
  });

  // Classify image
  var classifiedImage = medianCompositeClip.select(filter_bands).classify(svmModel);

  // Smooth classification
  if (smooth_classification === true) {

    var classifiedImage = classifiedImage.focal_mode({
      radius: 1,
      kernelType: 'square',
      units: 'pixels'
    });

  }

  // Add classified layer to map
  Map.addLayer(classifiedImage, {min: 0, max: 2, palette: ['blue', 'red', 'green']}, 'Classified Image');

} else if (model_type === 'naiveBayes') {

  // Naive Bayes

  // Define training features
  var trainFeatures = water_class.merge(bare_earth_class).merge(vegetation_class);

  // Define training data
  // Sample pixel values for each band at training data points
  var trainData = medianCompositeClip.sampleRegions({
    collection: trainFeatures,
    properties: ['land_cover'],
    scale: 10 // Resolution
  });

  // Train classification model
  var nbModel = ee.Classifier.smileNaiveBayes().train({
    features: trainData,
    classProperty: 'land_cover',
    inputProperties: filter_bands
  });

```

```

// Classify image
var classifiedImage = medianCompositeClip.select(filter_bands).classify(nbModel);

// Smooth classification
if (smooth_classification === true) {

  var classifiedImage = classifiedImage.focal_mode({
    radius: 1,
    kernelType: 'square',
    units: 'pixels'
  });

}

// Add classified layer to map
Map.addLayer(classifiedImage, {min: 0, max: 2, palette: ['blue', 'red', 'green']}, 'Classified Image');

} else if (model_type === 'gradientBoost') {

  // Gradient Boost

  // Define training features
  var trainFeatures = water_class.merge(bare_earth_class).merge(vegetation_class);

  // Define training data
  // Sample pixel values for each band at training data points
  var trainData = medianCompositeClip.sampleRegions({
    collection: trainFeatures,
    properties: ['land_cover'],
    scale: 10 // Resolution
  });

  // Train classification model
  var gbtModel = ee.Classifier.smileGradientTreeBoost({
    numberOfTrees: 50,
    shrinkage: 0.1,
    maxNodes: 3
  }).train({
    features: trainData,
    classProperty: 'land_cover',
    inputProperties: filter_bands
  });

  // Classify image
  var classifiedImage = medianCompositeClip.select(filter_bands).classify(gbtModel);

  // Smooth classification
  if (smooth_classification === true) {

    var classifiedImage = classifiedImage.focal_mode({
      radius: 1,
      kernelType: 'square',
      units: 'pixels'
    });

  }

  // Add classified layer to map
  Map.addLayer(classifiedImage, {min: 0, max: 2, palette: ['blue', 'red', 'green']}, 'Classified Image');

} else if (model_type === 'kNN') {

  // k Nearest Neighbours

  // Define training features
  var trainFeatures = water_class.merge(bare_earth_class).merge(vegetation_class);

  // Define training data
  // Sample pixel values for each band at training data points
  var trainData = medianCompositeClip.sampleRegions({
    collection: trainFeatures,
    properties: ['land_cover'],
    scale: 10 // Resolution
  });

```

```

});

// Train classification model
var knnModel = ee.Classifier.smileKNN({
  k: 5,
}).train({
  features: trainData,
  classProperty: 'land_cover',
  inputProperties: filter_bands
});

// Classify image
var classifiedImage = medianCompositeClip.select(filter_bands).classify(knnModel);

// Smooth classification
if (smooth_classification === true) {

  var classifiedImage = classifiedImage.focal_mode({
    radius: 1,
    kernelType: 'square',
    units: 'pixels'
  });

}

// Add classified layer to map
Map.addLayer(classifiedImage, {min: 0, max: 2, palette: ['blue', 'red', 'green']}, 'Classified Image');

} else if (model_type === 'minimumDistance') {

  // Minimum Distance

  // Define training features
  var trainFeatures = water_class.merge(bare_earth_class).merge(vegetation_class);

  // Define training data
  // Sample pixel values for each band at training data points
  var trainData = medianCompositeClip.sampleRegions({
    collection: trainFeatures,
    properties: ['land_cover'],
    scale: 10 // Resolution
  });

  // Train classification model
  var mindistModel = ee.Classifier.minimumDistance().train({
    features: trainData,
    classProperty: 'land_cover',
    inputProperties: filter_bands
  });

  // Classify image
  var classifiedImage = medianCompositeClip.select(filter_bands).classify(mindistModel);

  // Smooth classification
  if (smooth_classification === true) {

    var classifiedImage = classifiedImage.focal_mode({
      radius: 1,
      kernelType: 'square',
      units: 'pixels'
    });

  }

  // Add classified layer to map
  Map.addLayer(classifiedImage, {min: 0, max: 2, palette: ['blue', 'red', 'green']}, 'Classified Image');

}

}

// Assess classification accuracy
if (run_confusion_matrix === true) {

```



```

// Define test features
var testFeatures = water_test.merge(bare_earth_test).merge(vegetation_test);

// Define test data
// Sample pixel values for each band at test data points
var testData = classifiedImage.sampleRegions({
  collection: testFeatures,
  properties: ['land_cover'],
  scale: 10 // Resolution
}).map(function(feature) {
  return feature.set('predicted', feature.get('classification'));
});

// Test accuracy
var testAccuracy = testData.errorMatrix('land_cover', 'predicted');
print('Confusion Matrix: ', testAccuracy);
print('Overall Accuracy: ', testAccuracy.accuracy());
}

```

```

// Elevation for study area visualisation
// Flags
var filter_by_rwanda_or_district = "District";
var district_name = "Nyagatare";
var zoom_level = 11;
var export_elevation_data = false;
var export_resolution = 20;

// Colour palette
var colour_palette = ['3288bd', '66c2a5', 'abdda4', 'e6f598', 'ffffbf', 'fee08b', 'fdae61', 'f46d43', 'd53e4f'];
// ['0000ff', '00ffff', 'ffff00', 'ff0000', 'ffffff'];
// ['5e3c99', 'b2abd2', 'f5f5f5', 'fdb863', 'e66101'];
// ['000000', '1e9600', 'fff200', 'ff0000'];
// ['3288bd', '66c2a5', 'abdda4', 'e6f598', 'ffffbf', 'fee08b', 'fdae61', 'f46d43', 'd53e4f'];

if (filter_by_rwanda_or_district == "Rwanda") {

  // Import country borders
  var countries = ee.FeatureCollection('USDOS/LSIB_SIMPLE/2017');

  // Filter for Rwanda
  var study_area = countries.filter(ee.Filter.eq('country_co', 'RW'));

} else if (filter_by_rwanda_or_district === 'Region') {

  // Import region borders
  var regions = ee.FeatureCollection('users/aldasair-breasley/rwanda_regions');

  // Filter for specific region
  var regionFeature = ee.Feature(regions.filter(ee.Filter.eq("ADM1_EN", district_name)).first());
  var study_area = regionFeature.geometry();

} else if (filter_by_rwanda_or_district == "District") {

  // Import district borders
  var districts = ee.FeatureCollection('users/aldasair-breasley/rwanda_districts');

  // Filter for specified district
  var districtFeature = ee.Feature(districts.filter(ee.Filter.eq("ADM2_EN", district_name)).first());
  var study_area = districtFeature.geometry();

}

// Load SRTM elevation data and clip for study area
var srtm = ee.Image('USGS/SRTMGL1_003').clip(study_area);

// Calculate the min and max elevation within Rwanda
var elevationStats = srtm.reduceRegion({
  reducer: ee.Reducer.minMax(),
  geometry: study_area,
  scale: 30, // Match the resolution of the SRTM data
  maxPixels: 1e9
});

// Print elevation statistics
elevationStats.evaluate(function(result) {
  print('Min Elevation: ', result.elevation_min);
  print('Max Elevation: ', result.elevation_max);
  // Add elevation layer to map
  Map.layers().set(0, ui.Map.Layer(srtm, {
    min: result.elevation_min,
    max: result.elevation_max,
    palette: colour_palette
  }, 'SRTM Elevation'));
});

Map.centerObject(study_area, zoom_level);

if (export_elevation_data === true) {

  // Export SRTM data
  Export.image.toDrive({
    image: srtm,
    description: 'SRTM_Elevation_Nyagatare',
    scale: export_resolution,

```

```
    region: study_area,  
    fileFormat: 'GeoTIFF',  
    maxPixels: 1e9  
  });  
}
```

```
// Manual feature creation
// Combine feature collections
var combinedFeatures = Water
  .merge(Forest)
  .merge(Bare_Earth)
  .merge(Rice)
  .merge(Banana)
  .merge(Wetland)
  .merge(Grassland);

// Check
print('Combined Features:', combinedFeatures);

// Export as shapefile
Export.table.toDrive({
  collection: combinedFeatures,
  description: 'combined_features',
  fileFormat: 'SHP'
});
```

```

// Individual low cloud image selection
// Flags
var filter_by_rwanda_or_district = 'District';
var district_name = 'Nyagatare';
var zoom_level = 11;
var scene_or_composite = 'composite';
var filter_bands = false;
var bands = ['B2', 'B3', 'B4', 'B8'];
var export_reflectance_values = false;
var export_resolution = 20;

// Date range
var startDate = '2020-12-01';
var endDate = '2021-02-01';

if (filter_by_rwanda_or_district === "Rwanda") {

  // Import country borders
  var countries = ee.FeatureCollection('USDOS/LSIB_SIMPLE/2017');

  // Filter for Rwanda
  var study_area = countries.filter(ee.Filter.eq('country_co', 'RW')).geometry();

} else if (filter_by_rwanda_or_district === 'Region') {

  // Import region borders
  var regions = ee.FeatureCollection('users/aldasair-breasley/rwanda_regions');

  // Filter for specific region
  var regionFeature = ee.Feature(regions.filter(ee.Filter.eq("ADM1_EN", district_name)).first());
  var study_area = regionFeature.geometry();

} else if (filter_by_rwanda_or_district === "District") {

  // Import district borders
  var districts = ee.FeatureCollection('users/aldasair-breasley/rwanda_districts');

  // Filter for specific district
  var districtFeature = ee.Feature(districts.filter(ee.Filter.eq("ADM2_EN", district_name)).first());
  var study_area = districtFeature.geometry();

}

// Function to scale reflectance values
function scaleReflectance(image) {

  if (filter_bands === true) {

    return image.select(bands).divide(10000).copyProperties(image, ['system:time_start']);

  } else {

    return image.divide(10000).copyProperties(image, ['system:time_start']);

  }

}

// Load Sentinel-2 data
var S2 = ee.ImageCollection('COPERNICUS/S2_SR')
  .filterDate(startDate, endDate)
  .filterBounds(study_area)
  .map(function(image) { return image.clip(study_area); })
  .map(scaleReflectance);

// Visualisation parameters for RGB composite
var visParams = {
  min: 0.0,
  max: 0.3,
  bands: ['B4', 'B3', 'B2']
};

// Add each image in the collection to the map
S2 = S2.toList(S2.size());

```

```

var numImages = S2.size().getInfo(); // Get the number of images in the collection

for (var i = 0; i < numImages; i++) {

  var image = ee.Image(S2.get(i));
  var layerName = 'Image ' + (i + 1);
  Map.addLayer(image, visParams, layerName);

}

// Add border layer to the map
Map.addLayer(study_area, {color: 'FF0000'}, 'Border');
Map.centerObject(study_area, zoom_level);

// Select the 15th, 16th, 23rd, and 24th images
var image15 = ee.Image(S2.get(14));
var image16 = ee.Image(S2.get(15));
var image23 = ee.Image(S2.get(22));
var image24 = ee.Image(S2.get(23));

if (scene_or_composite === 'composite') {

  // Combine into a collection
  var selectedImages = ee.ImageCollection([image15, image16, image23, image24]);

} else if (scene_or_composite === 'scene') {

  // Combine into a collection
  var selectedImages = ee.ImageCollection([image23, image24]);

}

// Define a function that will add an NDVI band
var addNDVI = function(image) {

  var ndvi = image.normalizedDifference(['B8', 'B4']).rename('NDVI');

  return image.addBands(ndvi);

};

// Define a function that will add an NDSI band
var addNDSI = function(image) {

  var ndsi = image.normalizedDifference(['B3', 'B11']).rename('NDSI');

  return image.addBands(ndsi);

};

// Function to add BSI band - Not currently used
var addBSI = function(image) {

  var bsi = image.expression(
    '((SWIR1 + RED) - (NIR + BLUE)) / ((SWIR1 + RED) + (NIR + BLUE))',
    {
      SWIR1: image.select('B11'),
      RED: image.select('B4'),
      NIR: image.select('B8'),
      BLUE: image.select('B2')
    }
  ).rename('BSI');

  return image.addBands(bsi);

};

// Define a function that will add a QualityMetric band
var addQualityMetric = function(image) {

  var ndvi = image.select('NDVI');

```

```

var ndsi = image.select('NDSI');

// var bsi = image.select('BSI');

var quality = ndvi.subtract(ndsi).rename('QualityMetric');

return image.addBands(quality);

};

// Create a composite image selecting pixels with the least cloud
var cloudFreeComposite =
selectedImages.map(addNDVI).map(addNDSI).map(addQualityMetric).qualityMosaic('QualityMetric').clip(study_area);

// Add the composite image to the map
Map.addLayer(cloudFreeComposite, visParams, 'Cloud-free Composite');

if (export_reflectance_values === true) {

  // Export composite image to GEE Assets
  Export.image.toDrive({
    image: cloudFreeComposite,
    description: 'cloud_free_composite',
    scale: export_resolution,
    region: study_area,
    fileFormat: 'GeoTIFF',
    folder: 'GEE_Files',
    maxPixels: 1e13
  });
}

```

```

// Pregenerate optical images
// Flags
var filter_by_rwanda_or_district = "District";
var district_name = "Nyagatare";
var zoom_level = 11;
var scene_or_composite = 'scene';
var resolution = 10;
var reducer_type = 'median';
var mixed_pixel_sa = false;
var sample_all_pixels_sa = false;
var run_classification = false;
var smooth_classification = false;
var run_confusion_matrix = false;
var model_type = 'randomForest';
var filter_bands = false;
var bands = ['B2', 'B3', 'B4', 'B8'];
var export_reflectance_values = false;

if (scene_or_composite === 'composite') {

  // Load pregenerated image
  var pregenerated_image = ee.Image('users/aldasair-breasley/cloud_free_composite_15_16_23_24');

} else if (scene_or_composite === 'scene') {

  // Load pregenerated image
  var pregenerated_image = ee.Image('users/aldasair-breasley/cloud_free_composite_23_24');

}

// Visualisation parameters for RGB composite
var visParams = {
  min: 0.0,
  max: 0.3,
  bands: ['B4', 'B3', 'B2']
};

// Add image to the map
Map.addLayer(pregenerated_image, visParams, 'RGB Image');

// Define study area
if (filter_by_rwanda_or_district === "Rwanda") {

  // Import country borders
  var countries = ee.FeatureCollection('USDOS/LSIB_SIMPLE/2017');

  // Filter for Rwanda
  var study_area = countries.filter(ee.Filter.eq('country_co', 'RW'));

} else if (filter_by_rwanda_or_district === 'Region') {

  // Import region borders
  var regions = ee.FeatureCollection('users/aldasair-breasley/rwanda_regions');

  // Filter for specific region
  var regionFeature = ee.Feature(regions.filter(ee.Filter.eq("ADM1_EN", district_name)).first());
  var study_area = regionFeature.geometry();

} else if (filter_by_rwanda_or_district === "District") {

  // Import district borders
  var districts = ee.FeatureCollection('users/aldasair-breasley/rwanda_districts');

  // Filter for specific district
  var districtFeature = ee.Feature(districts.filter(ee.Filter.eq("ADM2_EN", district_name)).first());
  var study_area = districtFeature.geometry();

}

```



```

// Load plots data
var plots_2021_a = ee.FeatureCollection('users/alasdair-breasley/sas_plot_data_2021_a_nyagatare_classes');

// Filter plots to study area
if (filter_by_rwanda_or_district === "District" || filter_by_rwanda_or_district === "Region") {

  // Function to check if a feature is contained fully within the study area
  var isContained = function(feature) {

    var contained = study_area.contains(feature.geometry(), ee.ErrorMargin(1));

    return ee.Feature(feature).set('contained', contained);

  };

  // Apply the function to add contained property
  plots_2021_a = plots_2021_a.map(isContained);

  // Filter based on contained property
  plots_2021_a = plots_2021_a.filter(ee.Filter.eq('contained', true));

}

// Add plots layer to map
Map.addLayer(plots_2021_a, {color: 'FF0000'}, 'Season A 2021 Plots');
Map.centerObject(study_area, zoom_level);

// Debug: Print sample from plots_2021_a
print('Sample from plots_2021_a:', plots_2021_a.first());

// Filter for specific bands
if (filter_bands === true) {

  // Select bands from the composite
  var image = pregenerated_image.select(bands);

} else {

  // Select all bands from the composite
  var image = pregenerated_image;

}

// Reducer type
if (reducer_type === 'median') {

  var reducer = ee.Reducer.median();

} else if (reducer_type === 'mean') {

  var reducer = ee.Reducer.mean();

}

if (mixed_pixel_sa === true) {

  // Function to create mask of fully contained pixels
  var getFullyContainedPixels = function(image, geometry) {
    // Add small inward buffer
    var bufferedGeometry = geometry.buffer(-1);

    var mask = image.select(0).multiply(10000).toInt().reduceToVectors({
      geometry: bufferedGeometry,
      scale: resolution,
      crs: image.projection(),
      reducer: ee.Reducer.countEvery()
    }).geometry();
    return mask;
  };

```

```

if (sample_all_pixels_sa === true) {

  var fullyContainedPixels = plots_2021_a.map(function(plot) {
    var containedGeometry = getFullyContainedPixels(image, plot.geometry());
    return ee.Feature(containedGeometry, plot.toDictionary());
  });

  // Sample reflectance values for each fully contained pixel within each plot
  var sampledReflectance = fullyContainedPixels.map(function(feature) {
    var region = feature.geometry();

    // Check if the geometry is empty
    var isEmpty = region.area(1).eq(0);

    // Sample only if the region is not empty
    var samples = ee.Algorithms.If(isEmpty,
    ee.FeatureCollection([]), // Return empty collection if region is empty
    image.sample({
      region: region,
      scale: resolution,
      numPixels: 1e13,
      geometries: true,
    })
  );

  return ee.FeatureCollection(samples).map(function(sampledFeature) {
    return sampledFeature.set(feature.toDictionary());
  });
}).flatten();

  // Filter out any remaining null values
  sampledReflectance = sampledReflectance.filter(ee.Filter.notNull(image.bandNames()));

} else {

  // Sample reflectance values for each plot
  var sampledReflectance = plots_2021_a.map(function(plot) {
    var containedGeometry = getFullyContainedPixels(image, plot.geometry());
    var sampled = image.reduceRegion({
      reducer: reducer,
      geometry: containedGeometry,
      scale: resolution,
      maxPixels: 1e13
    });
    return plot.set(sampled);
  });

}

} else {

  if (sample_all_pixels_sa === true) {

    // Sample reflectance values for each pixel within each plot
    var sampledReflectance = image.sampleRegions({
      collection: plots_2021_a,
      scale: resolution,
      geometries: true
    });

  } else {

    // Sample reflectance values for each plot
    var sampledReflectance = image.reduceRegions({
      collection: plots_2021_a,
      reducer: reducer,
      scale: resolution
    });
  }
}

```

```
}  
  
}  
  
// Debug: Print total number of sampled pixels  
print("Total number of sampled pixels:", sampledReflectance.size());  
  
if (export_reflectance_values === true) {  
  
  // Export sampled reflectance values to Google Drive  
  Export.table.toDrive({  
    collection: sampledReflectance,  
    description: 'plot_reflectance_values',  
    fileFormat: 'CSV'  
  });  
  
}
```

```

// Pregenerate SAR images
// Flags
var filter_by_rwanda_or_district = 'District';
var district_name = 'Nyagatare';
var zoom_level = 11;
var resolution = 20;
var reducer_type = 'median';
var mixed_pixel_sa = false;
var sample_all_pixels_sa = false;
var clip_scenes = false;
var export_sar_scenes = false;
var export_backscatter_values = false;

// Date range
var startDate = '2021-01-25';
var endDate = '2021-01-26';

if (filter_by_rwanda_or_district === "Rwanda") {

  // Import country borders
  var countries = ee.FeatureCollection("USDOS/LSIB_SIMPLE/2017");

  // Filter for Rwanda
  var study_area = countries.filter(ee.Filter.eq('country_co', 'RW')).geometry();

} else if (filter_by_rwanda_or_district === 'Region') {

  // Import region borders
  var regions = ee.FeatureCollection('users/aldasair-breasley/rwanda_regions');

  // Filter for specific region
  var regionFeature = ee.Feature(regions.filter(ee.Filter.eq("ADM1_EN", district_name)).first());
  var study_area = regionFeature.geometry();

} else if (filter_by_rwanda_or_district === "District") {

  // Import district borders
  var districts = ee.FeatureCollection('users/aldasair-breasley/rwanda_districts');

  // Filter for specific district
  var districtFeature = ee.Feature(districts.filter(ee.Filter.eq("ADM2_EN", district_name)).first());
  var study_area = districtFeature.geometry();

}

// Load Sentinel-1 SAR collection for Interferometric Wide Swath mode
var sentinel1 = ee.ImageCollection("COPERNICUS/S1_GRD")
  .filterBounds(study_area)
  .filterDate(startDate, endDate)
  .filter(ee.Filter.eq('instrumentMode', 'IW'));

// Check if images are loaded
print('Number of images:', sentinel1.size());
print('Image collection:', sentinel1);

// Select VV, VH polarization bands
var sentinel1_vv = sentinel1.select('VV');
var sentinel1_vh = sentinel1.select('VH');
//var sentinel1_hh = sentinel1.select('HH');
//var sentinel1_hv = sentinel1.select('HV');

if (clip_scenes === true) {

  // Create median composite images for each band
  var sentinel1_vv_composite = sentinel1_vv.median().clip(study_area);
  var sentinel1_vh_composite = sentinel1_vh.median().clip(study_area);
  //var sentinel1_hh_composite = sentinel1_hh.median().clip(study_area);
  //var sentinel1_hv_composite = sentinel1_hv.median().clip(study_area);

```

```

} else {

var no_clip_nyagatare = /* color: #98ff00 */ ee.Geometry.Polygon(
  [[[30.05000302624871, -1.4991861720690347],
    [30.348007176639335, -1.6762726278811257],
    [30.78334042859246, -1.4978133507051719],
    [30.732528661014335, -1.4030866426157846],
    [30.493576024295585, -1.0268901798230325],
    [30.33015439343621, -1.0241440377301052],
    [30.00880429577996, -1.3989680011994687]]]);

// Create median composite images for each band
var sentinel1_vv_composite = sentinel1_vv.median().clip(no_clip_nyagatare);
var sentinel1_vh_composite = sentinel1_vh.median().clip(no_clip_nyagatare);
//var sentinel1_hh_composite = sentinel1_hh.median();
//var sentinel1_hv_composite = sentinel1_hv.median();

}

// Visualisation parameters
var visParams = {
  min: -25,
  max: 0,
  palette: ['blue', 'cyan', 'yellow', 'red']
};

// Add composite images to the map
Map.centerObject(study_area, zoom_level);

Map.addLayer(sentinel1_vv_composite, visParams, 'Sentinel-1 VV');
Map.addLayer(sentinel1_vh_composite, visParams, 'Sentinel-1 VH');
//Map.addLayer(sentinel1_hh_composite, visParams, 'Sentinel-1 HH');
//Map.addLayer(sentinel1_hv_composite, visParams, 'Sentinel-1 HV');

if (export_sar_scenes === true) {

  // Export SAR VV scene
  Export.image.toDrive({
    image: sentinel1_vv_composite,
    description: 'Sentinel1_SAR_VV',
    scale: resolution,
    region: sentinel1_vv_composite.geometry(),
    fileFormat: 'GeoTIFF',
    folder: 'GEE_Files',
    maxPixels: 1e13
  });

  // Export SAR VH scene
  Export.image.toDrive({
    image: sentinel1_vh_composite,
    description: 'Sentinel1_SAR_VH',
    scale: resolution,
    region: sentinel1_vh_composite.geometry(),
    fileFormat: 'GeoTIFF',
    folder: 'GEE_Files',
    maxPixels: 1e13
  });

}

// Load plots data
var plots_2021_a = ee.FeatureCollection('users/aldasair-breasley/sas_plot_data_2021_a_nyagatare_classes');

// Filter plots to study area
if (filter_by_rwanda_or_district === "District" || filter_by_rwanda_or_district === "Region") {

  // Function to check if a feature is contained fully within the study area
  var isContained = function(feature) {

```

```

var contained = study_area.contains(feature.geometry(), ee.ErrorMargin(1));

return ee.Feature(feature).set('contained', contained);

};

// Apply the function to add contained property
plots_2021_a = plots_2021_a.map(isContained);

// Filter based on contained property
plots_2021_a = plots_2021_a.filter(ee.Filter.eq('contained', true));

}

// Add plots layer to map
Map.addLayer(plots_2021_a, {color: 'FF0000'}, 'Season A 2021 Plots');
Map.centerObject(study_area, zoom_level);

// Debug: Print sample from plots_2021_a
print('Sample from plots_2021_a:', plots_2021_a.first());

// Function to create mask of fully contained pixels
var getFullyContainedPixels = function(image, geometry) {
  var bufferedGeometry = geometry.buffer(-1);
  var mask = image.select(0).toInt().reduceToVectors({
    geometry: bufferedGeometry,
    scale: resolution,
    crs: image.projection(),
    reducer: ee.Reducer.countEvery()
  }).geometry();
  return mask;
};

// Reducer type
if (reducer_type === 'median') {

  var reducer = ee.Reducer.median();
  var get_value = 'median';

} else if (reducer_type === 'mean') {

  var reducer = ee.Reducer.mean();
  var get_value = 'mean';

}

if (mixed_pixel_sa === true) {

  if (sample_all_pixels_sa === true) {

    // Create feature collection of fully contained pixels
    var fullyContainedPixels = plots_2021_a.map(function(plot) {
      var containedGeometry = getFullyContainedPixels(sentinel1_vv_composite, plot.geometry());
      return ee.Feature(containedGeometry, plot.toDictionary());
    });

    // Sample backscatter values for each fully contained pixel within each plot
    var sampledBackscatter_vv = fullyContainedPixels.map(function(feature) {
      var region = feature.geometry();
      var isEmpty = region.area(1).eq(0);
      var samples = ee.Algorithms.If(isEmpty,
        ee.FeatureCollection([]), // Return empty collection if region is empty
        sentinel1_vv_composite.sample({
          region: region,
          scale: resolution,
          numPixels: 1e13,
          geometries: true,
        })
      );
    });
  }
}

```

```

    return ee.FeatureCollection(samples).map(function(sampledFeature) {
      return sampledFeature.set(feature.toDictionary());
    });
  }).flatten();

var sampledBackscatter_vh = fullyContainedPixels.map(function(feature) {
  var region = feature.geometry();
  var isEmpty = region.area(1).eq(0);
  var samples = ee.Algorithms.If(isEmpty,
    ee.FeatureCollection([]), // Return empty collection if region is empty
    sentinel1_vh_composite.sample({
      region: region,
      scale: resolution,
      numPixels: 1e13,
      geometries: true,
    })
  );
  return ee.FeatureCollection(samples).map(function(sampledFeature) {
    return sampledFeature.set(feature.toDictionary());
  });
}).flatten();

} else {

  // Sample backscatter values for each plot
  var sampledBackscatter_vv = plots_2021_a.map(function(plot) {
    var containedGeometry = getFullyContainedPixels(sentinel1_vv_composite, plot.geometry());
    var sampled = sentinel1_vv_composite.reduceRegion({
      reducer: reducer,
      geometry: containedGeometry,
      scale: resolution,
      maxPixels: 1e13
    });
    return plot.set(sampled);
  });

  var sampledBackscatter_vh = plots_2021_a.map(function(plot) {
    var containedGeometry = getFullyContainedPixels(sentinel1_vh_composite, plot.geometry());
    var sampled = sentinel1_vh_composite.reduceRegion({
      reducer: reducer,
      geometry: containedGeometry,
      scale: resolution,
      maxPixels: 1e13
    });
    return plot.set(sampled);
  });

}

} else {

  if (sample_all_pixels_sa === true) {

    // Sample backscatter values for each pixel within each plot
    var sampledBackscatter_vv = sentinel1_vv_composite.sampleRegions({
      collection: plots_2021_a,
      scale: resolution,
      geometries: true
    });

    var sampledBackscatter_vh = sentinel1_vh_composite.sampleRegions({
      collection: plots_2021_a,
      scale: resolution,
      geometries: true
    });

  } else {

    // Sample backscatter values for each plot

```

```

var sampledBackscatter_vv = sentinel1_vv_composite.reduceRegions({
  collection: plots_2021_a,
  reducer: reducer,
  scale: resolution
}).map(function(feature) {
  return feature.set('VV', feature.get(get_value)).select(['.*'], null, false);
});

var sampledBackscatter_vh = sentinel1_vh_composite.reduceRegions({
  collection: plots_2021_a,
  reducer: reducer,
  scale: resolution
}).map(function(feature) {
  return feature.set('VH', feature.get(get_value)).select(['.*'], null, false);
});

}

}

// Debug: Check the size of the training data
print("Training data size VV:", sampledBackscatter_vv.size());

// Debug: Print the first training data
print("Training data sample VV:", sampledBackscatter_vv.first());

// Debug: Check the size of the training data
print("Training data size VH:", sampledBackscatter_vh.size());

// Debug: Print the first training data
print("Training data sample VH:", sampledBackscatter_vh.first());

if (export_backscatter_values === true) {

  // Export sampled reflectance values to Google Drive
  Export.table.toDrive({
    collection: sampledBackscatter_vv,
    description: 'plot_backscatter_values_vv',
    fileFormat: 'CSV'
  });

  Export.table.toDrive({
    collection: sampledBackscatter_vh,
    description: 'plot_backscatter_values_vh',
    fileFormat: 'CSV'
  });

}

```



```

// Combined optical and SAR data-fusion, sampling, and image for classification output
// Flags
var filter_by_rwanda_or_district = 'District';
var district_name = 'Nyagatare';
var zoom_level = 11;
var scene_or_composite = 'composite';
var export_combined_image = true;
var resolution = 20;
var reducer_type = 'median';
var mixed_pixel_sa = false;
var sample_all_pixels_sa = true;
var run_classification = false;
var seed = 123456;
var ntrees = 100;
var mtry = 3;
var smooth_classification = false;
var neighbourhood_size = 1;
var filter_bands = false;
var bands = ['B2', 'B3', 'B4', 'B8'];
var export_reflectance_values = false;
var export_resolution = 20;
var export_classified_map = false;

if (scene_or_composite === 'composite') {

  // Load pregenerated image
  var pregenerated_image = ee.Image('users/aldasair-breasley/cloud_free_composite_15_16_23_24_20m');

} else if (scene_or_composite === 'scene') {

  // Load pregenerated image
  var pregenerated_image = ee.Image('users/aldasair-breasley/cloud_free_composite_23_24_20m');

}

// Load pregenerated SAR
var pregenerated_sar_vv = ee.Image('users/aldasair-breasley/sar_vv_20m');
var pregenerated_sar_vh = ee.Image('users/aldasair-breasley/sar_vh_20m');

// Resample SAR data to match Sentinel-2 resolution
pregenerated_sar_vv = pregenerated_sar_vv.resample('bilinear').reproject({
  crs: pregenerated_image.projection().crs(),
  scale: resolution
});

// Resample SAR data to match Sentinel-2 resolution
pregenerated_sar_vh = pregenerated_sar_vh.resample('bilinear').reproject({
  crs: pregenerated_image.projection().crs(),
  scale: resolution
});

pregenerated_image = pregenerated_image.addBands(pregenerated_sar_vv).addBands(pregenerated_sar_vh);

// Visualisation parameters for RGB composite
var visParams = {
  min: 0.0,
  max: 0.3,
  bands: ['B4', 'B3', 'B2']
};

// Add image to the map
Map.addLayer(pregenerated_image, visParams, 'RGB Image');

// Define study area
if (filter_by_rwanda_or_district === "Rwanda") {

  // Import country borders
  var countries = ee.FeatureCollection("USDOS/LSIB_SIMPLE/2017");

  // Filter for Rwanda
  var study_area = countries.filter(ee.Filter.eq('country_co', 'RW'));

} else if (filter_by_rwanda_or_district === 'Region') {

```

```

// Import region borders
var regions = ee.FeatureCollection('users/aldasair-breasley/rwanda_regions');

// Filter for specific region
var regionFeature = ee.Feature(regions.filter(ee.Filter.eq("ADM1_EN", district_name)).first());
var study_area = regionFeature.geometry();

} else if (filter_by_rwanda_or_district === "District") {

// Import district borders
var districts = ee.FeatureCollection('users/aldasair-breasley/rwanda_districts');

// Filter for specific district
var districtFeature = ee.Feature(districts.filter(ee.Filter.eq("ADM2_EN", district_name)).first());
var study_area = districtFeature.geometry();

}

if (export_combined_image === true) {

  pregenerated_image = pregenerated_image.select(['B2', 'B3', 'B4', 'B5', 'B6', 'B7', 'B8', 'B8A', 'B11', 'B12', 'VV',
  'VH']).toFloat().clip(study_area);

  // Export combined image
  Export.image.toDrive({
    image: pregenerated_image,
    description: 'combined_composite',
    scale: resolution,
    region: study_area,
    fileFormat: 'GeoTIFF',
    folder: 'GEE_Files',
    maxPixels: 1e13
  });

}

// Load plots data
var plots_2021_a = ee.FeatureCollection('projects/ee-ab-rwanda-
dissertation/assets/sas_plot_data_2021_a_nyagatare_classes_new');

// Filter plots to study area
if (filter_by_rwanda_or_district === "District" || filter_by_rwanda_or_district === "Region") {

  // Function to check if a feature is contained fully within the study area
  var isContained = function(feature) {

    var contained = study_area.contains(feature.geometry(), ee.ErrorMargin(1));

    return ee.Feature(feature).set('contained', contained);

  };

  // Apply the function to add contained property
  plots_2021_a = plots_2021_a.map(isContained);

  // Filter based on contained property
  plots_2021_a = plots_2021_a.filter(ee.Filter.eq('contained', true));

}

// Add plots layer to map
Map.addLayer(plots_2021_a, {color: 'FF0000'}, 'Season A 2021 Plots');
Map.centerObject(study_area, zoom_level);

// Debug: Print sample from plots_2021_a
print('Sample from plots_2021_a:', plots_2021_a.first());

// Filter for specific bands
if (filter_bands === true) {

  // Select bands from the composite
  var image = pregenerated_image.select(bands);

} else {

```

```

// Select all bands from the composite
var image = pregenerated_image;

}

// Reducer type
if (reducer_type === 'median') {

  var reducer = ee.Reducer.median();

} else if (reducer_type === 'mean') {

  var reducer = ee.Reducer.mean();

}

if (mixed_pixel_sa === true) {

  // Function to create mask of fully contained pixels
  var getFullyContainedPixels = function(image, geometry) {
    // Add small inward buffer
    var bufferedGeometry = geometry.buffer(-1);

    var mask = image.select(0).multiply(10000).toInt().reduceToVectors({
      geometry: bufferedGeometry,
      scale: resolution,
      crs: image.projection(),
      reducer: ee.Reducer.countEvery()
    }).geometry();
    return mask;
  };

  if (sample_all_pixels_sa === true) {

    var fullyContainedPixels = plots_2021_a.map(function(plot) {
      var containedGeometry = getFullyContainedPixels(image, plot.geometry());
      return ee.Feature(containedGeometry, plot.toDictionary());
    });

    // Sample reflectance values for each fully contained pixel within each plot
    var sampledReflectance = fullyContainedPixels.map(function(feature) {
      var region = feature.geometry();

      // Check if the geometry is empty
      var isEmpty = region.area(1).eq(0);

      // Sample only if the region is not empty
      var samples = ee.Algorithms.If(isEmpty,
        ee.FeatureCollection([]), // Return empty collection if region is empty
        image.sample({
          region: region,
          scale: resolution,
          numPixels: 1e13,
          geometries: true,
        })
      );

      return ee.FeatureCollection(samples).map(function(sampledFeature) {
        return sampledFeature.set(feature.toDictionary());
      });
    }).flatten();

    // Filter out any remaining null values
    sampledReflectance = sampledReflectance.filter(ee.Filter.notNull(image.bandNames()));

  } else {

    // Sample reflectance values for each plot
    var sampledReflectance = plots_2021_a.map(function(plot) {
      var containedGeometry = getFullyContainedPixels(image, plot.geometry());
      var sampled = image.reduceRegion({
        reducer: reducer,
        geometry: containedGeometry,

```

```

        scale: resolution,
        maxPixels: 1e13
    });
    return plot.set(sampled);
});

}

} else {

    if (sample_all_pixels_sa === true) {

        // Sample reflectance values for each pixel within each plot
        var sampledReflectance = image.sampleRegions({
            collection: plots_2021_a,
            scale: resolution,
            geometries: true
        });

    } else {

        // Sample reflectance values for each plot
        var sampledReflectance = image.reduceRegions({
            collection: plots_2021_a,
            reducer: reducer,
            scale: resolution
        });

    }

}

// Debug: Print total number of sampled pixels
print("Total number of sampled pixels:", sampledReflectance.size());

if (export_reflectance_values === true) {

    // Export sampled reflectance values to Google Drive
    Export.table.toDrive({
        collection: sampledReflectance,
        description: 'plot_reflectance_values',
        fileFormat: 'CSV'
    });

}

```

Appendix D – R Code

```
---
title: "Exploratory Data Analysis - Nyagatare"
output: html_document
---

```{r setup, include = FALSE}

knitr::opts_chunk$set(echo = TRUE)

```

```{r include = FALSE}

Load packages
library(tidyverse)
library(ggplot2)
library(sf)
library(car)
library(caret)
library(randomForest)
library(openxlsx)
library(raster)
library(terra)
library(corrplot)

```

```{r}

Seed
seed = 42

Run options - data
scene_or_composite = "_composite" # "_scene" "_composite"
sa_option = "_all_pixel_sa" # "" "_mixed_pixel_sa" "_all_pixel_sa" "_mixed_pixel_all_pixel_sa"
resolution = "_20m" # "_20m"
reduction = "" # "" "_mean"
filter_bands = TRUE
bands = c("B2", "B3", "B4", "B5", "B6", "B7", "B8", "B11", "B12", "VV")
NDVI = TRUE
EVI = FALSE
NDWI = TRUE
SR = FALSE
SAVI = FALSE
GNDVI = FALSE
SAR = TRUE
bare_built_grouped = TRUE
new_classes = TRUE

Run options - script
full_run = FALSE
create_shapefile = FALSE
save_shapefile = FALSE
run_non_train_test_rfs = FALSE
test_split = 0.3
undersample_majority_classes = FALSE
div_by_grass_wet = 5
div_by_rice = 2
run_post_combination_summaries = TRUE
pure_crop_training = FALSE
run_hyperparameter_tuning = FALSE
limit_mtry = FALSE
mtry_max_value = 3
mtry_default = 7
save_outputs = FALSE
run_forward_selection = FALSE

Image classification options
run_image_classification = FALSE
save_classified_image = FALSE
```

```

...

```{r}

# Load crop data
# filepath = "C:/Users/alsd/Documents/MDS/Dissertation/Data/Seasonal Agricultural Survey Data/crop_data.csv"
# filepath = "C:/Users/ADMIN/Documents/MDS/Dissertation/Data/Seasonal Agricultural Survey Data/crop_data.csv"
crop_data = read.csv(filepath, header = TRUE)

...

```{r}

2021 Season A data - Nyagatare
crop_data_2021A_Nyagatare = crop_data %>%
 dplyr::filter(Season == "SeasonA2021") %>%
 dplyr::filter(X1.2.District.Name == "Nyagatare")

...

```{r}

if (full_run == TRUE) {

  # Counts of crop type instances
  crop_data_2021A_Nyagatare %>%
    dplyr::group_by(X2.1.1.Crop.name) %>%
    dplyr::summarise(Count = n()) %>%
    dplyr::arrange(desc(Count))

}

...

```{r}

if (full_run == TRUE) {

 # Counts of cropping system type plot instances
 crop_data_2021A_Nyagatare %>%
 dplyr::distinct(plot.unique.id, .keep_all = TRUE) %>%
 dplyr::group_by(X2.8..Cropping.system) %>%
 dplyr::summarise(Count = n()) %>%
 dplyr::arrange(desc(Count))

}

...

```{r}

if (full_run == TRUE) {

  # Counts of plot instances smaller and larger than 200m^2
  crop_data_2021A_Nyagatare %>%
    dplyr::distinct(plot.unique.id, .keep_all = TRUE) %>%
    dplyr::mutate(X2.3.Plot.size.square.meters. = if_else(X2.3.Plot.size.square.meters. > 200, "big", "small")) %>%
    dplyr::group_by(X2.3.Plot.size.square.meters.) %>%
    dplyr::summarise(Count = n()) %>%
    dplyr::arrange(desc(Count))

}

...

```{r}

if (full_run == TRUE) {

 # Sum of crop type instances areas
 crop_data_2021A_Nyagatare %>%
 dplyr::group_by(X2.1.1.Crop.name) %>%
 dplyr::summarise(Area = sum(X2.3.Plot.size.square.meters.)) %>%

```

```

dplyr::arrange(desc(Area))

}

...

```{r}

if (full_run == TRUE) {

  # Counts of non-agricultural land use type instances
  crop_data_2021A_Nyagatare %>%
    dplyr::group_by(X2.7..Non.agricultural.land.type) %>%
    dplyr::summarise(Count = n()) %>%
    dplyr::arrange(desc(Count))

}

...

```{r}

if (full_run == TRUE) {

 # Count of number of crops in a plot plot instances
 crop_data_2021A_Nyagatare %>%
 dplyr::distinct(plot.unique.id, .keep_all = TRUE) %>%
 dplyr::mutate(X2.9..Number.of.main.crops.in.the.plot = as.factor(X2.9..Number.of.main.crops.in.the.plot)) %>%
 dplyr::group_by(X2.9..Number.of.main.crops.in.the.plot) %>%
 dplyr::summarise(Count = n()) %>%
 dplyr::arrange(desc(Count))

}

...

```{r}

# Create dataset with intercropping combinations by plot
crop_data_2021A_Nyagatare_plots = crop_data_2021A_Nyagatare %>%
  dplyr::group_by(plot.unique.id) %>%
  dplyr::summarise(
    intercrops = if_else(
      any(X2.11.Crop.name != ""),
      paste(sort(unique(X2.11.Crop.name[X2.11.Crop.name != "]])), collapse = ", "),
      first(X2.7..Non.agricultural.land.type[X2.11.Crop.name == "]])),
    non_ag_flag = if_else(any(X2.11.Crop.name != ""), "Crop", "Non-agricultural") %>%
  dplyr::ungroup() %>%
  dplyr::mutate(intercrops = case_when(intercrops == "Forest/Bush" ~ "Forest",
    intercrops == "Water body" ~ "Water",
    intercrops == "Bare/Rocky soil" | intercrops == "Road/Path" ~ "Bare Earth",
    intercrops == "Buildings" ~ "Built Up",
    TRUE ~ intercrops)) %>%
  dplyr::filter(intercrops != "" & intercrops != "Others" & intercrops != "Unmanaged marshland")

...

```{r}

if (full_run == TRUE) {

 # Count of intercropping combination type plot instances
 crop_data_2021A_Nyagatare_plots %>%
 dplyr::filter(non_ag_flag != "Non-agricultural") %>%
 dplyr::group_by(intercrops) %>%
 dplyr::summarise(Count = n()) %>%
 dplyr::arrange(desc(Count))

}

...

```{r}

```

```

if (full_run == TRUE) {

  # Unique crop types
  sort(unique(crop_data_2021A_Nyagatare$X2.11.Crop.name))

}

```

```{r}

# Data processing to create classes for prediction
crop_data_2021A_Nyagatare_classes = crop_data_2021A_Nyagatare_plots %>%
  dplyr::mutate(crop_class = case_when(grepl("Paddy rice", intercrops) ~ "Rice",
    grepl("Cooking banana|Dessert banana|Banana for beer", intercrops) ~ "Banana",
    grepl("Maize|Sorghum|Millet", intercrops) ~ "Dryland Cereals",
    grepl("Bush bean|Climbing bean|French beans|Groundnut|Other pulse|Pea|Small red bean|Soybean",
intercrops) ~ "Legumes and Pulses",
    grepl("Cassava|Sweet potato|Irish potato|Taro|Yams", intercrops) ~ "Tubers",
    non_ag_flag == "Non-agricultural" ~ intercrops,
    TRUE ~ "Other Crop")) %>%
  dplyr::mutate(plot.unique.id = as.character(plot.unique.id))

```

```{r}

if (full_run == TRUE) {

  # Count of crop class instances
  crop_data_2021A_Nyagatare_classes %>%
  dplyr::group_by(crop_class) %>%
  dplyr::summarise(Count = n()) %>%
  dplyr::arrange(desc(Count))

}

```

```{r}

if (create_shapefile == TRUE) {

  # Load shapefile plot data
  shapefile_read_path = "C:/Users/Admin/Documents/MDS/Dissertation/Data/Plot Data/2021
a/sas_plot_data_2021_a_nyagatare.shp"

  shapefile = st_read(shapefile_read_path)

}

```

```{r}

if (create_shapefile == TRUE) {

  # Fix unique plot ID issue
  shapefile = shapefile %>%
  dplyr::mutate(plot_uid = substring(plot_uid, 1, nchar(plot_uid) - 2))

}

```

```{r}

if (create_shapefile == TRUE) {

  # Join chosen classes to shapefile
  shapefile_joined = shapefile %>%
  dplyr::inner_join(crop_data_2021A_Nyagatare_classes, by = c("plot_uid" = "plot.unique.id"))

}

```



```

...

```{r}

if (create_shapefile == TRUE) {

 # Count of crop class instances post shapefile join
 as.data.frame(st_drop_geometry(shapefile_joined)) %>%
 dplyr::group_by(crop_class) %>%
 dplyr::summarise(Count = n()) %>%
 dplyr::arrange(desc(Count))

}

...

```{r}

if (create_shapefile == TRUE) {

  # Load manually created features data
  shapefile_read_path = "C:/Users/Admin/Documents/MDS/Dissertation/Data/Plot Data/2021 a manual
features/combined_features_manual_nyagatare_new.shp"

  shapefile_features = st_read(shapefile_read_path)

  # Rename non_ag fla due to GEE Property name 10 character limit
  shapefile_features = shapefile_features %>%
  dplyr::rename(non_ag_flag = non_ag fla)

}

...

```{r}

if (create_shapefile == TRUE) {

 # Check both shapefiles are in the same CRS
 if (st_crs(shapefile_joined) != st_crs(shapefile_features)) {

 shapefile_features = st_transform(shapefile_features, st_crs(shapefile_joined))

 warning(paste("Features shapefile required a CRS transformation."))

 }

}

...

```{r}

if (create_shapefile == TRUE) {

  # Check for valid geometries in shapefile_joined and clean them
  valid_geometries1 = st_is_valid(shapefile_joined)

  if (sum(!valid_geometries1) > 0) {

    shapefile_joined = st_make_valid(shapefile_joined)

    warning(paste("shapefile_joined required geometry validation."))

  }

  # Check for valid geometries in shapefile_features and clean them
  valid_geometries2 = st_is_valid(shapefile_features)

  if (sum(!valid_geometries2) > 0) {

    shapefile_features = st_make_valid(shapefile_features)

  }

}

}

```

```

warning(paste("shapefile_features required geometry validation."))

}

# Verify all geometries are now valid
valid_geometries1 = st_is_valid(shapefile_joined)
valid_geometries2 = st_is_valid(shapefile_features)

print(sum(!valid_geometries1))
print(sum(!valid_geometries2))

# Print any invalid geometries
invalid_geometries1 = shapefile_joined[!valid_geometries1, ]
invalid_geometries2 = shapefile_features[!valid_geometries2, ]

print(invalid_geometries1)
print(invalid_geometries2)

}

```{r}

if (create_shapefile == TRUE) {

 # Add manual plot_uid to shapefile_features
 shapefile_features = shapefile_features %>%
 dplyr::mutate(plot_uid = row_number()) %>%
 dplyr::mutate(plot_uid = as.character(plot_uid))

}

```{r}

if (create_shapefile == TRUE) {

  # Join manually created features to shapefile_joined retaining all shapefile_joined columns
  # Get the column names for both shapefiles
  cols1 = colnames(shapefile_joined)
  cols2 = colnames(shapefile_features)

  # Add missing columns to shapefile_joined
  for (col in setdiff(cols2, cols1)) {

    shapefile_joined[[col]] = NA

  }

  # Add missing columns to shapefile_features
  for (col in setdiff(cols1, cols2)) {

    shapefile_features[[col]] = NA

  }

  # Ensure the columns are in the same order
  shapefile_features = shapefile_features[, colnames(shapefile_joined)]

  # Combine the rows
  shapefile_combined = rbind(shapefile_joined, shapefile_features)

}

```{r}

if (create_shapefile == TRUE) {

 # Remove columns
 shapefile_combined = shapefile_combined %>%

```

```

dplyr::select(-c(province_i, district, district_i, cluster_id, CreationDa, Creator, EditDate, Editor, created_da,
 Creator_1, Creation_1, Creator_12, EditDate_1, Editor_1, Creator_2, Creation_2, EditDate_2,
 Editor_12, Creator__1, Creation_3, EditDate_3, province_2, province_3, district_2, district_3,
 cluster_2, Creation_4, Creator_2, EditDate_4))

}

```{r}

if (create_shapefile == TRUE) {

  if (save_shapefile == TRUE) {

    # Save new combined shapefile
    shapefile_write_path = "C:/Users/Admin/Documents/MDS/Dissertation/Data/Plot Data/2021 a Nyagatare
classes/sas_plot_data_2021_a_nyagatare_classes_new.shp"

    st_write(shapefile_combined, shapefile_write_path)

  }

}

```{r}

Analysis With 10m Bands

```{r}

if (full_run == TRUE) {

  # Load sampled reflectance values data
  plot_reflectance_values_path = paste0("C:/Users/Admin/Documents/MDS/Dissertation/Data/Plot Data/2021 a Nyagatare addit
reflectance values/plot_reflectance_values", scene_or_composite, sa_option, resolution, reduction, ".csv")

  plot_reflectance_data = read.csv(plot_reflectance_values_path, header = TRUE)

  # Rename to unshorted names
  plot_reflectance_data = plot_reflectance_data %>%
    dplyr::rename(crop_class = crp_cls, intercroops = intrcrp, non_ag_flag = nn_g_fl, plot_uid = plot_ud)

}

```{r}

if (full_run == TRUE) {

 # Select required columns
 plot_reflectance_data = plot_reflectance_data %>%
 dplyr::select(B2, B3, B4, B8, crop_class, intercroops)

 # Check for NAs and remove rows with NAs
 row_count_before = nrow(plot_reflectance_data)
 plot_reflectance_data = plot_reflectance_data %>% na.omit()
 row_count_after = nrow(plot_reflectance_data)
 rows_removed = row_count_before - row_count_after

 # Print a warning with the number of rows removed
 if (rows_removed > 0) {

 warning(paste("Number of rows removed due to NAs:", rows_removed))

 }

}

```{r}

```

```

if (full_run == TRUE) {

  # Long format
  plot_reflectance_data_long = plot_reflectance_data %>%
    tidyr::pivot_longer(cols = starts_with("B"), names_to = "Band", values_to = "Reflectance") %>%
    dplyr::select(crop_class, Band, Reflectance)

}

...

```{r}

if (full_run == TRUE) {

 # Plot band reflectance values by crop_class
 ggplot(plot_reflectance_data_long, aes(x = Reflectance, colour = crop_class, fill = crop_class)) +
 geom_density(alpha = 0.25) +
 facet_wrap(~ Band, scales = "free")

}

...

```{r}

if (full_run == TRUE) {

  # ANOVA test to check if there are statistically significant differences in the mean reflectance between crop classes for each band
  anova_results = plot_reflectance_data_long %>%
    group_by(Band) %>%
    summarise(ANOVA = list(anova(lm(Reflectance ~ crop_class, data = pick(everything())))))

  anova_results$ANOVA

}

...

```{r}

if (full_run == TRUE) {

 # Process data for modelling
 plot_reflectance_data_m = plot_reflectance_data %>%
 dplyr::select(crop_class, B2, B3, B4, B8) %>%
 dplyr::filter(!crop_class %in% c("", "Other Crop")) %>%
 dplyr::mutate(crop_class = as.factor(crop_class))

}

...

```{r}

if (full_run == TRUE) {

  if (run_non_train_test_rfs == TRUE) {

    # Set seed
    set.seed(seed)

    # Random Forest model
    rf_model = randomForest(crop_class ~ B2 + B3 + B4 + B8, data = plot_reflectance_data_m)

    print(rf_model)

  }

}

...

```{r}

```

```

Load sampled reflectance values data
plot_reflectance_values_path = paste0("C:/Users/Admin/Documents/MDS/Dissertation/Data/Plot Data/2021 a Nyagatare addit
reflectance and backscatter values/plot_reflectance_values", scene_or_composite, sa_option, resolution, reduction, "_new.csv")

plot_reflectance_data = read.csv(plot_reflectance_values_path, header = TRUE)

Rename to unshorted names
plot_reflectance_data = plot_reflectance_data %>%
 dplyr::rename(crop_class = crp_cls, intercrops = intrcrp, non_ag_flag = nn_g_fl, plot_uid = plot_ud)

Group Bare Earth and Built Up as they are spectrally similar
if (bare_built_grouped == TRUE) {

 plot_reflectance_data = plot_reflectance_data %>%
 dplyr::mutate(crop_class = case_when(crop_class == "Bare Earth" | crop_class == "Built Up" ~ "Bare Earth and Built Up",
 TRUE ~ crop_class),
 intercrops = case_when(intercrops == "Bare Earth" | intercrops == "Built Up" ~ "Bare Earth and Built Up",
 TRUE ~ intercrops))

}

...

```{r}

# Select required columns
plot_reflectance_data = plot_reflectance_data %>%
  dplyr::select(B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12, VV, VH, crop_class, intercrops, non_ag_flag, plot_uid)

# Check for NAs and remove rows with NAs
row_count_before = nrow(plot_reflectance_data)
plot_reflectance_data = plot_reflectance_data %>% na.omit()
row_count_after = nrow(plot_reflectance_data)
rows_removed = row_count_before - row_count_after

# Print a warning with the number of rows removed
if (rows_removed > 0) {

  warning(paste("Number of rows removed due to NAs:", rows_removed))

}

...

```{r}

if (SAR == TRUE) {

 plot_backscatter_data_prelong = plot_reflectance_data %>%
 dplyr::select(c(VV, VH))

} else {

 plot_reflectance_data_prelong = plot_reflectance_data %>%
 dplyr::select(-c(VV, VH))

}

...

```{r}

if (full_run == TRUE) {

  # Long format
  plot_reflectance_data_long = plot_reflectance_data_prelong %>%
    tidyr::pivot_longer(cols = starts_with("B"), names_to = "Band", values_to = "Reflectance") %>%
    dplyr::select(crop_class, Band, Reflectance)

}

...

```

```

```{r}

if (full_run == TRUE) {

 # Plot band reflectance values by crop_class
 ggplot(plot_reflectance_data_long, aes(x = Reflectance, colour = crop_class, fill = crop_class)) +
 geom_density(alpha = 0.25) +
 facet_wrap(~ Band, scales = "free")

}

```

```{r}

if (full_run == TRUE) {

 # ANOVA test to check if there are statistically significant differences in the mean reflectance between crop classes for each band
 anova_results = plot_reflectance_data_long %>%
 group_by(Band) %>%
 summarise(ANOVA = list(anova(lm(Reflectance ~ crop_class, data = pick(everything())))))

 anova_results$ANOVA

}

```

```{r}

Process data for modelling
plot_reflectance_data_m = plot_reflectance_data %>%
 dplyr::select(plot_uid, crop_class, B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12, VV, VH, intercroops, non_ag_flag) %>%
 dplyr::filter(!crop_class %in% c("", "Other Crop")) %>%
 dplyr::mutate(crop_class = as.factor(crop_class))

if (NDVI == TRUE) {

 plot_reflectance_data_m = plot_reflectance_data_m %>%
 dplyr::mutate(NDVI = (B8 - B4) / (B8 + B4))

}

if (EVI == TRUE) {

 plot_reflectance_data_m = plot_reflectance_data_m %>%
 dplyr::mutate(EVI = 2.5 * ((B8 - B4) / (B8 + 6 * B4 - 7.5 * B2 + 1)))

}

if (NDWI == TRUE) {

 plot_reflectance_data_m = plot_reflectance_data_m %>%
 dplyr::mutate(NDWI = (B3 - B8) / (B3 + B8))

}

if (SR == TRUE) {

 plot_reflectance_data_m = plot_reflectance_data_m %>%
 dplyr::mutate(SR = B8 / B4)

}

if (SAVI == TRUE) {

 L = 0.5 # Soil adjustment factor

 plot_reflectance_data_m = plot_reflectance_data_m %>%
 dplyr::mutate(SAVI = ((B8 - B4) * (1 + L)) / (B8 + B4 + L))

}

if (GNDVI == TRUE) {

```

```

plot_reflectance_data_m = plot_reflectance_data_m %>%
 dplyr::mutate(GNDVI = (B8 - B3) / (B8 + B3))
}

if (SAR != TRUE) {

 plot_reflectance_data_m = plot_reflectance_data_m %>%
 dplyr::select(-c(VV, VH))

}

Check for NAs and remove rows with NAs
row_count_before = nrow(plot_reflectance_data_m)
plot_reflectance_data_m = plot_reflectance_data_m %>% na.omit()
row_count_after = nrow(plot_reflectance_data_m)
rows_removed = row_count_before - row_count_after

Print a warning with the number of rows removed
if (rows_removed > 0) {

 warning(paste("Number of rows removed due to NAs:", rows_removed))

}

if (filter_bands == TRUE) {

 all_bands = c("B2", "B3", "B4", "B5", "B6", "B7", "B8", "B8A", "B11", "B12", "VV", "VH")

 cols_to_remove = setdiff(all_bands, bands)

 plot_reflectance_data_m = plot_reflectance_data_m %>%
 dplyr::select(-all_of(cols_to_remove))

}

if (undersample_majority_classes == TRUE) {

 # Grassland
 set.seed(seed)

 plot_reflectance_data_m_grassland = plot_reflectance_data_m %>%
 dplyr::filter(crop_class == "Grassland")

 n_grassland = nrow(plot_reflectance_data_m_grassland) / div_by_grass_wet

 grassland_indices = sample(1:nrow(plot_reflectance_data_m_grassland), size = n_grassland, replace = FALSE)

 plot_reflectance_data_m_grassland_sampled = plot_reflectance_data_m_grassland[grassland_indices,]

 # Wetland
 set.seed(seed)

 plot_reflectance_data_m_wetland = plot_reflectance_data_m %>%
 dplyr::filter(crop_class == "Wetland")

 n_wetland = nrow(plot_reflectance_data_m_wetland) / div_by_grass_wet

 wetland_indices = sample(1:nrow(plot_reflectance_data_m_wetland), size = n_wetland, replace = FALSE)

 plot_reflectance_data_m_wetland_sampled = plot_reflectance_data_m_wetland[wetland_indices,]

 # Rice
 set.seed(seed)

 plot_reflectance_data_m_rice = plot_reflectance_data_m %>%
 dplyr::filter(crop_class == "Rice")

 n_rice = nrow(plot_reflectance_data_m_rice) / div_by_rice

 rice_indices = sample(1:nrow(plot_reflectance_data_m_rice), size = n_rice, replace = FALSE)

 plot_reflectance_data_m_rice_sampled = plot_reflectance_data_m_rice[rice_indices,]

```

```

plot_reflectance_data_m = plot_reflectance_data_m %>%
 dplyr::filter(!(crop_class %in% c("Grassland", "Wetland", "Rice")))

plot_reflectance_data_m = rbind(plot_reflectance_data_m, plot_reflectance_data_m_grassland_sampled)

plot_reflectance_data_m = rbind(plot_reflectance_data_m, plot_reflectance_data_m_wetland_sampled)

plot_reflectance_data_m = rbind(plot_reflectance_data_m, plot_reflectance_data_m_rice_sampled)

}

...

```{r}

if (run_post_combination_summaries == TRUE) {

  # Process combined modelling data for summary tables
  plot_reflectance_data_m_summary = plot_reflectance_data_m %>%
    dplyr::mutate(intercrops = if_else(intercrops == "", crop_class, intercrops)) %>%
    dplyr::mutate(sas_flag = if_else(nchar(plot_uid) > 3, 1, 0))

}

...

```{r}

if (run_post_combination_summaries == TRUE) {

 # Pixels by crop type
 plot_reflectance_data_m_summary %>%
 dplyr::group_by(crop_class) %>%
 dplyr::summarise(Count = n()) %>%
 dplyr::arrange(desc(Count))

}

...

```{r}

if (run_post_combination_summaries == TRUE) {

  # Pixels by crop type SAS
  plot_reflectance_data_m_summary %>%
    dplyr::filter(sas_flag == 1) %>%
    dplyr::group_by(crop_class) %>%
    dplyr::summarise(Count = n()) %>%
    dplyr::arrange(desc(Count))

}

...

```{r}

if (run_post_combination_summaries == TRUE) {

 # Pixels by crop type manually created
 plot_reflectance_data_m_summary %>%
 dplyr::filter(sas_flag == 0) %>%
 dplyr::group_by(crop_class) %>%
 dplyr::summarise(Count = n()) %>%
 dplyr::arrange(desc(Count))

}

...

```{r}

if (run_post_combination_summaries == TRUE) {

```



```

# Plots by crop type
plot_reflectance_data_m_summary %>%
  dplyr::group_by(plot_uid) %>%
  dplyr::slice(1) %>%
  dplyr::ungroup() %>%
  dplyr::group_by(crop_class) %>%
  dplyr::summarise(Count = n()) %>%
  dplyr::arrange(desc(Count))

}

...

```{r}

if (run_post_combination_summaries == TRUE) {

 # Plots by crop type SAS
 plot_reflectance_data_m_summary %>%
 dplyr::filter(sas_flag == 1) %>%
 dplyr::group_by(plot_uid) %>%
 dplyr::slice(1) %>%
 dplyr::ungroup() %>%
 dplyr::group_by(crop_class) %>%
 dplyr::summarise(Count = n()) %>%
 dplyr::arrange(desc(Count))

}

...

```{r}

if (run_post_combination_summaries == TRUE) {

  # Pixels by crop type manually created
  plot_reflectance_data_m_summary %>%
    dplyr::filter(sas_flag == 0) %>%
    dplyr::group_by(plot_uid) %>%
    dplyr::slice(1) %>%
    dplyr::ungroup() %>%
    dplyr::group_by(crop_class) %>%
    dplyr::summarise(Count = n()) %>%
    dplyr::arrange(desc(Count))

}

...

```{r}

if (run_post_combination_summaries == TRUE) {

 # Plots by cropping practice SAS
 plot_reflectance_data_m_summary %>%
 dplyr::filter(sas_flag == 1 & non_ag_flag == "Crop") %>%
 dplyr::mutate(intercropping_flag = if_else(grepl(",", intercrops), 1, 0)) %>%
 dplyr::group_by(plot_uid) %>%
 dplyr::slice(1) %>%
 dplyr::ungroup() %>%
 dplyr::group_by(intercropping_flag) %>%
 dplyr::summarise(Count = n()) %>%
 dplyr::arrange(desc(Count))

}

...

```{r}

if (run_post_combination_summaries == TRUE) {

  # Plots by intercropping combination SAS

```

```

plot_reflectance_data_m_summary %>%
  dplyr::filter(sas_flag == 1 & non_ag_flag == "Crop") %>%
  dplyr::mutate(crop_count = str_count(inter crops, ",") + 1) %>%
  dplyr::group_by(plot_uid) %>%
  dplyr::slice(1) %>%
  dplyr::ungroup() %>%
  dplyr::group_by(crop_count) %>%
  dplyr::summarise(Count = n()) %>%
  dplyr::arrange(desc(Count))

}

...

```{r}

if (run_post_combination_summaries == TRUE) {

 # Plots by intercropping combination SAS
 plot_reflectance_data_m_summary %>%
 dplyr::filter(sas_flag == 1 & non_ag_flag == "Crop") %>%
 dplyr::group_by(plot_uid) %>%
 dplyr::slice(1) %>%
 dplyr::ungroup() %>%
 dplyr::group_by(inter crops) %>%
 dplyr::summarise(Count = n()) %>%
 dplyr::arrange(desc(Count))

}

...

```{r}

if (run_non_train_test_rfs == TRUE) {

  # Set seed
  set.seed(seed)

  if (sa_option == "_all_pixel_sa" | sa_option == "_mixed_pixel_all_pixel_sa") {

    # List of columns to exclude
    exclude_columns = c("plot_uid", "inter crops", "non_ag_flag", "pixel_id")

  } else {

    # List of columns to exclude
    exclude_columns = c("plot_uid", "inter crops", "non_ag_flag")

  }

  # Names of predictor columns
  predictor_columns = setdiff(names(plot_reflectance_data_m), c("crop_class", exclude_columns))

  # Create model formula
  formula = as.formula(paste("crop_class ~ ", paste(predictor_columns, collapse = " + ")))

  # Random Forest model
  rf_model = randomForest(formula, data = plot_reflectance_data_m)

  print(rf_model)

}

...

```{r}

if (pure_crop_training == TRUE) {

 # Add pure cropping flag based on
 plot_reflectance_data_m_pure_flag = plot_reflectance_data_m %>%
 dplyr::mutate(inter crops = if_else(inter crops == "", crop_class, inter crops)) %>%

```

```

dplyr::mutate(pure_flag = case_when(crop_class == "Rice" & grepl("^(Paddy rice|Rice)(\\s*(Paddy rice|Rice))*$", intercrops) ~ 1,
 crop_class == "Rice" & !grepl("^(Paddy rice|Rice)(\\s*(Paddy rice|Rice))*$", intercrops) ~ 0,
 crop_class == "Banana" & grepl("^(Cooking banana|Dessert banana|Banana for beer|Banana)(\\s*(Cooking
banana|Dessert banana|Banana for beer|Banana))*$", intercrops) ~ 1,
 crop_class == "Banana" & !grepl("^(Cooking banana|Dessert banana|Banana for beer|Banana)(\\s*(Cooking
banana|Dessert banana|Banana for beer|Banana))*$", intercrops) ~ 0,
 crop_class == "Dryland Cereals" & grepl("^(Maize|Sorghum|Millet)(\\s*(Maize|Sorghum|Millet))*$",
intercrops) ~ 1,
 crop_class == "Dryland Cereals" & !grepl("^(Maize|Sorghum|Millet)(\\s*(Maize|Sorghum|Millet))*$",
intercrops) ~ 0,
 crop_class == "Legumes and Pulses" & grepl("^(Bush bean|Climbing bean|French beans|Groundnut|Other
pulse|Pea|Small red bean|Soybean)(\\s*(Bush bean|Climbing bean|French beans|Groundnut|Other pulse|Pea|Small red
bean|Soybean))*$", intercrops) ~ 1,
 crop_class == "Legumes and Pulses" & !grepl("^(Bush bean|Climbing bean|French beans|Groundnut|Other
pulse|Pea|Small red bean|Soybean)(\\s*(Bush bean|Climbing bean|French beans|Groundnut|Other pulse|Pea|Small red
bean|Soybean))*$", intercrops) ~ 0,
 crop_class == "Tubers" & grepl("^(Cassava|Sweet potato|Irish potato|Taro|Yams)(\\s*(Cassava|Sweet
potato|Irish potato|Taro|Yams))*$", intercrops) ~ 1,
 #
 crop_class == "Tubers" & !grepl("^(Cassava|Sweet potato|Irish potato|Taro|Yams)(\\s*(Cassava|Sweet
potato|Irish potato|Taro|Yams))*$", intercrops) ~ 0,
 crop_class == intercrops ~ 1,
 crop_class != intercrops ~ 0))

Note: Not used in current form with manually running sections to create train and test sets for intercropping analysis for
dissertation
Instances of pure cropped plots by crop_type
plot_reflectance_data_m_pure %>%
dplyr::group_by(crop_class) %>%
dplyr::summarise(Count = n()) %>%
dplyr::arrange(desc(Count))
#
Set seed
set.seed(seed)
#
Sample indices based on the test split proportion using stratified random sampling at the crop_class level
This ensures the test dataset includes a representative proportion of each type1 class from the full dataset
indices = createDataPartition(plot_reflectance_data_m_pure$crop_class, p = test_split, list = FALSE, times = 1)
#
#indices = sample(1:nrow(plot_reflectance_data_m), size = round(nrow(plot_reflectance_data_m) * test_split))
#
Create training set
training_data = plot_reflectance_data_m_pure[-indices,]
#
Null dataframe
plot_reflectance_data_m_pure = NULL
#
Create test set
test_data_pure = plot_reflectance_data_m[indices,]
#
test_data_mixed = plot_reflectance_data_m_pure_flag %>%
dplyr::filter(pure_flag == 0) %>%
dplyr::select(-pure_flag)
#
test_data = rbind(test_data_pure, test_data_mixed)
#
Null dataframes
test_data_pure = NULL
test_data_mixed = NULL
} else {

Set seed
set.seed(seed)

Sample indices based on the test split proportion using stratified random sampling at the crop_class level
This ensures the test dataset includes a representative proportion of each type1 class from the full dataset
indices = createDataPartition(plot_reflectance_data_m$crop_class, p = test_split, list = FALSE, times = 1)

#indices = sample(1:nrow(plot_reflectance_data_m), size = round(nrow(plot_reflectance_data_m) * test_split))

Create test set
test_data = plot_reflectance_data_m[indices,]
Create training set
training_data = plot_reflectance_data_m[-indices,]

```

```

}
...

```{r}

# Remove categorical variables
plot_reflectance_data_m_pairs = plot_reflectance_data_m %>%
  dplyr::select(-c(plot_uid, crop_class, intercroops, non_ag_flag))

# Prepare data
# Use Spearman's Rank Correlation Coefficient because the data is not normally distributed
plot_reflectance_data_m_pairs_matrix = cor(plot_reflectance_data_m_pairs, method = "spearman")

# Create correlation heatmap
corrplot(plot_reflectance_data_m_pairs_matrix, method = "color", type = "upper", order = "hclust",
  tl.col = "black", tl.srt = 90, addCoef.col = "black", number.cex = 0.8)

...

```{r}

if (run_hyperparameter_tuning == TRUE) {

 # Set mtry max
 if (limit_mtry == TRUE) {

 mtry_max = mtry_max_value

 } else if (sa_option == "_all_pixel_sa" | sa_option == "_mixed_pixel_all_pixel_sa") {

 mtry_max = ncol(plot_reflectance_data_m) - 4

 } else {

 mtry_max = ncol(plot_reflectance_data_m) - 4

 }

 # Define mtry values to run for tuning
 tuneGrid = expand.grid(mtry = c(1:(mtry_max)))

 # Specify 5-fold cross-validation
 fitControl = trainControl(method = "cv",
 number = 5)

 # Set seed
 set.seed(seed)

 if (sa_option == "_all_pixel_sa" | sa_option == "_mixed_pixel_all_pixel_sa") {

 # List of columns to exclude
 exclude_columns = c("plot_uid", "intercroops", "non_ag_flag", "pixel_id")

 } else {

 # List of columns to exclude
 exclude_columns = c("plot_uid", "intercroops", "non_ag_flag")

 }

 # Names of predictor columns
 predictor_columns = setdiff(names(plot_reflectance_data_m), c("crop_class", exclude_columns))

 # Create model formula
 formula = as.formula(paste("crop_class ~ ", paste(predictor_columns, collapse = " + ")))

 # Perform 5-fold cross-validation and mtry tuning
 model_rf = train(formula,
 data = training_data,
 method = "rf",
 tuneGrid = tuneGrid,
 trControl = fitControl,

```

```

 metric = "Accuracy")

 print(model_rf)
}
...

```{r}

if (run_hyperparameter_tuning == TRUE) {

    # Predict crop_class values for test data
    predictions = predict(model_rf, newdata = test_data)
    # Print confusion matrix
    confusionMatrix(predictions, test_data$crop_class)

}
...

```{r}

This runs as ntrees hyperparameter tuning only when run_hyperparameter_tuning is TRUE and any non all pixel sensitivity
analysis option is selected
When run_hyperparameter_tuning is FALSE this runs simply as model training
When any all pixel sensitivity analysis option is selected this runs simply as model training due to constraints on compute
TODO: For final model hyperparameter tuning change this and run

if (run_hyperparameter_tuning == TRUE) {

 # Best mtry value
 mtry = model_rf$bestTune[[1]]

} else {

 # Default mtry value
 mtry = mtry_default

}

if (sa_option == "_all_pixel_sa" | sa_option == "_mixed_pixel_all_pixel_sa") {

 # List of columns to exclude
 exclude_columns = c("plot_uid", "intercrops", "non_ag_flag", "pixel_id")

} else {

 # List of columns to exclude
 exclude_columns = c("plot_uid", "intercrops", "non_ag_flag")

}

Names of predictor columns
predictor_columns = setdiff(names(plot_reflectance_data_m), c("crop_class", exclude_columns))

Create model formula
formula = as.formula(paste("crop_class ~ ", paste(predictor_columns, collapse = " + ")))

best_accuracy = 0

if (run_hyperparameter_tuning == FALSE | sa_option == "_all_pixel_sa" | sa_option == "_mixed_pixel_all_pixel_sa") {

 ntrees_vector = c(500)

} else {

 ntrees_vector = c(500, 1000, 2000, 4000)

}

Loop through ntrees values and tune
Functions as only model training if not running hyperparameter tuning
for (ntrees in ntrees_vector) {

```

```

Set seed
set.seed(seed)

Define mtry values to run for tuning
tuneGrid = expand.grid(mtry = c(mtry))

Specify 5-fold cross-validation
fitControl = trainControl(method = "cv",
 number = 5)

Create model version based on ntrees value
model_rf_nt = randomForest(formula, data = training_data, ntree = ntrees, mtry = mtry)

Perform 5-fold cross-validation
model_rf_nt = train(formula,
 data = training_data,
 method = "rf",
 tuneGrid = tuneGrid,
 trControl = fitControl,
 metric = "Accuracy",
 ntree = ntrees)

Print accuracy for each model version
print(model_rf_nt$results$Accuracy)

if (model_rf_nt$results$Accuracy > best_accuracy) {

 ntrees_best = ntrees

 model_rf_best = model_rf_nt

}

}

```{r}

# Define suffixes for automated filepath generation
# Filepaths for every output record the settings ran with to generate that output
if (NDVI == TRUE) {

  NDVI_suffix = "_NDVI"

} else {

  NDVI_suffix = ""

}

if (EVI == TRUE) {

  EVI_suffix = "_EVI"

} else {

  EVI_suffix = ""

}

if (GNDVI == TRUE) {

  GNDVI_suffix = "_GNDVI"

} else {

  GNDVI_suffix = ""

}

if (SAVI == TRUE) {

```

```

SAVI_suffix = "_SAVI"
} else {
  SAVI_suffix = ""
}
if (SAR == TRUE) {
  SAR_suffix = "_SAR"
} else {
  SAR_suffix = ""
}
test_split_suffix = paste0("_", as.character(test_split))
if (pure_crop_training == TRUE) {
  pure_crop_training_suffix = "_PCT"
} else {
  pure_crop_training_suffix = ""
}
if (run_hyperparameter_tuning == TRUE) {
  run_hyperparameter_tuning_suffix = "_HPT"
} else {
  run_hyperparameter_tuning_suffix = ""
}
if (limit_mtry == TRUE) {
  limit_mtry_suffix = "_LMT"
} else {
  limit_mtry_suffix = ""
}
ntrees_suffix = paste0("_", as.character(ntrees_best))
mtry_suffix = paste0("_", as.character(mtry))

# Concatenate final output filepath suffix
output_filepath_suffix = paste0(scene_or_composite, sa_option, resolution, reduction, NDVI_suffix, EVI_suffix, GNDVI_suffix,
SAVI_suffix,
                                SAR_suffix, test_split_suffix, pure_crop_training_suffix, run_hyperparameter_tuning_suffix, limit_mtry_suffix,
                                ntrees_suffix, mtry_suffix)

...

```{r}

Predict crop_class values for test data
predictions = predict(model_rf_best, newdata = test_data)
Print confusion matrix
cm = confusionMatrix(predictions, test_data$crop_class)

cm

if (save_outputs == TRUE) {
 # Create excel workbook

```

```

wb = createWorkbook()

Extract overall statistics
overall_stats = as.data.frame(t(cm$overall))

Extract statistics by class
by_class_stats = as.data.frame(cm$byClass)

by_class_stats = tibble::rownames_to_column(by_class_stats, var = "Class") %>%
 dplyr::mutate(Class = substring(Class, 8))

Extract confusion matrix table
cm_table = as.data.frame(cm$table)

cm_table = cm_table %>%
 tidyr::spread(Reference, Freq)

Add to excel tabs
addWorksheet(wb, "Overall Statistics")
writeData(wb, sheet = "Overall Statistics", x = overall_stats)

addWorksheet(wb, "Statistics by Class")
writeData(wb, sheet = "Statistics by Class", x = by_class_stats)

addWorksheet(wb, "Confusion Matrix")
writeData(wb, sheet = "Confusion Matrix", x = cm_table)

addWorksheet(wb, "Precision")
writeData(wb, sheet = "Precision", x = as.data.frame(macro_precision))

addWorksheet(wb, "Recall")
writeData(wb, sheet = "Recall", x = as.data.frame(macro_recall))

addWorksheet(wb, "Macro F1")
writeData(wb, sheet = "Macro F1", x = as.data.frame(macro_f1))

addWorksheet(wb, "Weighted F1")
writeData(wb, sheet = "Weighted F1", x = as.data.frame(weighted_f1))

addWorksheet(wb, "Precision by Class")
writeData(wb, sheet = "Precision by Class", x = as.data.frame(precision))

addWorksheet(wb, "Recall by Class")
writeData(wb, sheet = "Recall by Class", x = as.data.frame(recall))

addWorksheet(wb, "F1 Score by Class")
writeData(wb, sheet = "F1 Score by Class", x = as.data.frame(f1_score))

}

...

```{r}

# Assess feature importance
imp = importance(model_rf_best)

imp

if (save_outputs == TRUE) {

  imp_df = data.frame(Band = rownames(imp), Importance = imp[,1])

  addWorksheet(wb, "Feature Importance")
  writeData(wb, sheet = "Feature Importance", x = imp_df)

}

...

```{r}

if (save_outputs == TRUE) {

```



```

Save excel
saveWorkbook(wb, paste0("results/", output_filepath_suffix, ".xlsx"), overwrite = TRUE)

}

...

```{r}

if (run_forward_selection == TRUE) {

# Note: Forward selection was only used to check for bias created by resampling 20m bands to 10m and is not detailed in the
dissertation
training_data = training_data %>%
  dplyr::select(-c(plot_uid, intercroops, non_ag_flag))

# Set up cross-validation
fitControl = trainControl(method = "cv", number = 5)

evaluate_model = function(predictors, data) {

  # Ensure mtry is within valid range
  mtry_value = min(sqrt(length(predictors)), length(predictors))

  # Set seed
  set.seed(seed)

  model = train(crop_class ~ ., data = data[, c(predictors, "crop_class")],
    method = "rf",
    tuneGrid = expand.grid(mtry = mtry_value),
    trControl = fitControl,
    metric = "Accuracy")

  return(model$results$Accuracy)

}

# Initialise variables
selected_predictors = c()
remaining_predictors = setdiff(names(training_data), "crop_class")
best_accuracy = 0
overall_best_accuracy = 0

# Perform forward selection
for (i in 1:length(remaining_predictors)) {

  accuracies = sapply(remaining_predictors, function(predictor) {

    evaluate_model(c(selected_predictors, predictor), training_data)

  })

  # Find best predictor
  best_predictor = remaining_predictors[which.max(accuracies)]
  best_accuracy = max(accuracies)

  # Add best predictor to selected list
  selected_predictors = c(selected_predictors, best_predictor)

  # Remove best predictor from the remaining list
  remaining_predictors = setdiff(remaining_predictors, best_predictor)

  # Print current step results
  cat("Step:", i, "Selected predictor:", best_predictor, "Accuracy:", best_accuracy, "\n")

  if (best_accuracy > overall_best_accuracy) {

    overall_best_accuracy = best_accuracy

    best_selected_predictors = selected_predictors

  }

}

```

```

# Final selected predictors
print(selected_predictors)

print(best_selected_predictors)

}

...

```{r}

if (run_image_classification == TRUE) {

 # Load Nyagatare image
 image = rast("C:/Users/Admin/Documents/MDS/Dissertation/Data/Nyagatare
Images/combined_composite_float32_converted_export.tif")

 image_df = as.data.frame(image, xy = TRUE, na.rm = FALSE)

 image_colnames = c("B2", "B3", "B4", "B5", "B6", "B7", "B8", "B8A", "B11", "B12", "VV", "VH")

 colnames(image_df)[-1:2] = image_colnames

}

...

```{r}

if (run_image_classification == TRUE) {

  image_df = image_df %>%
    dplyr::mutate(NDVI = (B8 - B4) / (B8 + B4))

  image_df = image_df %>%
    dplyr::mutate(NDWI = (B3 - B8) / (B3 + B8))

}

...

```{r}

if (run_image_classification == TRUE) {

 predicted_classes = predict(model_rf_best, newdata = image_df)

 image_df = image_df %>%
 dplyr::select(x, y)

 image_df$predicted_class = predicted_classes

}

...

```{r}

if (run_image_classification == TRUE) {

  area_estimates = image_df %>%
    dplyr::group_by(predicted_class) %>%
    dplyr::summarise(Count = round(n() * 400 / 10000, 0))

}

...

```{r}

if (run_image_classification == TRUE) {

 classification_raster = rast(nrows = nrow(image),

```

```

 ncols = ncol(image),
 ext = ext(image),
 crs = crs(image),
 resolution = res(image))

n_cells_raster = ncell(classification_raster)
n_rows_df = nrow(image_df)

if (n_cells_raster != n_rows_df) {

 warning(paste("Classification raster does not have the same number of pixels as the initial image."))

}

values(classification_raster) = as.numeric(predicted_classes)

}

'''

'''{r}

if ((run_image_classification == TRUE) & (save_classified_image == TRUE)) {

 output_filepath = "C:/Users/Admin/Documents/MDS/Dissertation/Data/Nyagatare Images/classified_image_1.tif"

 writeRaster(classification_raster, output_filepath, overwrite = FALSE)

}

'''

```

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